

Kidney Replacement Therapy (Dialysis and Transplantation) in Children

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KEY POINTS

- End-stage kidney failure (ESKF) in children has multiple and far-reaching clinical consequences, including a major impact on growth and development.
- The most common causes of ESKF in children are congenital anomalies of kidneys and urinary tract, acquired glomerular disease, and genetic diseases.
- Although kidney transplantation is the preferred modality of kidney replacement therapy (KRT), most children require dialysis before transplantation. Many children will require multiple KRT modalities before transitioning to adult services.
- Choice of dialysis modality depends on the child's age, underlying kidney disease and presence of comorbidities, geographic and center-specific factors, and social considerations.
- Some complications are unique to or more severe in pediatric dialysis patients, such as catheter malfunction, vascular damage and stenosis, and infections.
- Advances in kidney transplantation over the past decades have transformed the field, enabling children with ESKF to live close to normal lives, with constantly improving outcomes.
- Infections after kidney transplantation may be more common in children, particularly viral infections.
- Clinical data on many medications and interventions are not available in the pediatric ESKF population and require further study.

BACKGROUND

Until the middle of the 20th century, end-stage kidney failure (ESKF) was a lethal condition, with a life expectancy of at most weeks to months. The developments over the subsequent 2 decades in both dialysis and transplantation have transformed the lives of patients with chronic kidney disease (CKD), dramatically reducing mortality and morbidity. As technologies and surgical techniques developed, these treatments became available for children and later infants as well. In this chapter we discuss various kidney replacement therapy (KRT) modalities—for long-term dialysis (hemodialysis [HD] and peritoneal dialysis [PD]), acute dialysis (including continuous kidney replacement therapy [CKRT]), and transplantation in children.

DEFINITION OF CHRONIC KIDNEY DISEASE

CKD is defined as abnormalities of kidney structure or function, present for >3 months, with implications for health (Table 72.1).¹

For children, the same criteria generally apply as for adults, with the following notable exceptions:

- (a) The criteria for duration >3 months do not apply to newborns or infants <3 months of age.
- (b) The criteria of a glomerular filtration rate (GFR) <60 mL/min/1.73 m² do not apply to children <2 years of age in whom pediatric-specific formulas (modified Schwartz: estimated GFR [eGFR] = height [in cm] × 40.5 [coefficient]/serum creatinine level [in μmol/L]) should apply. More detailed consideration of GFR estimation in children is provided in Chapter 71.
- (c) In children with a low muscle mass or in those with a metabolic disorder that limits dietary protein

restriction, creatinine levels may be spuriously low. Cystatin-C, a nonglycosylated, 13.3-kDa protein belonging to cystatin protease inhibitors, is not reabsorbed in the proximal renal tubule and not affected by gender, age, population ancestry or ethnicity, protein intake, and muscle mass, unlike serum creatinine, and may be a better marker of eGFR.^{2,3}

- (d) A urinary total protein or albumin excretion rate above the normal value for age may be substituted for albuminuria ≥30 mg/24 hours.
- (e) All electrolyte abnormalities are to be defined considering age-normative values.

The KDOQI classification system identified five stages of CKD, from mild disease (stage 1) to ESKF (stage 5), based primarily on the level of GFR and extent of albuminuria, which is an indicator of glomerular dysfunction.

Treating pediatric ESKF poses unique challenges to the medical team and health care system, which must address not only the disease itself but also the many manifestations that affect patients' lives and families. Reynolds and colleagues⁴ reported 20 years ago that compared with the general population, adult survivors of pediatric ESKF were less socially mature and had lower educational qualifications. Fewer had intimate relationships outside the family, and more were unemployed. Two decades later, changing this unfavorable state remains the major goal for providers treating children with ESKF.⁴

EPIDEMIOLOGY OF END-STAGE KIDNEY DISEASE

Approximately 80% of pediatric KRT is provided to patients who live in high-income countries, who account for only 12% of the global population. Registries, such as the North American Pediatric Renal Trials and Collaborative Studies (NAPRTCS), the European Society of Pediatric Nephrology, and the European Renal Association (ESPN/ERA), which can provide robust information, exist only in higher resource countries. The information from lower-resource countries is derived from surveys conducted among health care providers or from reports of patients admitted to tertiary centers. Both methods result in underestimation of the ESKF burden in lower-resource settings because much of the population has limited access to health services. Information from high-income countries shows an incidence of ESKF of 15 per million age-related population (pmarp) in the United States, 9.5 pmarp in most West-European countries and Australia, and only 4.0 pmarp in Japan.⁵⁻¹⁰ This is less than 10% of the incidence of ESKF in adults in the same countries. At the far end of the spectrum, Nepal, Nigeria, and some Eastern European countries report an incidence of less than 1 pmarp,^{9,11} reflecting underdiagnosis rather than true rarity of ESKF. The incidence, in all registries, is higher in adolescents due to new cases of acquired kidney disease typical to this age group, as well as the accelerated progression of CKD during adolescence. Rapid physical growth and possibly also hormonal factors are responsible for the decline in kidney function observed in adolescents with CKD. Higher incidence in males, due to the higher prevalence of congenital anomalies of kidneys and urinary

Table 72.1 Criteria for Defining Chronic Kidney Disease (CKD)

Criteria for CKD (Either of the Following Must Be Present for >3 Months)	
Markers of kidney damage	• Albuminuria (>30 mg/24 h) or albumin:creatinine ratio >30 mg/g (3 mg/mmol) • Ultrasound evidence of structural abnormalities • Disease-specific mutations detected by genetic analysis • Electrolyte abnormalities and acidosis/alkalosis due to tubular disorders • Abnormalities detected on histology • History of kidney transplantation
Decreased GFR	GFR <60 mL/min/1.73 m ² (GFR G3a–G5)

tract (CAKUT) in boys compared with girls, is also a finding common to all registries and reports.¹²⁻¹⁶

Over the years there has been an increase in both incidence and prevalence of pediatric ESKF in Western countries (U.S. Renal Data System [USRDS], 2017).¹⁷ This may not indicate a real biological change in disease incidence but instead may result from the inclusion of individuals (such as infants), who in the past would not have been thought to be eligible for RRT. In addition, improved childhood RRT survival rates may increase the ESKF prevalence.

The European Society of Paediatric Nephrology/European Renal Association Registry reports that the overall prevalence of KRT in European children has increased by nearly 2% per year, from 2120 children (29.5 per million age-related population [pmarp]) in December 2007 to 2675 (35.6 pmarp) in December 2016. The prevalence of patients on all KRT modalities increased during the same study period.¹⁸ These data are in line with reports from the Australian registry, while in the United States and New Zealand, the trend has remained stable over the same time period.

The ancestry composition of each one of the registries may also affect the incidence of ESKF reported, suggesting yet undefined genetic susceptibility factors. In the United States, African-American children have a twofold increased incidence of ESKF compared with Caucasian children.⁶ High-risk variants in the *APOL1* gene may partially account for this observation, as they appear in about 16% of African-American children and not in other ethnic groups. CKD patients with high-risk *APOL1* variants present with later-onset focal segmental glomerulosclerosis (FSGS; 89% of CKD patients with high-risk variants, compared with 25% with low-risk genotype), with lower GFR at diagnosis, rapid decline in GFR, and higher risk of uncontrolled hypertension (HTN).^{19,20} Differences related to population ancestry and ethnic affiliation are also seen in other countries. In the United Kingdom the incidence is higher in patients who originate from Southeast Asia, and in Australia and New Zealand, the incidence was higher in indigenous populations.^{21,22} In Kuwait, the incidence of CKD is high (17 pmarp), probably due to the increased frequency of autosomal recessively inherited diseases among populations with a high rate (40%–50%) of consanguineous marriages.²³

ETIOLOGY

A strictly defined and universally accepted CKD etiologic classification is often lacking. In spite of these caveats and lack of concise etiologic definitions, CAKUT remains the leading primary diagnosis in pediatric patients with ESKF worldwide, responsible for 24% to 39.5% of patients with ESKF.^{6,8,10,12,17,24,25} Of note, this is proportionately lower compared with the incidence of CAKUT at earlier CKD stages, reflecting the relatively slow progression rate of this subgroup of diseases, at least during childhood and early adolescence. Hypoplastic-dysplastic kidneys with or without vesicoureteral reflux (VUR) and obstructive uropathies are the main categories of CAKUT, and the differences in listed diagnoses between various reports and registries are often semantic rather than biological in nature. In most countries,

the second most frequent category is glomerulonephritis (8.3%–30.4%), of which the most common entity is FSGS, especially in Japan and the United States, where it is the most common cause of ESKF among adolescent African-Americans. Inherited diseases comprise the next category, although there are variations in which diseases are classified under this heading in different reports. The frequency varies from 19% (Australia and New Zealand) to 47.5% (Iran).^{5,24} In countries where there is a high rate of consanguineous marriages, autosomal recessive inherited diseases comprise a much higher proportion of ESKF etiology. Specific diagnoses include cystinosis, familial steroid-resistant nephrotic syndrome (SRNS; mostly due to mutations in *NPHS1*, *NPHS2*, and *PLCE1*), autosomal recessive polycystic kidney disease, PH1, nephronophthisis, and others.

A major obstacle to accurately comparing the various etiologies leading to ESKF worldwide is the definition of a given entity, which may vary by reporting center. This may be due to the introduction of a more accurate diagnosis based on genetic or other diagnostic study, which may be expensive and thus not widely available, or to adherence to previous dogmas that are now regarded as inaccurate. These include FSGS, which is a histologic diagnosis encompassing different disease processes, or reflux nephropathy, which may be more accurately described as hypodysplastic kidneys with VUR. In addition, a specific patient's diagnosis may occasionally fit more than one etiologic category; for example, autosomal-dominant FSGS may be recorded under FSGS or as a hereditary disorder. For a more detailed consideration of etiologies, the reader is referred to Chapter 71.

CLINICAL CONSEQUENCES OF PEDIATRIC END-STAGE KIDNEY DISEASE

The key aspects of CKD management are summarized in Table 72.2 and discussed in detail as follows.

NUTRITION AND GROWTH

Nutrition has a critical role in childhood. Multiple factors can impair linear growth and neurocognitive development as described in Table 72.3. The Kidney Disease Outcomes Quality Initiative (KDOQI) guidelines²⁶ and Pediatric Renal Nutrition Taskforce, a team of doctors and dietitians from Europe and North America, have produced clinical practice recommendations on the assessment²⁷ of nutritional status and management of various aspects of the nutritional prescription,²⁸⁻³³ as well as the role of enteral tube feeding.³⁴ Detailed guides for health care professionals, as well as children and their caregivers, are available at <https://www.espn-online.org/nutrition-taskforce>.

Malnutrition, which is the result of poor appetite, vomiting, and decreased intestinal absorption, is common in children with ESKF. Close monitoring is crucial, especially when the child is younger than 1 year (every 2–4 weeks), decreasing in frequency as the child grows.³⁵ Height and weight are plotted on appropriate growth curves, and Z-scores are calculated. Head circumference is measured regularly in children younger than 3 years. Most younger children fail to

Table 72.2 Clinical Consequences and Key Aspects of the Management of Chronic Kidney Disease (CKD) in Children

Hypertension	Proteinuria
- Individual BP targets are based on age, height, and comorbidities - Low-salt diet and normalize weight (children with salt-losing disease such as CAKUT do not require a low-salt diet, especially in the early stages of CKD) - Regular physical exercise - Minimize screen time	- RAAS inhibition with angiotensin-converting enzyme inhibitors (ACEis) or angiotensin receptor blockers (ARBs) - Newer antiproteinuric treatments including sodium-glucose cotransporter 2 inhibitors (SGLT2i) are under investigation in children
Chronic Hyperkalemia	Metabolic Acidosis
Dietary potassium restriction Loop diuretics and potassium-binding resins should be administered or dose adjustments of RAAS inhibitors and aldosterone antagonists must be considered	- Oral bicarbonate - Maintain bicarbonate levels within normal range (>21 mmol/L)
Mineral and bone disease - Dietary restriction of phosphate from early stages of CKD - Vitamin D supplementation while monitoring of 25-hydroxyvitamin D levels - Phosphate binders. In CKD G3a-G5 (including those on dialysis), lower elevated phosphate levels toward the normal range but avoid hypercalcemia - For patients on dialysis, PTH-lowering therapy, vitamin D analogs and calcimimetics are recommended	Anemia - Hb should be maintained within normal range for age - Erythropoiesis-stimulating agents (ESAs) - Iron supplementation, oral or intravenous - B₁₂ and folate supplementation - Avoid blood transfusions, especially in potential transplant recipients to avoid sensitization

meet the energy and protein requirements by oral feeding alone and need supplementation by nasogastric tube or gastrostomy.³⁴

Initial recommended caloric intake is the age-related dietary reference intake (DRI) but is adjusted according to weight gain. Protein intake recommendation is 100% to 120% of the age-related DRI, with somewhat higher intake in PD patients to compensate for peritoneal losses.³⁶ In contrast to adults, a Cochrane review found that protein restriction has no impact on delaying progression of renal failure.^{37,38} Dietary restrictions may include fluid, sodium, phosphorous, and potassium, and special infant formulas are adapted to the needs of patients with CKD. Human milk also has the adequate electrolyte composition for the infant with CKD,

Table 72.3 Risk Factors for Growth Failure in Children with Chronic Kidney Disease (CKD)

Risk Factor	Pathophysiology
Degree of CKD	Standardized height worsens with the progression of CKD. A strong correlation is present between eGFR and height SDS (−3.2, −1.9, −1.5 and −0.9 for GFR <10, 10–25, 25–50, and >50 mL/min/1.73 m ² , respectively) ⁹
Malnutrition	Due to a combination of altered taste sensation, altered clearance of cytokines that affect appetite and satiety, electrolytes, and fluid imbalance
CKD-MBD	Results from high FGF23 levels, hyperphosphatemia, 25-hydroxyvitamin D deficiency, reduced production of 1,25-dihydroxyvitamin D, hypocalcemia, hyperparathyroidism, malnutrition and drug toxicity (corticosteroids, calcineurin inhibitors)
Metabolic acidosis	Due to the decrease in total renal ammonium excretion when the GFR falls to <40–50 mL/min/1.73 m ² (CKD G3)
Anemia	Low erythropoietin production, reduced lifespan of red cells, and impaired intestinal iron absorption
Endocrine dysfunction	Resistance to growth hormone (GH), modifications of the GH insulin-like growth factor 1 (IGF-1) axis
Underlying syndrome	Genetic disorders such as trisomy 21, Turner syndrome, and multiple inherited disorders that affect kidneys
Treatments	Example: corticosteroids

as it is relatively low in potassium and phosphorus. However, because infants’ diet is fluid based, it may ultimately cause volume overload. This requires in some cases concentrating the formula to achieve a higher caloric intake in a smaller volume (80–120 Kcal/100 mL vs. the usual 67 Kcal/100 mL). It may also influence the dialytic modality selected or the frequency of the dialysis sessions. Conversely, in several renal diseases (e.g., dysplastic kidneys, nephronophthisis, and cystinosis) patients may be polyuric due to a renal concentrating defect even when they reach ESKF. This makes water handling and nutrition possible but will require careful attention to electrolyte replacement, too.

LINEAR GROWTH

Growth is severely impaired in children with ESKF. A longer period with ESKF and short stature at disease onset is associated with decreased final height. Short stature affects body image and may have an impact on self-esteem. In

the context of chronic illness, this leads to overprotection, low expectations, and delayed independence and may subsequently affect future socioeconomic status and formation of intimate relationships.

When assessing patient height, it should be remembered that there are differences between ethnicities and growth curves, with normal values available only for some (higher income) countries.

Many factors contribute to impaired linear growth:

- Low energy and protein intake: This is especially significant in early childhood.
- Impaired growth hormone (GH)–IGF-1 (insulin-like growth factor 1) axis.
- Additional factors include water and electrolyte imbalance, anemia, mineral bone disease, and metabolic acidosis.³⁹ Correcting these abnormalities may improve the likelihood of achieving normal growth.

Beyond correction of metabolic abnormalities and providing the best possible nutritional support, treatment includes recombinant human GH (rhGH) therapy.

The use of GH in kidney failure is based on experimental and clinical evidence demonstrating that GH insensitivity can be overcome by supraphysiologic doses of exogenous GH.⁴⁰ This treatment was proved to be effective in a study based on NAPRTCS data: Of 5122 dialysis patients, 33% received rhGH treatment. Catch-up growth was achieved in 11% of the treated dialysis patients (compared with 27% of patients in earlier stages of CKD). The mean increase was 0.5 standard deviation score (0.8 standard deviation score in earlier CKD stages).⁴¹ In addition, Cochrane database meta-analysis found increased height velocity (+3.9 cm/year after the first year), without advance in bone age.⁴² In the NAPRTCS 2011 report, children younger than 6 years of age had improvement in Z score of +0.63 compared with −0.1 in the untreated patients. Older children had milder improvement of +0.26 Z-scores.⁴³ A consensus document from ESPN describes the indications and management of GH therapy in ESKF.⁴⁴

HYPERKALEMIA

Hyperkalemia is commonly found in children with ESKF, due to decreased urinary excretion. Treatment includes dietary potassium restriction, particularly the restriction of processed foods that are likely to contain potassium additives,³⁰ and the use of potassium binders such as Patiromer,⁴⁵ sodium zirconium cyclosilicate,⁴⁶ or sodium polystyrene sulfonate. Formula adapted for infants with CKD, low in potassium, phosphorous, and calcium, is commercially available.

METABOLIC ACIDOSIS

Chronic metabolic acidosis is treated in children because of its impact on enzymatic functions, bone demineralization, and linear growth. The effect of acidosis on bone health and growth is multifactorial including increased osteoclastic and decreased osteoblastic activity; blunted action of the GH–IGF-1 axis on bones; decreased 1,25-dihydroxyvitamin D production, and hence less intestinal calcium absorption;

and alteration of homeostatic relationships among vitamin D, parathyroid hormone (PTH), and ionized calcium. Metabolic acidosis is treated with bicarbonate salts with target bicarbonate serum concentrations of 20 to 22 mEq/L.

ANEMIA

Hemoglobin starts to decline in children when eGFR is lower than approximately 50 mL/min/1.73 m².⁴⁷ Diagnosis of iron deficiency may be challenging because many patients with ESKF have low transferrin levels due to malnutrition or chronic inflammation, causing the calculated transferrin saturation to be seemingly normal. Ferritin concentrations, by contrast, may be high as ferritin is an acute-phase reactant and increases in inflammatory states. The physiologic hemoglobin concentrations vary with age and gender, as do the recommended target levels: above the fifth percentile of the specific age. Treatment includes iron supplementation and erythropoiesis-stimulating agents (ESAs). Target iron levels are higher than in the healthy population: transferrin saturation greater than 20% and ferritin greater than 100 ng/mL (to optimize the effect of ESA). The relative ESA dose (units/kg/week) is higher in children and may even reach 1250 erythropoietin units/kg/week in infants.⁴⁸ Other causes for failure of ESAs include osteitis fibrosa cystica, chronic inflammation, malnutrition, hemolysis, and rarely carnitine deficiency or aluminum toxicity. Studies in adults on dialysis found that excessive correction of anemia (even within the physiologic hemoglobin level) is associated with increased mortality; therefore it is recommended not to exceed hemoglobin levels of above 12 g/dL. The pathophysiology of this phenomenon is unclear and has not been validated in children.

CHRONIC KIDNEY DISEASE–MINERAL BONE DISEASE

Although the pathophysiology of CKD-mineral bone disease (MBD) is similar to that in adults, both the management and the consequences differ, as linear growth and rapid accrual of calcium into the growing skeleton is an important factor in children.^{49,50}

Monitoring of various CKD–MBD-associated parameters includes the serial measurement of serum calcium, phosphorous, PTH, and alkaline phosphatase levels every 1 to 3 months^{51,52} and vitamin D every 3 to 12 months.⁵³ Bone mineral density may be measured by dual-energy x-ray absorptiometry (DXA); however, this modality is limited in differentiating between trabecular and cortical bone in CKD–MBD and thus does not predict fracture risk well in this population.⁵⁴ Peripheral quantitative computed tomography is another modality to assess bone mineralization, which can distinguish between cortical and trabecular bone.⁵⁵ Adjustment of DXA spine and whole-body measures for height Z-scores can provide clinically relevant data on bone health that can be monitored over time.⁵⁶

Because normal blood calcium and phosphorous concentrations and dietary requirements²⁹ are age dependent, the target levels and nutritional recommendations also vary with age.²⁹ Age-adjusted hypophosphatemia should be avoided as it can result in hypophosphatemic rickets, often seen in preterm infants with insufficient phosphorous intake,

in children with phosphaturia due to proximal tubulopathy, or in those on intensified dialysis regimens. In CKD stage 5 the KDOQI 2017 guidelines recommend phosphorous restriction depending on blood levels.⁵⁷ Phosphorous restriction can be challenging to achieve and maintain, particularly in teenagers who may indulge in more processed and ultraprocessed foods that have a high phosphate content and high bioavailability of the phosphate salts used in these. Of note, a person on a strict phosphate-controlled diet is likely to become calcium deficient, as a similar range of foods are rich in both calcium and phosphate.^{58,59}

A controlled phosphate intake is expected to result in lower PTH and increased 1,25-dihydroxyvitamin D (calcitriol) levels, as well as better bone morphology and reduced risk for CKD-associated vasculopathy.^{57,60-62} The first-line phosphate binders in children remain calcium-based phosphate binders.^{51,52} The non-calcium-based phosphate binder, sevelamer carbonate or sevelamer hydrochloride, can be used as a second-line agent in those who are hypercalcemic and have comparable phosphate-binding efficacy.⁶³⁻⁶⁵ The use of lanthanum carbonate is not recommended in children because lanthanum deposits in bones, including growth plates, and the consequences on the developing bone are yet unknown.

Vitamin D monitoring and supplementation are important in patients at all CKD stages. The use of activated vitamin D analogs, such as alfacalcidol and calcitriol, in adults is limited to CKD stage 5D, but as recommended by KDIGO and the European Society for Pediatric Nephrology, in children their use may be considered to maintain serum calcium levels in the normal range in CKD stages 2 to 5D.^{53,57} Treatment with activated vitamin D analogs should be started if hyperparathyroidism is persistent despite therapy with native vitamin D, aimed at normalizing serum calcium concentration and decreasing PTH (although the target PTH values at each CKD stage are not widely agreed^{66,67}). Finally, calcimimetics, which increase the sensitivity of the parathyroid calcium receptors, may have a role in the management of persistent hyperparathyroidism that is resistant to conventional treatments with calcium and vitamin D and should be considered before parathyroidectomy.⁶⁸

Deformities of the growing bone and impaired linear growth are complications that are typical in the pediatric age group (Fig. 72.1), in addition to vascular calcifications (VCs) that are also well described in the adult patient population. Skeletal deformities in CKD, like the phenotype of vitamin D-deficient rickets, can be debilitating but are amenable to medical therapy with marked improvement and even complete resolution.

CARDIOVASCULAR DISEASE

Cardiovascular disease (CVD) in children with CKD is the most important comorbidity affecting long-term survival, with a 500- to 1000-fold greater cardiovascular mortality compared with the general population.⁶⁹ Different factors that contribute to developing CVD include hypertension, disorders in mineral metabolism, dyslipidemia, hyperuricemia, abnormal glucose metabolism, obesity, systemic inflammation, and oxidative stress.⁷⁰ Uncontrolled hypertension affects approximately 25% of patients with CKD

stages 3 and 4 and 47% with CKD stage 5.⁷⁰ Hypertension is most commonly due to fluid overload, especially in patients with advanced stages of CKD and on dialysis, but other contributing factors are hyperactivation of the renin-angiotensin-aldosterone system (RAAS), catecholamine secretion and hypervolemia, and in some cases is caused by corticosteroids or calcineurin inhibitors used to treat the underlying kidney disease.^{71,72} Atherosclerotic disease that involves calcification of the tunica media⁷³ may be seen even in young children on dialysis,⁷⁴ eventually contributing to left ventricular hypertrophy (LVH). Hypertension along with sustained high levels of FGF23, hypervolemia, and anemia are the drivers of LVH, either concentric or eccentric, which leads to systolic and diastolic cardiac dysfunction.^{75,76} Finally, carotid intima-media thickness, a surrogate marker of CVD, has been found to be elevated in 41.6% of children with CKD stages 3 to 5.⁷⁷ Further details are described under transplant outcomes later.

NEURODEVELOPMENT

Numerous reasons account for neurodevelopmental compromise in children with ESKF. Some pediatric kidney diseases are a part of a syndrome or systemic disease, which may have an effect on the central nervous system and psychomotor development (e.g., syndromes such as Bardet-Biedl, Joubert, trisomy 21, Galloway Mowat, Wilms tumor, aniridia, genitourinary anomalies, and intellectual disability).

Chromosomal microarrays to detect genomic imbalances were performed in children enrolled in the Chronic Kidney Disease in Children study.⁷⁸ Genomic imbalances were detected in 7.4% of children with CKD. In the vast majority of these cases, the genomic lesion was unsuspected based on clinical assessment.⁷⁸ Other pediatric kidney diseases can be associated with prematurity (Finnish-type nephrotic syndrome) or oligohydramnios and hypoplastic lungs and hence possible perinatal hypoxia (autosomal recessive polycystic kidney disease). Uremic toxins, HTN (or BP fluctuations during HD), and chronic anemia can cause further damage to the developing brain. Motor skills development can also be delayed because the sick infant is not free to move and play while connected to the dialysis or feeding machine. Various catheters including feeding tubes and gastrostomy may interfere with rolling over and crawling, and there is less sensory stimulation during hospitalization or dialysis. Muscle tone is diminished, and renal osteodystrophy may be painful and prevent the needed physical activity for normal development. Some medications frequently given to CKD patients, including aminoglycoside antibiotics (limited use, only in anuric patients) and furosemide, are potentially ototoxic.

Several studies over the past 30 years showed that patients with CKD have compromised cognitive functions. This can be seen even in the early stages of CKD.⁷⁹ Specifically, studies found lower intelligence quotient (IQ) levels (~10 points), verbal or written language difficulties, impaired memory, decreased executive functions, and inefficiency in key neurocognitive domains.⁸⁰⁻⁸⁴ Particularly affected are children who reached ESKF at a younger age,^{81,85} children with longer dialysis vintage,^{76,80,81} more severe

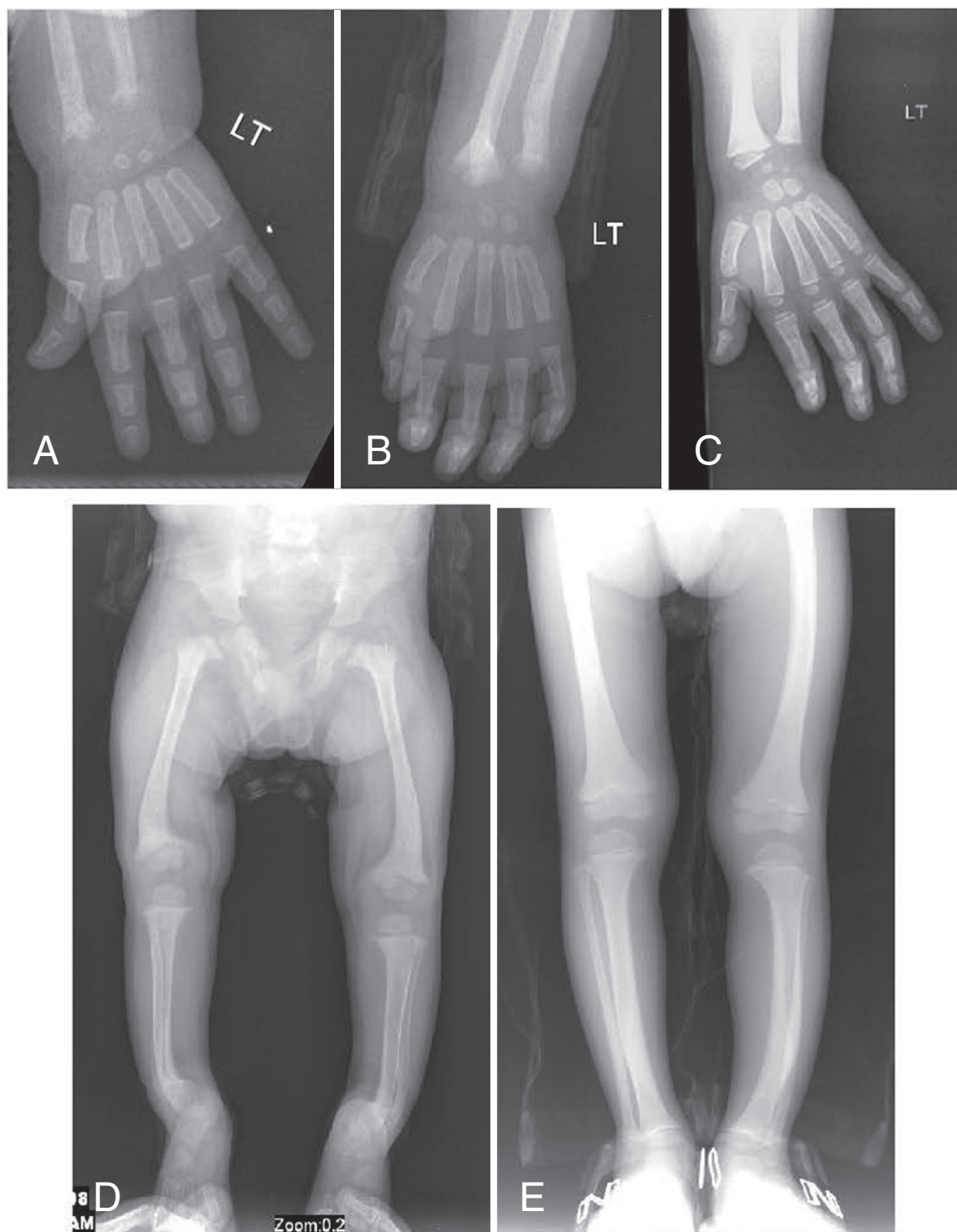


Fig. 72.1 Severe renal osteodystrophy in a boy with end-stage renal disease with resolution obtained with medical treatment. (A) Age 1 year: severe rickets, absorption of the metaphyseal edges of the radius and ulna. Delayed bone age (6 months). (B) Age 2 years: deformation in the radius and ulna bones, widening of the metaphyseal edges, healing rickets. (C) Age 4 years: normal bone structure. Bone age is 3 1/2 years. (D) Age 2 years: deformation with angulations of the femur, tibia, and fibula bones—healing phase. (E) Age 4 years: marked improvement, although some angulation can still be seen.

HTN or hypertensive crisis events,^{86,87} and children with comorbidities.⁸⁸ Sensorineural hearing loss was found in 14% to 18%, and ischemic lesions of variable severity on magnetic resonance imaging (MRI) were seen in 18% to 33% of the patients.^{86,88} Despite this, 61% to 79% of the

children with ESKF attend regular school and have academic achievement comparable with their siblings.^{82,83,86} Long-term follow-up in adults treated since childhood for ESKF found impaired schooling and lower IQ (9.2–10.4 points less than controls).⁸⁹ Fortunately, the current ESKF treatment

modality (dialysis or transplantation) was not associated with decreased IQ score. A study in children and adults younger than 30 years of age assessed neurocognitive functioning using a comprehensive battery of tests and brain structure by MRI. It showed that a combination of risk factors (including time since kidney transplantation, longer dialysis duration and late CKD onset) was significantly associated with lower intelligence and/or worse processing speed and working memory and that disrupted white matter integrity may contribute to these neurocognitive impairments.⁹⁰ A further study from the CKiD cohort has shown that despite high rates of high school graduation, nearly 20% of patients with CKD are unemployed or receiving disability at long-term follow-up.⁹¹ Tailored interventions may benefit patients with CKD with lower kidney function and/or executive function deficits to optimize educational/employment outcomes in adulthood.

QUALITY OF LIFE

Quality of life of children on RRT was assessed in only a few studies.⁹² Quality of life scores are lower than the general population norms in all domains. Generally, patients perceive their quality of life as better than their parents do.⁹³ This may be because patients are unaware of an alternative reality, developing a defense mechanism, or reflecting parental overprotection. The disadvantage is that parents' lower rating may decrease the patients' or their parents' motivation and thus may adversely affect their outcomes. A recent study has shown that despite the presence of a chronic disease, children with CKD endorse a positive self-concept, which may predict academic success in this population.⁹⁴

MORTALITY

Mortality rate in pediatric patients with ESKF is 30 to 74 times higher than healthy age-adjusted population,^{89,95} with overall survival (in Australia) of 79% at 10 years and 66% at 20 years.⁹⁶ Factors associated with higher death rates are earlier era, younger age at initiation of RRT, and mode of treatment (dialysis mortality rate higher compared with transplantation). These factors were repeatedly confirmed in other studies.^{6,7,25} Most deaths occurred within the first year of life. Analysis of the USRDS data by type of RRT found 5-year survival of 97% for transplanted patients, 81% for HD patients, and 83% for PD patients (all somewhat better than results achieved in the period 2001–2005). Finally, there are also racial disparities, with Hispanic children having the lowest mortality rates, followed by non-Hispanic white children, and non-Hispanic black children with the highest mortality rates.⁹⁷

The main causes of death were documented as cardiovascular disease and infection, responsible for 45% and 21%, respectively, according to the Australian and New Zealand Dialysis and Transplant Registry (ANZDATA).⁹⁶ Similar findings in the United States showed death from cardiovascular disease of 27.1% to 39.3% and from infection of 9.7% to 22.5%.⁹⁸ According to the USRDS, the 1-year adjusted cardiovascular mortality rates were lower and declining over time: 7.5 per 1000 patients from 2010 to 2014, compared with 14 per 1000 patient years in 2005 to

2010 (47% reduction). This was noted in all age groups and for all treatment modalities. A similar trend was found for infection-related mortality: a decline from seven per 1000 patient years in 2005 to 2009 to four in 2010–2014. The decrease in the cardiovascular mortality is (at least in part) due to better definition of the cause of death. A different point of view was presented by Kramer and colleagues,⁹⁹ studying young adults who started RRT as children. The 5-year survival from the 18th birthday was 95.1% (confidence interval [CI] 93.9–96.0), with an average life expectancy of 63 years for young adults with a functioning graft and 38 years for those remaining on dialysis.⁹⁹ This was confirmed in a study by Gansevoort and colleagues¹⁰⁰ that showed a significant decrease in expected life-years in patients with CKD stage 5 compared with those with CKD stages 1 to 2.

CARDIOVASCULAR MORTALITY

Cardiovascular mortality is the most frequently reported cause of death: 14% to 41% of all pediatric deaths, depending on age and on population and ethnic affiliation.^{93,96,99,101,102} To underscore the high risk for morbidity and mortality, the American Heart Association classified pediatric CKD patients in the same high-risk group as children with homozygous familial hypercholesterolemia, type 1 diabetes mellitus, and postorthotopic heart transplantation.¹⁰³ In a single study where the authors directly reviewed all patients' files, the report includes under the heading of CVS death, cerebrovascular accident (58% of cardiac deaths), congestive heart failure (15%), and other diagnoses that may be the consequence of HTN, heparinization, and volume overload.¹⁰⁴ In addition, most of the data available are on children who reached ESKF 2 or 3 decades ago. There is marked decrease in not only the absolute death rate but also the percentage of deaths attributed to cardiac causes over the years: from 44.4% in 1972 to 1981 to 33.3% in 1992 to 1999. This was also shown from 1990 to 2010: In patients younger than 5 years at initiation of dialysis, CVS mortality decreased from 35.3% in 1990 to 1994 to 22.6% in 2005 to 2010⁹⁶ or, according to the USRDS, from 29% in 2005 to 2009 to 15% in 2010 to 2014 (same trend, lower percentages, in other age groups). A study from The Netherlands points to a shift from cardiovascular disease to infections as the main cause of death at long-term follow-up in patients with CKD—all these may be reflecting a more accurate diagnosis.¹⁰⁵ However, more specific causes of death must be defined. This is not merely semantic, as accurate determination of causes of death can help focus efforts aimed at improving survival.

KIDNEY REPLACEMENT THERAPY

All available KRT modalities can be used in children. Transplantation is the preferred long-term modality, but it is not always initially possible.

WHEN TO START DIALYSIS

There is no recommended eGFR for initiation of dialysis in the absence of symptomatic uremia, volume overload, or growth failure refractory to medical therapy. The timing

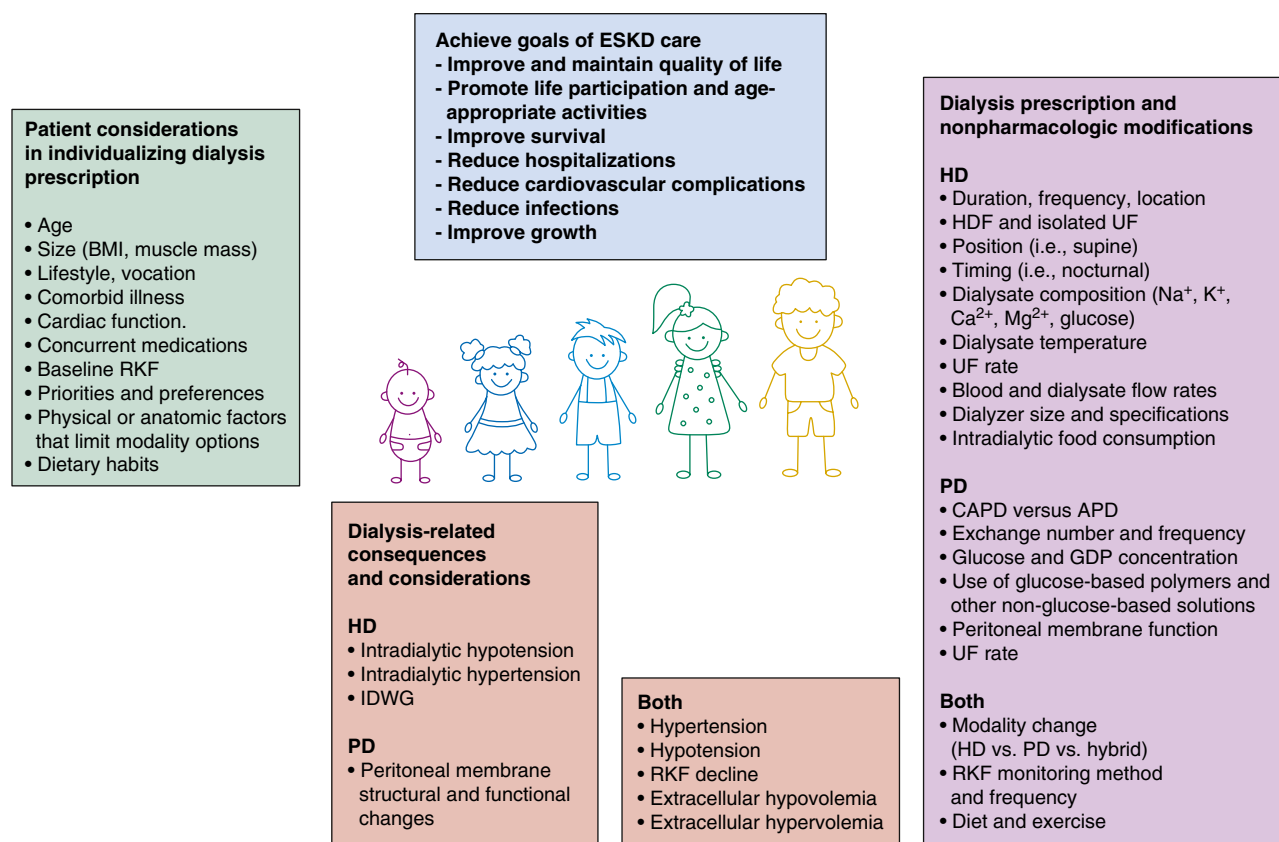


Fig. 72.2 The factors to consider when selecting a dialysis modality.

of initiation of dialysis is guided by a composite assessment of the child's physical and psychosocial well-being, fluid status, growth, level of kidney function, and biochemical abnormalities. The median eGFR at dialysis start is currently 8.2 and 7.8 mL/min/1.73 m² for children in registries from Europe and the United States, respectively.^{18,106} Prospective data from large international registries in more than 21,000 children show that clinical outcomes do not improve in patients starting dialysis early.¹⁰⁷ While young children with structural abnormalities of the kidney can have stable kidney function and adequate urine output for many years, uremic symptoms frequently begin when the GFR falls below 15 mL/min/1.73 m². Therefore use of available formulas (based on eGFR, proteinuria, and/or diagnosis) to prognosticate progression to ESKF and planning for initiation of dialysis when the GFR is <15 to 20 mL/min per 1.73 m² is pragmatic.¹⁰⁸

CHOOSING THE OPTIMAL KIDNEY REPLACEMENT THERAPY MODALITY

The optimal KRT modality in children with ESKF is preemptive transplantation, given the lower morbidity, higher survival, and better quality of life after transplantation. If preemptive transplantation is not feasible, home dialysis therapies are most appropriate for children as they will allow for the child's schooling and social life. No studies have suggested that either PD or HD has superior outcomes in children with ESKF. Rather, different dialysis modalities can be considered complementary to each other and the

most suitable type of dialysis for a given patient at the time should be used. Because children with ESKF have a lifetime of KRT ahead of them, preserving dialysis access including peritoneum and vascular access is paramount. There are few medical contraindications for either HD or PD, so the choice of dialysis modality must be largely based on patient (and caregiver) choice and the availability and expertise of the treating center. The factors to consider when selecting a dialysis modality are shown in Fig. 72.2 and the advantages and disadvantages of different dialysis modalities for children are shown in Table 72.4.

Although PD is the most commonly available home dialysis modality, some pediatric centers also offer home HD. There are international differences in choice of dialysis modality with PD being twice as common than HD in Europe, whereas the ratio is reversed in Asia and the United States^{18,106}; overall ~30% to 50% of children begin dialysis by HD. There is no evidence to suggest a difference either in morbidity or transplant outcome based on the initial dialysis modality in children and adolescents. Given the complexity of dialysis treatment in children, a multidisciplinary team including pediatric nephrologists, dialysis nurses, dietitians, pediatric surgeons, social workers, and psychologists are an important part of the team.

Sometimes, PD may be the only choice in locations where there are no available HD centers at all, or if the patient lives in a remote area where pediatric HD service is lacking, even in high-income countries.^{109,110} The treatment burden of dialysis must be carefully considered. Parents of patients on KRT become high-level health care providers,

Table 72.4 Advantages and Disadvantages of Different Dialysis Modalities

Modality	Advantages	Disadvantages
Peritoneal dialysis	Increased patient freedom Longer preservation of residual kidney function Absence of vascular access Improved school attendance Decreased dietary restrictions Technically more feasible in smaller children Possible for patients who live far away from HD center Lower cost	Dependent on parental/caregiver participation Caregiver burden Medicalization of the home Infection Risk of peritoneal membrane failure if on PD for >5 years
In-center hemodialysis	Long-term technique survival Suitable if patients/parents/caregivers unable to perform dialysis at home Decreased treatment time	Limited by availability of vascular access Availability dependent on local resources Increased cost when compared with other modalities
Home hemodialysis	Increased flexibility with dialysis times Improved school attendance Reduced intradialytic symptoms and hypotensive episodes Improved cardiovascular outcomes compared with ICHD More cost-effective compared with ICHD Improved appetite, growth and energy	Dependent on parental/caregiver participation Caregiver burden More rapid loss of residual renal function Increased fistula complications Medicalization of the home Increased home utility bills (water, electricity)

HD, Hemodialysis; ICHD, in-center hemodialysis; PD, peritoneal dialysis.

are required to solve problems and seek information, and provide financial resources and practical skills. Caregivers are in continuous need of emotional, psychological, and financial support.⁹² These stresses cause fatigue, as well as disruption of work and social life.¹¹¹ Thus once a patient approaches ESKF, the family's abilities and socioeconomic and psychological background must be carefully evaluated by a multidisciplinary team including a nephrologist, dialysis nurse, social worker, and psychologist.

The choice of selection of HD versus PD is greatly influenced by patient age: According to the NAPRTCS 2011 report, of 927 patients initiating dialysis at the age of 0 to 1 year, 857 (92%) were on PD compared with 552 of 727 (76%) in the 2- to 5-year age group, 1373 of 2125 patients (65%) in the 6- to 12-year age group, and 1648 of 3250 (51%) in the 13- to 18-year age group who were on this dialysis mode. In Australia and New Zealand, similar proportions are seen: 89.8% of infants were initially treated with PD, with declining percentage to 34.5% of adolescents.¹⁷ Similar trends were noted in the United Kingdom: at age 0 to 2 years, 80% were on PD and at age 16 to 18 years only 10.4% were on PD. In the United States, there is a steady trend of increased use of HD instead of PD, from 34% of patients treated with HD in 1991 to 54% in 2010. This trend was not found in the United Kingdom, where a relatively constant percentage, around 30%, of the pediatric dialysis patient population has been on HD since 1996. Similarly, in Australia and New Zealand, 30% to 40% of all dialysis patients were on HD during the period 2006 to 2015.

Infants are a unique group for whom the widespread recommendation is to preferentially treat with PD and

to initiate HD only if PD is not feasible. An ESPN/ERA registry study has shown that of the 1063 infants, 13.7% were initiated on HD rather PD. The HD and PD groups had similar characteristics at initiation, and prospective mortality and time to transplantation were the same.¹¹² The proportion of patients younger than the age of 1 year at initiation of dialysis has steadily increased and doubled from 10% to around 20% of all pediatric dialysis patients from 1990 to 2010.¹¹³

HEMODIALYSIS

The principles of HD, physiology of fluid and toxin clearances, and several aspects of the dialysis prescription are similar in children and adults; these have been described in Chapter 62. Here we focus on key technologic differences, modifications of the HD procedure and complications that are distinct for children, and several specific medical issues that must be addressed in addition to tailoring the dialysis prescription for every child according to size.

VASCULAR ACCESS

A well-functioning vascular access is crucial for dialysis. Recent guidelines suggest that children requiring chronic HD should start with a functioning arteriovenous fistula (AVF) where appropriate.¹¹⁴ AVFs have better patency rates, lower access-related infection rates, and decreased need for access revision compared to central venous lines (CVCs). However, CVCs remain the most common access type in all

pediatric age groups, due to the perceived ease of access creation.¹¹⁵ The median access survival time is 3.14 years for AV fistula (95% CI 1.22–5.06) compared with 0.6 year for a CVC (95% CI 0.2–1.00).¹¹⁶ Additional advantages attributed to AV fistulas including better clearance and higher albumin and hemoglobin levels are less convincing.¹¹⁷ CVCs cause more damage to blood vessels and the resulting central venous stenosis may preclude future CVC insertion or ipsilateral AVF creation. This is important in children, who have a lifetime of KRT ahead of them, often also for a second dialysis period after failure of kidney transplant, and therefore blood vessel preservation is crucial.¹¹⁸ Therefore AVF is the preferred vascular access, except for children weighing less than 15 to 20 kg, but this depends on the skills of the vascular surgeons.

Clinical practice recommendations promote the use of AVFs over CVCs,¹¹⁴ but pediatric registries suggest a different picture. A survey in Europe and the International Pediatric Hemodialysis Registry found that the use of CVCs was more frequent than recommended in prevalent pediatric HD patients: 40% had an AVF or a graft versus 57% who had a CVC.^{115,119} Only in children older than 15 years of age were there more patients with AVF as an access.

CENTRAL VENOUS LINE

A subcutaneously tunneled, cuffed CVC may be preferred in small children or those with an anticipated short dialysis course before transplantation or transition to PD.¹¹⁴ The type and size of catheter and site of insertion are individualized for each patient. CVCs should be adjusted to patient size, aiming for the largest-bore catheter that can be accommodated in the vessel without the risk of vascular damage and stenosis (Table 72.5). The internal jugular veins (IJVs) are the first choice; the subclavian and femoral routes are associated with increased vessel stenosis and infection, respectively. CVCs are inserted in the IJV so that the tip is in the right atrium using ultrasound guidance. Ultrasound and fluoroscopic guidance are used to prevent catheter malposition.

Table 72.5 Temporary and Tunneled Hemodialysis Central Venous Lines (CVCs)

Patient Weight	Temporary CVCs	Tunneled CVCs
<10 kg	6.5 Fr 10 cm	8 Fr 18 cm (double-lumen)
10–20 kg	8.0 Fr 10 cm	10 Fr 15 cm (split); 8 Fr 18 cm (double-lumen)
10–20 kg	8.0 Fr 12.5 cm	10 Fr 18 cm (split); 14.5 Fr 13 cm (double-lumen)
20–30 kg	11 Fr 12.5 cm	14 Fr 24 cm; 14.5 Fr 19 cm (double-lumen)
>30 kg	11 Fr 15 cm	14 Fr 24 cm; 14.5 Fr 23 cm (double-lumen)

The main CVC complications are infections and malfunction. Infections are more common in younger patients, possibly due to close proximity of the exit site to the infection source: diapers and gastrostomy tubes.^{120,121} Infection rate is usually 1.5 to 4.8/1000 CVC days, although an exceptionally favorable report documented a rate of 0.5/1000 CVC days.^{121–125} Malfunction may be the result of thrombosis, fibrin sheath formation, vascular stenosis, or mechanical damage to the CVC. If local fibrinolysis fails, the CVC must be replaced. A major complication with long-term sequelae is central venous stenosis. It occurs more often in younger patients, with small-diameter veins, with prolonged use of CVCs, with many reinsertions, and possibly also depending on CVC locations. The worst outcome is for subclavian veins, followed by left IJ vein, whereas the right IJ vein has the smallest chance of becoming obstructed. Clinically, stenosis is often asymptomatic until an ipsilateral AV fistula is constructed, but sometimes superior vena cava syndrome may develop.¹²⁶

VASCULAR ACCESS–RELATED INFECTION

Infection in vascular catheters may be at the exit site, in the subcutaneous tunnel, or in the catheter.¹²⁷ The development of biofilm within the catheter makes bacterial eradication particularly difficult. Line sepsis may present as a rigor soon after starting dialysis, fever with raised C-reactive protein, or septicemic collapse. Factors increasing the risk of catheter infection include exit site and/or tunnel infection or contamination of the hub, failure of aseptic technique, frequent need to access the catheter during dialysis and long duration of use, use of nontunneled rather than tunneled lines, immunosuppression, hypoalbuminemia, diabetes, and nasal and cutaneous colonization with *Staphylococcus aureus*. Postdialysis heparin or alteplase into the line decreases infection risk by decreasing clot formation, which predisposes to infection.¹²⁸

ARTERIOVENOUS FISTULA

An AVF may be created at the wrist (radiocephalic or radiobasilic), the elbow (brachiocephalic), or by basilic vein transposition to create a brachioasilic fistula. Distal vessels of the nondominant arm are preferred; however, the larger the vessels, the greater the chance of success (intraluminal venous diameter >3 mm is preferred). A child who has had previous central lines may need upper limb venography to confirm patency of the central vessels because ultrasound examination of the proximal vessels can be misleading, and veins cannot easily be seen under the clavicle. The average AV fistula blood flow is 800 to 1200 mL/min. In adults, this is significant compared with the normal cardiac output (CO): normal adult CO is 5.2 to 8.6 L/min and the flow through an AV fistula will increase CO by 9% to 23%. A similar flow in a child, required for adequate dialysis, may increase the CO of a child with a 0.7-m² body surface area (BSA), whose normal CO is 2.1 to 3.5 L/min, by 23% to 57%. Theoretically, this may lead to high-output heart failure, especially in face of the high cardiac morbidity in dialysis patients; in practice, this rarely becomes clinically significant.

AVFs show an average maturation time of 8 to 10 weeks^{114,115,129} and can be cannulated by rope-ladder or

button-hole technique. Because fistula thrombosis/malfunction is the main complication, and this is usually the result of vessel stenosis, routine surveillance of the fistula by Doppler ultrasound performed by an experienced professional should be performed. Vascular surgeons and invasive radiologists should be available to detect and treat stenosis or thrombosis promptly to salvage the AVF.

GRAFTS AND SHUNTS

These are rarely used in children and usually a last resort option. An artificial conduit can be inserted between an artery and vein subcutaneously (graft) where it can be needled, or it may be brought out externally (shunt), where the loop can be disconnected to attach to dialysis lines. Grafts and shunts are, like AVF, also liable to stenosis (particularly at the anastomosis site) and clotting.

COMPONENTS OF THE DIALYSIS PRESCRIPTION

1. Hemodialysis machine: Pediatric hemodialysis machines must be capable of low blood flow speeds, use lines of varying blood volumes, and have precise ultrafiltration (UF) control so that they can be used even in infants; small inaccuracies in volumes removed that are negligible in adults can have a major hemodynamic effect on the small child. Automated volumetric UF systems that precisely monitor pressure across the dialysis membrane (transmembrane pressure, TMP) allow accurate fluid removal. The inherent level of UF inaccuracy mandates careful assessment of the patient’s volume status, especially in smaller children. Several options are now available for continuous online monitoring of changes in hematocrit and blood viscosity (blood volume), oxygen saturation, blood pressure, and pulse, as well as for urea clearance. Modern dialysis machines allow for both HD

and hemodiafiltration (HDF) and are suitable for children above 10- to 17-kg body weight.

2. Dialyzer: The dialyzer (dialysis filter) is selected on the basis of its surface area and the priming volume (Table 72.6).¹³⁰ The dialyzer membrane surface area should be approximately equal to the patient’s BSA. Conventional hemodialysis uses a low-flux (small pore size) membrane, and solute removal is primarily by diffusion. High-efficiency hemodialysis refers to a more rapid removal of urea, potassium, and other small solutes. This is achieved by using a low-flux membrane with a high efficiency (KoA) for removal of small solutes. It is also achieved by using a larger surface area membrane and a high blood flow. Even when using high-flux dialyzers, conventional HD is limited in clearing middle-sized molecules, which are better removed by convection using HDF.

3. Extracorporeal circuit: The tubings and hemodialyzer are selected on the basis that the child can tolerate 8% (up to a maximum of 10%) of their total blood volume (TBV, 80 mL/kg estimated dry weight) in the extracorporeal circuit (Table 72.7).¹³¹ Lines are primed with saline. However, in the very young/small, even the smallest circuit may exceed the safe extracorporeal volume and blood priming of the circuit is needed for each dialysis session, which increases the risks of infection and sensitization before transplantation.¹³¹

4. Water: A water treatment unit purifies municipal water containing potentially hazardous contaminants to dialysis-grade water that is safe if diffused into the bloodstream. A water treatment unit comprises filters to remove particulate matter, activated carbon for chlorine and chloramines, and water softeners to remove calcium and magnesium. Finally, a reverse osmosis unit purifies softened filtered water before it is of a quality suitable for dialysate production.

Table 72.6 Dialyzer Characteristics, Options, and Considerations for Pediatric Hemodialysis

Dialyzer Characteristic	Range of Options	Considerations
Surface area	0.2 m ² -2.5 m ²	Larger surface gives greater capacity for clearance and ultrafiltration but requires high blood flow to achieve these advantages
Priming volume	18 mL-140 mL	Consider along with tubing set volume to determine maximum safe extracorporeal volume, especially for smaller patients
Membrane material	Cellulosic, modified cellulose, artificial	Greater biocompatibility with modified cellulose and further with artificial (plastic) membranes
Sterilization	Ethylene oxide (ETO), γ irradiation, steam, electron beam	Allergic reactions associated with ETO; fewer reactions with γ and steam
Flux	Low flux vs. high flux	Higher-flux dialyzers have higher permeability (e.g., larger pores), permitting higher rates of middle molecule clearance and higher UF rate at a given prescription
Clearance (typically described as K _D for urea [mL/min] at blood flow of 200 mL/min)	K _{DA} Urea 170 mL/min-1700 mL/min	Varies with surface area, flux, membrane material
Ultrafiltration capability (K _{UF} mL/h/mm Hg)	K _{UF} 7 mL/h/mm Hg-111 mL/h/mm Hg	Varies with surface area, flux, membrane material

Table 72.7 Dialyzer and Tubings with Extracorporeal Priming Volumes*

Name	Company	Surface Area (m ²)	Volume (mL)
Dialyzers			
FX Paed	Fresenius	0.2	18
Sureflux 03L	Nipro	0.3	25
Sureflux 05E	Nipro	0.5	35
FX40	Fresenius	0.6	32
Sureflux 07E	Nipro	0.7	45
Sureflux 09E	Nipro	0.9	55
FX50	Fresenius	1.0	53
Sureflux 11E	Nipro	1.1	65
Sureflux 13E	Nipro	1.3	78
FX60	Fresenius	1.4	74
FX80	Fresenius	1.8	95
Tubings			
Neonatal	Gambro		33
Pediatric	Gambro		85
Adult	Gambro		132
Paed AV set ONLINEplusBVM	Fresenius		111
Paed SN set ONLINEplusBVM	Fresenius		142
Adult HDF	Gambro		151
Adult AV set + BVM	Fresenius		192

*Several different makes and sizes of dialyzers are available globally; this is an example only.
 AV, Arteriovenous; BVM, blood volume monitoring; HDF, hemodiafiltration; SN, single needle.

5. Blood flow rate (Q_B) and dialysis flow rate (Q_D): The blood pump flow rate is an important determinant of solute clearance, allowing maximum diffusion and convection; the vascular access determines the maximum achievable Q_B . Wider-bore CVCs and AVF allow higher Q_B . The blood flow rate should be up to body weight (kg) $\times$ 5 to 8 mL/min. Dialysis fluid flow rate (Q_D) is required to be 1.5 to 2 times that of Q_B .

ESTIMATION OF DRY WEIGHT AND FLUID REMOVAL

UF is targeted to achieve dry weight, which is the post-HD weight below which the child will become symptomatically hypotensive. Current guidelines recommend clinical assessment of fluid status, dry weight, and dietetic assessment, at least monthly, and more often in small children.¹³² Objective assessment of dry weight using body composition analysis is a useful adjunct to clinical assessment.^{132,133} The safe limit of UF is no more than 10 mL/kg/hour, and no more than 5% to 8% of body weight should be removed in one session. Adequate UF is difficult to manage for patients who have significant volume overload (a high interdialytic weight gain), have poor vascular tone (critically ill, sepsis), have cardiac dysfunction (risk for myocardial hypoperfusion), and in young children due to small intravascular volume. Careful bedside monitoring, with assistance of technology such as noninvasive blood volume monitors (that monitor circulating blood volume by determining changes in hematocrit), can help safe achievement of dry weight with appropriate UF goals. Regular assessment of UF tolerance,

using frequent HD or long dialysis times to avoid excessive UF rates, is required.¹³²

FREQUENCY OF SESSIONS

Conventional HD is 4 hours 3 times a week. There is emerging evidence of augmented dialysis schedules in children.¹³⁴ Intensified dialysis has been reported from several pediatric centers, with improvement in height, blood pressure, and enhanced control of levels of phosphate and PTH.¹³⁵ Guidelines suggest augmented schedules for patients with chronic fluid overload and cardiac dysfunction or for infants, whose predominantly liquid diet requires removal of relatively large fluid volumes.¹³²

Anticoagulation heparin is the standard anticoagulant used during HD.¹³⁶ It can be infused slowly and continuously throughout the session to prevent blood clotting in the circuit. It is given at a rate of 5 to 50 units/kg/hour through the arterial side of the circuit. Some units use low-molecular-weight heparin (such as dalteparin), given as a bolus of 1 mg/kg at the beginning of the session.

THE HEMODIALYSIS PRESCRIPTION

Once dialysis is started, the recommendation states that solute clearance should be greater than that recommended for adults ($Kt/V > 1.2$, urea reduction ratio $> 65\%$), given that it should support optimal growth and development.¹³⁷ There is little observational information and no randomized studies regarding the correct dose of dialysis for children. Hence HD is arbitrarily delivered thrice weekly, for 3 to 4 hours each session. However, the true adequacy test is based on clinical parameters, which include the short term, optimal

Table 72.8 Components of Hemodiafiltration Prescription

1.	Dialyzer	High-flux membrane $K_{uf} > 20$ mL/h/mm Hg transmembrane pressure/ m^2 and a sieving coefficient for β_2 -microglobulin of >0.6 ; surface area equal to the child's body surface area
2.	Extracorporeal circuit	Volume <10 mL/kg body weight
3.	Blood flow	5-8 mL/min/kg; suggested to increase blood flow rate from 100 mL/min in the first HDF sessions up to 200-250 mL/ m^2 /min, increasing by 10 mL/min per week
4.	Dialysate flow	Twice the blood flow rate
5.	Replacement fluid	Generated online Ultrapure (<0.1 colony-forming unit/mL and <0.03 endotoxin unit/mL) Microbiologic purity (bacterial count and endotoxin level) should be determined regularly at intervals of 3 months and chemical composition checked once a year
6.	Convective flow	Sum of the desired ultrafiltration volume and replacement fluid <u>Postdilution HDF</u> : The convective flow is set at $\sim 30\%$ of the blood flow and is limited by the risk of filter clotting. It typically decreases over the dialysis session to maintain TMP <300 mm Hg. <u>Predilution HDF</u> : The convective flow is set at 100% of blood flow. This can be done despite the dilution of the blood potentially impacting negatively on urea clearance. β_2 -m and phosphate dialytic removal is optimized, as is the clearance of uremic protein-bound toxins.
7.	Dialysate	Produced online; composition similar to that used in conventional HD Dialysate sodium concentration required is lower than in conventional HD. Sodium and UF profiling can help to correct fluid and sodium overload and maintain intradialytic hemodynamic stability and dialysis tolerance
8.	Anticoagulation	Continuous heparin infusion or single dose of low-molecular-weight heparin

balance of electrolytes, BP, and volume and in the long term, adequate weight gain, linear growth, neurodevelopment, and quality of life. Several small studies suggest that delivering more dialysis can improve BP control, hyperphosphatemia, appetite, general well-being, linear growth, and even school attendance and obviate dietary restriction.^{87,138-143} Intensive dialysis may also attenuate vascular disease, which is a major concern.¹⁴⁴ This goal can be achieved by either nocturnal

home HD, performing five to seven sessions per week, hemodiafiltration, or a combination of both. The theoretical basis for these observations was established in a study by Daugirdas and colleagues.¹³⁷ In the standard calculation, the dose of dialysis is scaled to urea distribution volume (V), which is similar to total body water. If, however, dialysis is scaled to BSA, similar to calculation of GFR, because BSA is relatively higher in children, this would require increased dialysis time for smaller children. When these calculations were applied to two of the studies of intensive dialysis, the patients were indeed found to have a much higher weekly dialysis dose.^{141,143}

HEMODIAFILTRATION

HDF combines both diffusive and conductive solute removal by UF of 20% or more of the blood volume processed through a high-flux dialyzer and maintenance of fluid balance by sterile substitution fluid.^{145,146} Essential requirements for performing HDF include (i) “ultrapure” water for substitution fluid, (ii) high-flux dialyzer, and (iii) dialysis machines that allow careful regulation of UF. During HDF, an iso-osmotic substitution fluid is given as replenishment for the removal of extra fluid. The substitution fluid is generated online by filtering dialysis fluid through bacteria- and endotoxin-retentive filters and is immediately infused into the patient's blood. Therefore ultrapure water (sterile and nonpyrogenic; containing <0.1 colony-forming unit/mL and <0.03 endotoxin unit/mL) is essential for online HDF since even low levels of endotoxin in the water can cause cytokine-mediated inflammation.¹⁴⁶

Highly permeable membranes are suitable for HDF to provide greater removal of middle-molecular-weight and protein-bound uremic retention solutes than conventional low- or high-flux HD. In practice, the K_{uf} should be high enough to allow 2 mL/min/kg body weight convective flow in postdilution HDF, but membranes with 20 mL/min/mmHg or greater are also used for HDF.

CONVECTIVE VOLUME

The convective volume is the sum of the net desired UF volume and amount of substitution fluid infused. A high convective blood volume is fundamental for improving long-term survival in patients on HDF. In the HDF, Hearts and Height (3H) study (described in detail later), median convection volumes of 13.4 L/ m^2 were achieved in children, which is comparable to the 23 L per 1.73 m^2 per session that proved beneficial in the pooled adult studies.¹⁴⁷ A UF volume of 30% to 35% of total blood volume can be achieved by modern HDF machines and should be aimed for in order to obtain optimal clearance.

The components of an HDF prescription are shown in Table 72.8. Today almost all new dialysis machines allow for both HD and HDF, are suitable for children from 12 to 17 kg body weight, and require a pediatric circuit with low extracorporeal volumes.

ADVANTAGES OF HEMODIAFILTRATION OVER CONVENTIONAL HEMODIALYSIS

HDF has several advantages over conventional HD through improved dialysis efficiency, clearance of

toxins across a wide molecular weight range, improved hemodynamic stability, reduced inflammation by use of “ultrapure” dialysate, and increased removal of inflammatory cytokines. A multicenter observational study tested the hypothesis that HDF improves the cardiovascular risk profile, growth and nutritional status, and health-related quality of life outcomes in children compared with conventional HD: the HDF, Hearts and Height (3H) Study.¹⁴⁵ Children receiving HDF had improved blood pressure and hemodynamic stability, impressive catch-up growth with a projected final height approaching target midparental height with daily HDF.¹⁴⁷ There was a high prevalence of subclinical cardiovascular disease in children on dialysis, but an attenuated progression of carotid intima-media thickness was noted in children receiving HDF compared with those receiving conventional HD. Importantly, children treated with HDF reported a reduction in the frequency and/or severity of headaches, dizziness, and cramps on dialysis, as well as a reduction in the postdialysis recovery time, leading to an improvement in school attendance and physical activity. Lower interdialytic weight gain on HDF, implying lower UF rates per session and greater hemodynamic stability, was strongly associated with fewer symptoms. There was no reduction in serum albumin levels with HDF and no difference in the rate of change of residual renal function in children on either dialysis modality, implying that HDF is a safe treatment. The 3H study showed reduced inflammatory markers and lower β -2 microglobulin compared with children on HD. Moreover, a significant improvement in inflammation, antioxidant capacity, and endothelial risk profile is achieved within 3 months in another study.^{145,147}

HOME HEMODIALYSIS

Home HD (HHD) transfers dialysis treatments from hospital settings to patients' homes.^{132,134} One of the key characteristics of all HHD practices is augmented HD prescriptions, owing to a higher frequency of dialysis sessions per week or longer dialysis treatments. Typically, HHD regimens are subdivided according to their duration (treatment hours per session), frequency (alternate days or four to seven times per week), and timing (daytime or overnight). Strict criteria for *patient selection* including a medically stable patient, suitable weight, commitment of caregivers, well-functioning vascular access, and suitable home conditions are required. The NxStage System One (NxStage Medical Inc., Lawrence, Massachusetts, USA) is a portable home dialysis machine that functions without home water modifications. Premixed dialysis fluid is available for home patients in sterile, 5-L bags, with a variety of fluid compositions. HHD prescriptions have the potential to deliver a higher dialysis dose and allow for lower UF rates. Reported improvements include reduced antihypertensive medication burden and lower incidence of LVH; improved anemia control and a lower average erythropoietin dose; a reduction in the use of phosphate binders and improved nutrition; and improved quality of life. Pediatric data are scarce with mostly single-center reports; findings are largely consistent with the

adult experiences.¹³⁵ Daily or nocturnal home HD is not available in most countries for children.

COMPLICATIONS OF HEMODIALYSIS

1. Hypotension is a common complication during dialysis. It is associated with excessive or rapid fluid removal. Hypotension may appear without warning, particularly in infants, and manifest as pallor, irritability, vomiting, or altered mental status. Accurate measurement of UF volume is crucial in small children, as small volumes constitute a relatively large percentage of their blood volume. In patients with repeated hypotensive episodes, sodium profiling, osmotic agents such as mannitol, lowering dialysate temperature, or (off-label) use of the α_1 receptor agonist, midodrine, can sometimes be helpful. More frequent dialysis sessions may be necessary for large interdialytic weight gain, to reach dry weight, and avoid hypotension during dialysis.¹⁴⁸

Immediate stepwise action includes (i) temporary suspension of UF; (ii) isotonic fluid bolus (5 ml/kg); and (iii) early discontinuation of dialysis. It is important to reassess the dry weight in case this has been underestimated, as well as the daily salt intake and fluid allowance, which may be too high so that excessive fluid needs to be removed. Measures to improve hemodynamic stability are (i) withholding antihypertensive medications on dialysis days; (ii) avoiding food during dialysis; (iii) use of noninvasive blood volume monitors that provide real-time measure of the changes in relative blood volume (RBV) during dialysis; changes in heart rate and a steeper gradient of the RBV curve in the first hour predict hemodynamic decompensation; (iv) manipulating dialysate characteristics (use of decreasing dialysate sodium concentration during a session, cooling to at least 0.5°C below the patient's temperature, use of bicarbonate buffers, adjusting dialysate calcium and potassium to avoid intradialytic hypocalcemia and large changes in serum potassium; and (v) use of midodrine or switching to HDF or augmented dialysis in refractory patients.

2. Disequilibrium syndrome is now a rare yet potentially dangerous complication of HD. It is usually precipitated by overly rapid urea removal causing a discrepancy between the osmolality of the plasma and that of the brain cells, causing fluid shift into the brain.¹⁴⁹ Seizures are more common in children with disequilibrium syndrome than in adults. Prevention is aimed at a gradual reduction of urea, especially in new HD patients and when urea is high. This is achieved by selection of an appropriately small dialyzer and limiting blood flow and session length for the first few treatments. Mannitol infusion or, in cases of HTN and hypervolemia, high dialysate glucose or slightly increased dialysate sodium concentration may also be helpful, by preventing the rapid decrease in plasma osmolality. Blood flow rate should be slowed if mild symptoms such as nausea, vomiting, or headache appear, and dialysis should be stopped if more significant neurologic manifestations appear.¹⁵⁰
3. Hemorrhage. While disconnection of the blood circuit tubing set from the vascular access puts the patient at risk for rapid exsanguination, the blood pump is automatically stopped while sounding an alarm in this scenario.

A more common scenario is clotting of the blood circuit precluding the return of blood in the extracorporeal circuit to the patient. Volume expansion with crystalloid or, in severe cases, blood transfusion is required if patient is symptomatic.

4. Blood-borne viruses. Hemodialysis unit patients and staff are at risk of blood-borne viruses, particularly hepatitis B, hepatitis C, and HIV. Transmission may result from percutaneous exposure to blood or other fluids, via droplets or through contaminated equipment. Universal precautions should be followed for all patients, and the entire dialysis circuit should be decontaminated after each use by heat or chemical disinfection. External surfaces should be wiped over between patients using a chlorine-based disinfectant. All staff and patients should be immunized and/or show immunity to hepatitis B.

Although screening for hepatitis C or HIV is not universally recommended at present, many units do so. Hepatitis C can be spread nosocomially, so a separate room is recommended for the patient who is hepatitis C positive, but a dedicated machine is not necessary. HIV is less infectious, but the same criteria apply.

5. Hypothermia may occur if the dialysate is not warmed, particularly in small children, as the dialysate flow is generally constant regardless of patient size.

6. Hypophosphatemia may be caused by HD, particularly in younger children, when kidney function is normal (e.g., when HD is done as emergency treatment for inborn errors of metabolism) and when intensive dialysis is performed (for fluid removal purposes or daily dialysis for ESKF due to primary hyperoxaluria). Hypophosphatemia is attributed to excessive phosphate clearance due to high flow of dialysate relative to patient weight.¹⁵¹ This can be treated by either adding phosphate (e.g., sodium phosphate in enema preparations) to the dialysate concentrate or slowing the dialysate flow.

7. Hypoglycemia may occur during HD and is associated with the use of glucose-free dialysate and high dialysate flow, leading to excessive glucose clearance. Infants, especially if catabolic, may be at a higher risk for hypoglycemia, and plasma glucose should be monitored during dialysis if suspected.

8. Other rare complications are hemolysis, air embolism, anaphylaxis, and long-term complications (dialysis-related amyloidosis, carnitine disorder, and hyperhomocysteinemia).

PERITONEAL DIALYSIS

PD has been used in children for acute and chronic kidney failure since the 1970s and has seen significant advances in technology over the decades. In many parts of the world, PD is often the dialysis modality of choice, particularly for younger children,^{6,11,43} given its almost universal applicability, cost-effectiveness, and possibility of a home-based treatment. In recent years improved survival and outcomes are largely due to advances in dialysis technology and clinical expertise, particularly with the development of home-based PD programs, improved dialysis fluids, and management of dialysis-related complications.¹⁵² Some contraindications to performing PD are specific to the pediatric patient (Table 72.9).^{153 154}

Table 72.9 Absolute and Relative Contraindications to Peritoneal Dialysis

Absolute Contraindications	Relative Contraindications
Omphalocele or gastroschisis	Impending abdominal surgery
Bladder extrophy	Presence of ileostomies and colostomies
Diaphragmatic hernia	Significant organomegaly such as with polycystic kidney disease
Peritoneal membrane failure	Lack of suitable caregiver
Obliterated peritoneal cavity	Inadequate living situation for home dialysis

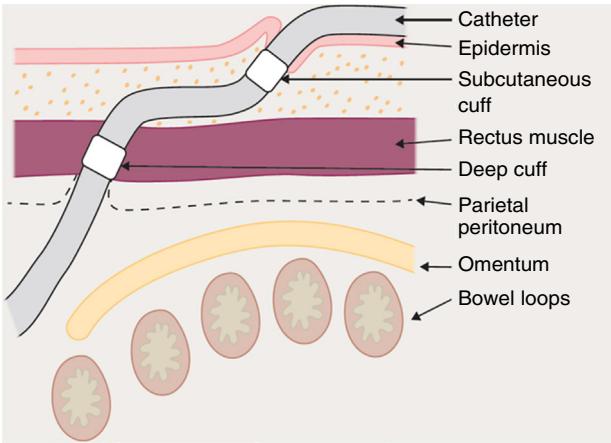


Fig. 72.3 A schematic representation of a peritoneal dialysis catheter and cuff placement in the abdominal wall.

ACCESS

The success of PD depends on a reliably functioning PD catheter. There are many different catheter designs with different intraperitoneal configurations (curled or straight), number of cuffs, and tunnel configurations. The most used catheter for children is the Tenckhoff catheter, which is surgically positioned in the pelvis via a subcutaneous tunnel by open surgical or laparoscopic technique. Pediatric PD guidelines recommend a double-cuff Tenckhoff catheter with a downward or lateral subcutaneous tunnel configuration and downward-pointing exit site. A Cochrane review has shown that there is no specific catheter insertion technique or catheter design that prevents or reduces the risk of peritonitis. However, randomized trials suggest that a downward-pointing exit site for the PD catheter is the single most important factor in preventing peritonitis, although this may not be relevant for the youngest children. A schematic representation of the catheter and cuff placement in the abdominal wall is shown in Fig. 72.3.

Before catheter insertion, a detailed clinical examination is required to determine the position of the exit site, particularly in those who may have had prior surgery and those with abdominal stomas, and also to check for hernias, as these may need to be repaired at the same time as the PD catheter insertion. Constipation is a common issue

in patients with CKD and must be addressed before PD catheter insertion; constipation is the most common cause for catheter migration and nonfunction. Preoperative antibiotic prophylaxis reduces the incidence of early-onset peritonitis. The choice of antibiotic should depend on local antibiotic sensitivity patterns: most centers in the United Kingdom use intravenous vancomycin, but routine prophylaxis with vancomycin carries a risk of vancomycin-resistant enterococci emergence, and European Best Practice guidelines recommend that a single dose of first- or second-generation cephalosporin is used just before catheter placement.

Ideally, the catheter should be left to rest for 10 to 14 days before PD initiation in order for the tunnel to heal. However, if the patient requires urgent dialysis, the catheter may be used immediately; in this situation, small dialysate volumes help to minimize intraperitoneal pressure. Standardized and rigorous exit-site care is essential for reducing the risk of exit-site infection and peritonitis. As shown by the quality improvement project, Standardizing Care to improve Outcomes in Pediatric End-stage renal disease (SCOPE), the implementation of standardized follow-up care led to a significant reduction in the average monthly peritonitis rates.¹⁵⁵ Specifically in infants or children on PD, the catheter must be placed as far as possible from sources of infection such as diapers or gastrostomy tubes.¹⁵³

MODALITIES OF PERITONEAL DIALYSIS AND THE PERITONEAL DIALYSIS PRESCRIPTION

The PD prescription can be delivered by several different techniques. The PD modality is selected depending on the patient's needs for solute and fluid clearance; the residual kidney function, if any; and its suitability for the child and family's lifestyle and daily routine. General considerations related to KRT by PD are provided in Chapter 63. The different PD modalities are shown in Fig. 72.4, and a brief summary of the technique and its relative advantages and disadvantages are discussed in Table 72.10.

Although the goals of adequate solute and water removal are universal, the PD prescription must be tailored to individual needs and a "one-size-fits-all" approach cannot be considered. For all PD prescriptions, the child's size (weight and BSA), residual kidney function, biochemistry, and peritoneal membrane function should be considered. After selecting the appropriate dialysis modality, the key aspects of the PD prescription that need to be addressed are (1) type of dialysis fluid, (2) the volume of dialysate per PD exchange (cycle), and (3) the total therapy time and number of cycles in that period. Details follow:

1. **Peritoneal dialysis solutions** used in children are identical to those used in adults and described in Chapter 63, although biocompatible PD fluids are favored in children. Recommendations by the European Pediatric Dialysis Working Group include the use of the lowest glucose concentration possible and fluids with reduced glucose degradation products (GDPs) content whenever possible.¹⁵⁴ In the face of a paucity of studies prospectively comparing low GDP solutions or various buffers, evidence-based recommendations cannot yet be given,

but it seems that correction of metabolic acidosis with pH-neutral bicarbonate-based fluids is superior to single-chamber, acidic, lactate-based solutions.¹⁵⁶

2. **Volume of dialysate exchanged in each cycle** is calculated on the basis of the child's BSA; this is approximately 600 to 800 mL/m² in those younger than 2 years and 1000 to 1200 mL/m² (up to a maximum of 1400 mL/m²) in older children.¹⁵³ The maximum safe dialysate volume per fill is determined based on intraperitoneal pressure at less than 14 cm H₂O in children older than 2 years and 8 to 10 cmH₂O in younger children.¹⁵⁴ A high intraperitoneal pressure may impede lymphatic absorption (causing reduced UF), physical discomfort, dyspnea, hernia formation, and gastroesophageal reflux.
3. **Total therapy time and number and duration of cycles** depend on the child's age, diet, and amount of urine passed as shorter cycles promote more UF. Infants and younger children, because of their mainly liquid diet and the need for longer dialysate drain times, need longer dialysis treatment times, usually approximately 12 hours, whereas older children do well on 8 to 10 hours of dialysis.

The duration of each PD exchange (cycle) will depend on the peritoneal membrane function. This can be measured by the Peritoneal Equilibration Test, but as a rule, infants and young children are usually high transporters (rapid small solute equilibration), whereas older children and young adults are usually low transporters (good UF with minimal reabsorption). High transporters require more frequent exchanges with shorter dwell times to avoid reabsorption and the use of icodextrin for the daytime dwell (see Fig. 72.4). Low transporters may require longer dwells with higher-volume exchanges to get adequate clearance.

Importantly, peritoneal membrane characteristics differ between individuals and can also change in any given individual depending on their dialysis prescription, complications of dialysis such as peritonitis, and the duration of time spent on dialysis. After a long duration of PD, particularly if highly osmolar dialysate fluids are used or if the child has suffered from peritonitis, the peritoneum can undergo sclerosis and even fibrosis so that it becomes less effective as a dialysis membrane. An understanding of peritoneal membrane characteristics allows clinicians to adjust the PD prescription to obtain optimal UF and solute clearance.

Daily monitoring of the child's PD is now possible through use of remote patient monitoring (RPM) platforms with some dialysis machines. RPM gives the PD team an overview of how effective the dialysis program is in terms of the daily UF achieved and the child's weight and BP. Problems with PD catheter drainage, as well as reduced UF, can be detected and the PD program is manipulated remotely to address these issues. When used effectively, RPM can lead to more individualized PD prescriptions for patients and proactive nursing care that can reduce hospital visits.

The development of new PD modalities, such as adapted automated PD, may also help achieve optimal sodium and water removal.¹⁵⁷ Adapted automated PD aims to improve solute and water removal by using initial cycles with short dwells and small fill volumes to increase sodium-free UF,

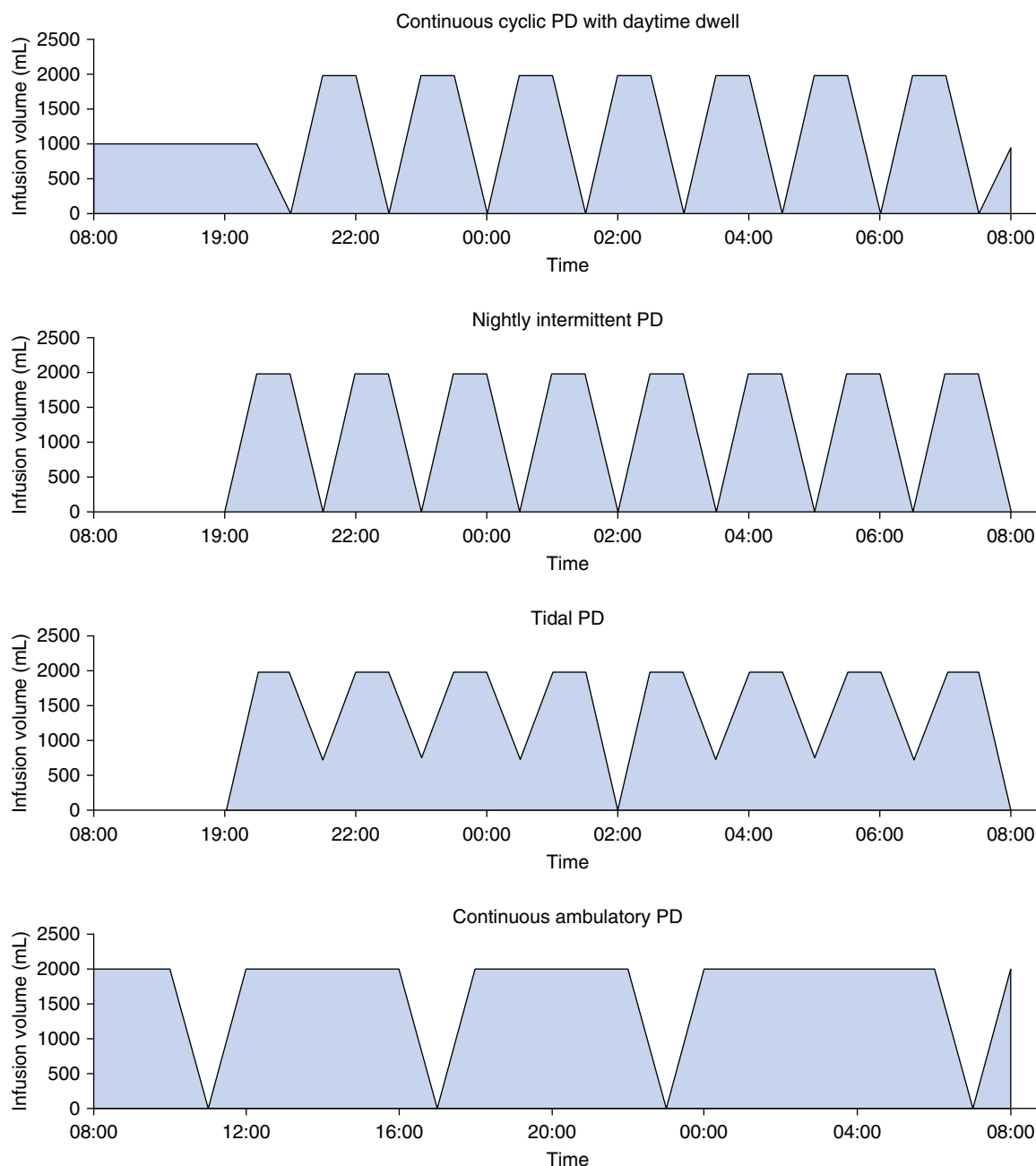


Fig. 72.4 Schematic representation of various peritoneal dialysis regimens based on a standard adult fill volume of 2000 mL of dialysis fluid. The fill volume in children is adjusted to the child's body surface area, with an optimal fill volume being 800 to 1000 mL/m² in those younger than 2 years and 1000 to 1400 mL/m² in older children. The shaded area represents the fill volume and dwell time, and the x-axis represents a 24-hour period. PD, Peritoneal dialysis.

followed by cycles with longer dwells and larger fill volumes, which favor clearance of solutes (sodium) and uremic toxins. Pediatric studies are ongoing to confirm the advantages of this regimen.

ADEQUACY OF DIALYSIS

No large-scale prospective studies that assess the correlation between solute removal and clinical outcomes have been performed in children. This is the result of the rarity of ESKF in children and the relatively shorter time on dialysis in

children compared with adult patients. Furthermore, longer life expectancy in pediatric KRT precludes the assessment of the effect of dialysis prescription changes on survival. In areas where there are no pediatric data, adult clinical practice guidelines serve as a minimal standard. Clinical and laboratory assessments should be done at least once a month (more often if clinically necessary). Kt/V should be measured 1 month after initiating PD and subsequently at least every 6 months, and RRF is assessed by 24-hour urine collection every 3 months. Adequacy is evaluated clinically, seeking signs and symptoms of uremia: HTN, pulmonary congestion, pericarditis, hyperkalemia, hyperphosphatemia,

Table 72.10 Different Types of Peritoneal Dialysis (PD) Modalities, with their Advantages, Disadvantages and Indications in Pediatric Patients

Modality	Technique	Advantages	Disadvantages	Indication
APD	All APD regimens	Cycling machine	Personalized regimen Remote control software programs	PD modality of choice
		Increased flexibility PD program Fewer connections/disconnections No interruption of schooling by dialysis procedures during the day	Cost Dependant on parental/caregiver participation/understanding of machine	
	CCPD	24 hr dialysis	Increased solute and fluid removal	Negligible residual kidney function or anuric patients
		Nocturnal PD program ~ 5-10 cycles	Possible additional daytime exchange to improve solute and fluid removal	
		Daytime dwell with ~ 60% fill volume Daytime dwell drained at bedtime when cyclor reconnected	Equilibration during the day (reabsorption sodium and water) Damage to the peritoneum	
	NIPD	Nocturnal PD program: a number (~ 5-10) short cycles	Improved appetite	High-transport peritoneal membrane
		Patient connected ~ 9-14 hours	Reduced risk of hernias	
		Abdomen empty during the daytime	Preservation peritoneal membrane No glucose absorption during the day	
			No loss of protein during the day	
	Tidal	Nocturnal PD program: only a proportion drained at each cycle	Reduced abdominal pain due to drainage issues	Drainage pain
		Full drain generally initial, mid and last drain	Avoids alarms due to low flow if there is catheter dysfunction	
		+/- daytime dwell	Increased clearance larger molecules	
CAPD	24 hr dialysis	No machine required/ease of use	Increased risk PD-related peritonitis	Low/middle-income countries Residual kidney function
		3-5 manual drains per 24 hr	Less dialysate used	
		Long dwell overnight	Less expensive Older children/adolescents who wish to socialize in the evening	

APD, Automated peritoneal dialysis; CAPD, continuous ambulatory peritoneal dialysis; CCPD, continuous cyclic peritoneal dialysis; NIPD, nightly intermittent peritoneal dialysis.

worsening school performance, and general well-being. Additional laboratory data may indicate inadequacy of dialysis (serum solute concentrations and hemoglobin and albumin levels) and, finally, calculating Kt/V. Target Kt/V is 1.8 or greater per week, while V, or total body water, should be calculated using age- and gender-specific nomograms.¹⁵³ Prescription should be modified if needed, taking into account also preservation of RRF, patient convenience (IPP, fitting with the child's daily routine), and lowest possible dialysate dextrose concentration to achieve sufficient fluid removal.¹⁵⁸

COMPLICATIONS OF PD

INFECTION

The diagnosis of peritonitis in a child on PD should be considered in the presence of cloudy peritoneal effluent, fever, chills and rigors, abdominal pain, vomiting, and in late cases, septic shock. The child's caregivers are trained to have a low threshold for suspecting peritonitis if any of these symptoms are present, immediately contacting their dialysis unit, and collecting a sample of peritoneal effluent that is sent for cell count, differential count, and culture to confirm the diagnosis of peritonitis. An empiric diagnosis of peritonitis is made if the dialysate effluent white blood cell (WBC) count is greater than 100/mm³ and at least 50% of the WBCs are polymorphonuclear leukocytes (4).

In PD-related peritonitis, instillation of antibiotics into the peritoneum is the preferred route of administration because high bactericidal concentrations can be rapidly established at the site of infection and most antibiotics are absorbed from the peritoneal cavity into the bloodstream. Intravenous antibiotics may be necessary in patients with severe systemic involvement or if the patient is immunosuppressed. Antibiotic therapy must begin with a broad spectrum and include cover for both gram-positive cutaneous flora and gram-negative enteric pathogens, and choice depends on local sensitivity patterns. Once PD fluid cultures are obtained, antibiotics are adjusted according to the final culture and antibiotic sensitivity results. It is recommended that all patients receive oral antifungal therapy (nystatin) while on antibiotics. Updated guidelines on the diagnosis and management of infectious complications of PD in children have been recently published by the International Society for Peritoneal Dialysis,¹⁵⁹ closely reflecting similar guidelines in adults on PD.¹⁶⁰

PD catheter removal should be considered in case of fungal peritonitis, severe sepsis, relapsing PD-related peritonitis, or a persistently raised PD fluid WBC count despite appropriate antibiotic therapy.

Peritonitis rates are higher in infants (1/15.3 months) compared with adolescents (1/21.2 months; NAPRTCS data), probably due to close proximity to contamination sources such as diapers or gastrostomy tubes in the young ages.¹⁶¹ Higher peritonitis rates have been reported in gastrostomy-fed children on PD; hence whenever possible, a gastrostomy should be inserted before PD catheter insertion and commencing PD.^{34,162} When this cannot be avoided, antibiotic and antifungal prophylaxis at the time of PD catheter insertion in children with a gastrostomy is essential.³⁴ A report of the Italian Registry of Paediatric

Chronic Dialysis found somewhat better results: 1/20.7 months in infants and 1/28.3 in older children.¹⁶³ In a survey including 47 centers in 14 countries, 44% of the peritonitis events were due to gram-positive bacteria, 25% due to gram-negative bacteria, and 31% were culture negative.¹⁶⁴ Less than half of the patients had fever higher than 38°C, less than half had abdominal pain, and 70% had marked effluent cloudiness. Of 482 children, 420 (89%) made a full recovery and 39 (8.1%) had to discontinue PD permanently due to UF problems, adhesions, uncontrolled infection, and secondary fungal infection. As such, infections are the most common reason for modality change for patients on PD. In another study, including 501 peritonitis events from 44 pediatric dialysis centers in 14 countries, significant regional variability was found for both bacteria type and sensitivity.¹⁶⁵ This makes recommendations for global treatment guidelines more complex. Permanent discontinuation of PD varied considerably, from 0/18 episodes (Argentina) to 9/46 episodes (20%) in Eastern Europe. Of note, a recent Cochrane study assessing evidence in PD-associated peritonitis (including adults) concluded that the data available are poor. In general, review conclusions were based on a small number of studies with few events, in which risk of bias was generally high; interventions were heterogeneous, and outcome definitions were often inconsistent. There were no randomized controlled trials evaluating optimal timing of catheter removal and data for APD were absent.¹⁶⁶

Other infections include exit site and tunnel infections. Preventive measures include aseptic handling of the catheter, daily cleansing, immobilization, and applying topical antibiotics and intranasal mupirocin application. Interestingly, an alternative measure of preventing exit-site infection with antibacterial honey applied to the exit site found noninferior results compared with intranasal mupirocin.¹⁶⁷

TECHNIQUE FAILURE

Catheter Malfunction

Catheter malfunction due to its migration or occlusion may preclude effective dialysis. Technical solutions such as the use of specific types of catheters and omentectomy at the time of catheter insertion may reduce the incidence.

Ultrafiltration Failure

UF failure is the main cause of technique failure and leads to volume overload. High solute transport, as detected by PET, causes rapid dissipation of the osmotic gradient needed for UF. Peritonitis events and long-term exposure to GDPs in the standard PD solutions are the main causes. Filling volume that is too high can, in addition, lead to UF failure due to lymphatic absorption, whereas low exchange volumes may falsely seem to be peritoneal failure because with low volumes there is rapid solute equilibration.

Fluid Leaks

Fluid leaks secondary to high IPP are more common in infants, often manifesting as hernias. Recurrent hydrothorax is due to pleura-peritoneal connections and the negative pressure produced on inhalation. Pericatheter exit-site leak is usually an early event after insertion and is usually managed by postponing start of PD or using smaller volumes initially. Leak into the abdominal wall causes subcutaneous edema.¹⁶⁸

Encapsulating Peritoneal Sclerosis

Encapsulating peritoneal sclerosis is a rare complication with high fatality rate, characterized by extensive intraperitoneal fibrosis, which causes UF failure occasionally with symptoms of bowel obstruction. It was found in 14 of 712 (1.9%) pediatric PD patients registered in the Italian registry of chronic dialysis.¹⁶⁹ Eleven of the patients were treated with PD for longer than 5 years. Because the disease is insidious and progresses slowly, five (36%) were diagnosed when no longer on dialysis. In 71% of the patients, high dextrose solutions were used over the 6 months before diagnosis. Peritonitis rate was not higher than in patients without this complication. A second study found the same rate of encapsulating peritoneal sclerosis, but a higher infection rate relative to those without this complication, and no correlation with dialysate type.¹⁷⁰ Data from the ESPN are more optimistic, reporting a low prevalence and a good response to dialysis modality change.^{170,171}

PERITONEAL DIALYSIS IN INFANTS

Initiating dialysis in infancy is rare and remains a difficult medical and ethical situation. Most infants born with CKF have been diagnosed antenatally, so parents may have developed preconceptions about the outcome for their child, which may or may not be appropriate. These expectations, alongside the high frequency of prematurity and comorbidities, may affect the decision and technical ability to proceed with dialysis. Nonetheless, the survival of infants who initiated dialysis in the first year of life has improved, with nonrenal comorbidities being the main cause of death. PD is the KRT modality of choice for infants in whom vascular access is likely to be more difficult and serves as an essential bridge to kidney transplantation, which is only possible in most centers once the child weighs >10 kg.

Several aspects of infant PD present a technical challenge, both in terms of gaining an effective dialysis access and the day-to-day management of the dialysis prescription. PD catheter insertion in infants can be difficult and more prone to leakage or blocking by omentum. Positioning the catheter exit site above the nappy line but away from a potential future gastrostomy site, as well as other abdominal stomas, is important to prevent cross-infections and peritonitis. Infants are at a higher risk of developing gastroesophageal reflux, hernias, and peritoneal fluid leaks, so the PD prescription requires careful adjustment to achieve adequate solute and fluid clearance without causing a significant increase in the intraabdominal pressure. In addition, excess fluid removal is particularly problematic in infants, especially for those who are oligoanuric, because an infant's diet is liquid. To achieve this lower dialyzate, fill volumes are used with short, low(er) volume exchanges and a longer total PD therapy time per 24-hour period.

Achieving optimal nutrition and growth is particularly important in infants on dialysis. Enteral tube feeding via a nasogastric tube or gastrostomy is often necessary to support nutrition and medication administration. Careful attention to the management of CKD-mineral bone disorder (CKD-MBD) is required as the growing bones of children rapidly accrue calcium and phosphate. Complication rates in infants, especially those with comorbidities, remain high with an increased risk of PD-related peritonitis, significant

neurocognitive and developmental delay, poor nutrition, and cardiovascular complications.

CONTINUOUS KIDNEY REPLACEMENT THERAPY

In recent decades, continuous KRT (CKRT) has become the preferred modality for the management of children with AKI and volume overload, especially for those who may be hemodynamically unstable.¹⁷²⁻¹⁷⁴ The most common indications for CKRT include sepsis, stem cell transplantation, cardiac disease, liver disease, and malignancies,¹⁷³ and less commonly for inborn errors of metabolism, primarily hyperammonemia, or intoxications/overdose.¹⁷⁵ CKRT allows provision of diffusive (continuous venovenous hemodialysis; CVVH) and convective (continuous venovenous hemodiafiltration; CVVHDF) clearance separately or in combination. CVVHDF combines both diffusive and convective clearance by countercurrent infusion of dialysate and net UF to maintain euvolemia. The choice of a given modality is often center dependent. Advantages of CKRT include the ability to dialyze patients who have cardiovascular instability since it allows for slow fluid removal without any major fluid shifts. Similarly, episodes of hypotension are less likely, thereby preventing any further ischemic insult to the kidneys, and therapeutic drug monitoring is more easily achieved.

TECHNICAL CONSIDERATIONS FOR CHRONIC KIDNEY REPLACEMENT THERAPY

The principles and technical aspects of CKRT are similar in children and adults. They are described in detail in Chapters 62 and 64. Dedicated machines, dialyzers, and replacement fluid are used for CKRT. The CKRT prescription is similar to that for intermittent hemodialysis but includes a set UF rate that is decided on the basis of the child's fluid status and the volume of fluid administered. It is usually expressed in mL/kg body weight/hour or as a percentage of the total body weight. There are no robust studies in children exploring optimal CKRT dosing, but the consensus in adults is that there is no benefit to delivering high-volume CKRT reaching >25 to 30 mL/kg/hour clearance.¹⁷⁶ Due to the continuous nature of CKRT, children may develop hypokalemia or hypophosphatemia, and these may require replacement in the dialysate or IV fluid.

Although CKRT shares similar principles with HD, both the blood and dialysate flows are significantly slower, resulting in a lower hourly clearance rate. This is compensated for by extending the clearance time (over 24 hours, CKRT provides solute clearance comparable with a 4-hour HD session), using hemodialysis or hemodiafiltration in the CKRT circuit. The main advantage of CKRT in the critically ill child is the ability to maintain hemodynamic stability. CKRT and PD are continuous in nature, but the former provides much greater and more controlled daily clearance rates.

Double-lumen catheters are often used, and the diameter size is selected on the basis of the patient's BSA. One must reconcile the need for adequate blood pump flow rates with the desire to limit vessel trauma. Although femoral catheters

are four times more often used than IJ ones, the latter result in a far better circuit survival.¹⁷⁷ Furthermore, femoral catheters, unless located in the inferior vena cava (IVC), are significantly affected by patient movement and may require the patient to be sedated or even paralyzed for proper use. Taken together, the preferred site for vascular access is in the right IJ vein with the distal tip of the catheter placed in the right atrium of the heart. The recommended blood flow is 3 to 10 mL/kg/h with a relatively higher flow in the very young child if an adult-size device is used. Although no prospective study has randomized pediatric patients to different effluent dose targets, the common practice is to use an effluent dose of approximately 2000 mL/1.73 m²/hour to achieve good metabolic control within 24 hours of CKRT initiation.¹⁷⁴ One study demonstrated that hypotension after connection to CKRT occurred in 49.7% of connections a median of 5 minutes after beginning therapy; therefore blood flow should be increased slowly.¹⁷⁸ If the circuit volume is large relative to the child's size, albumin or blood priming may be necessary. In previous years, lactate-based fluids were used, resulting in lactic acidosis, cardiac dysfunction, and hypotension. Citrate or bicarbonate-based dialysate and replacement fluids are currently considered standard of care.¹⁷⁹ Large volumes of replacement fluid, containing varying amounts of electrolytes, are administered. This requires close monitoring of the patient's laboratory profile to ensure stable acid-base and electrolyte balance.

Activation of the clotting cascade, impacting circuit longevity, results from the contact of blood with the artificial filter, as well as from slow blood flow and small catheters used. This mandates the administration of anticoagulants: Although a single pediatric study demonstrated that heparin and citrate are equally efficacious with more bleeding episodes occurring with the former,¹⁸⁰ most adult studies favor regional citrate, together with systemic calcium infusion, for better longevity of the circuit and fewer bleeding episodes. Citrate is metabolized to bicarbonate in the liver and should be cautiously used in patients with liver failure and lactic acidosis. In a multicenter pediatric survey, it was shown that citrate was used in 56% of children and heparin in 37%.¹⁷⁵ In a recent retrospective study it was demonstrated that hemofilter survival was higher in the citrate group than in the heparin group. The risk of hemofilter clotting was significantly increased when heparin was used, regardless of hemofilter size and pump flow.¹⁸¹

OUTCOMES OF CHRONIC KIDNEY REPLACEMENT THERAPY TREATMENTS

Patient outcomes are largely dependent on the underlying disease state and comorbid conditions. According to the CKRT Registry, overall mortality was 42%, with higher rates in children with liver failure or liver transplant, pulmonary disease, or lung transplant and in stem cell transplant recipients, ranging from 55% to 69%.¹⁸² Fluid overload at the initiation of CKRT is an independent risk factor for mortality: patients with >20% fluid overload were 8.5-fold more likely to die.¹⁸³ Of note, this incremental increased risk of mortality was greater than that conferred by multiorgan failure (defined as receiving one vasoactive medication and invasive mechanical ventilation), sepsis, or malignancy, none of which were modifiable risk factors. Patients with >20%

fluid overload demonstrated similar illness severity to those with 10% to 20%, inferring that the most severely fluid-overloaded group was not sicker and the excessive fluid load may have been iatrogenic. A prospective multicenter study in children, addressing the effect of early initiation of CKRT on outcome measures, including mortality, is currently required.

PEDIATRIC KIDNEY TRANSPLANTATION

Dialysis and kidney transplantation developed in parallel. They gradually became available for pediatric patients, offering long-term survival to children with ESKF. The first case series of kidney transplantation in children was described in the mid-1960s by Starzl and colleagues.¹⁸⁴ Over the subsequent decades, tremendous advances in pretransplant and post-transplant patient care, operative techniques, immunosuppression, and infection prophylaxis and treatment have transformed the field, enabling children with ESKF to undergo transplantation and live a normal life with constantly improving outcomes.¹⁸⁵

Although chronic dialysis enables long-term survival of patients with ESKF, renal transplantation offers significant advantages, particularly in children. A well-functioning kidney graft can provide all the functions of a normal kidney, in addition to efficient clearance of metabolic waste products and fine-tuning of water and electrolyte balance, as well as secretion of hormones involved in systemic and renal hemodynamics, erythropoietin and calcitriol. There is a significant survival advantage for children with a functioning graft, compared with children remaining on chronic dialysis, with expected life span close to the general age-matched population (Fig. 72.5).^{186,187}

Several national and international databases collect information on pediatric patients undergoing kidney transplantation. The NAPRTCS database has been collecting data on pediatric kidney transplantation recipients since 1987 and was later expanded to include children on dialysis or with CKD.^{188,189} The Organ Procurement and Transplant Network (OPTN) has been collecting data on all organ transplantations in the United States since 1986, including pediatric renal transplantation.¹⁹⁰ The USRDS is the U.S. database on ESKF including kidney transplantation.¹⁸⁶ These registries provide current information about the status of pediatric renal transplantation in the United States and extensive analysis has been performed on the data collected. Other registries providing information on pediatric renal transplantation include ESPN/ERA-EDTA in Europe, with the latest annual report providing data from 35 countries, the CERTAIN registry,¹⁹¹ the United Kingdom renal registry, the ANZDATA in Australia/New Zealand, and others.^{187,192,193}

INCIDENCE, PREVALENCE, AND ALLOCATION

As of January 2024, 1208 children were listed in the United States for a kidney, out of a total of 88,695 patients listed (1.4%).¹⁹⁰ Fig. 72.6 and Table 72.11 show data on age, race, gender, and ESKF etiology at time of transplant in the United States. Of the 27,332 kidney transplants performed in 2023, 789 were in children (2.8%), a significantly higher

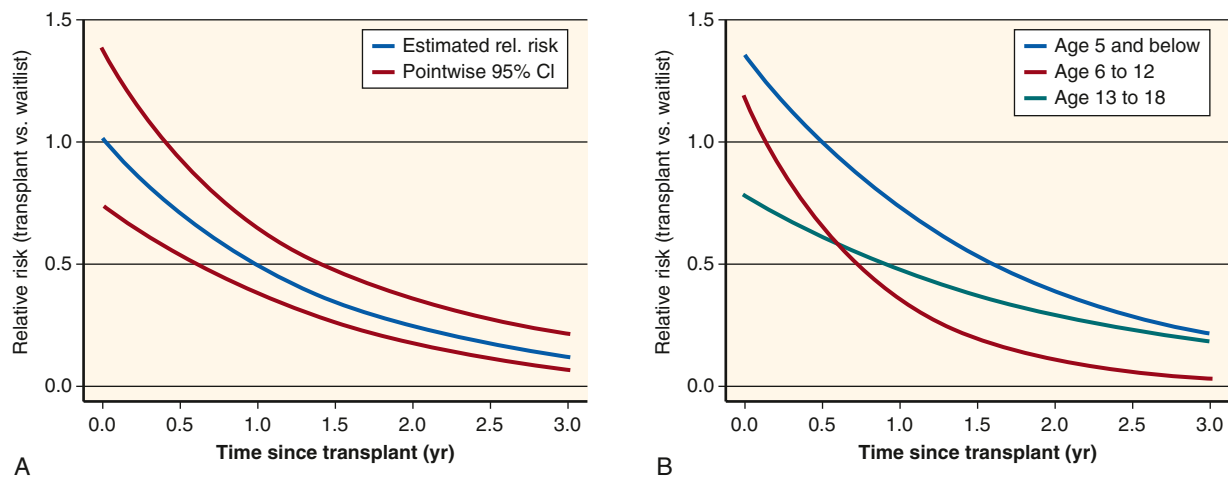


Fig. 72.5 Estimated relative risk of mortality by time since transplantation. (A) The figure is estimated from a model including all ages. (B) Model-based estimates stratified by age. *CI*, Confidence interval. (Modified from Gillen DL, Stehman-Breen CO, Smith JM, et al. Survival advantage of pediatric recipients of a first kidney transplant among children awaiting kidney transplantation. *Am J Transplant*. 2008;8:2600–2606.)

proportion than those waitlisted. Whereas there was an increase in living kidney donation to pediatric patients between 1987 and 2001, with a peak of 64% of kidney transplants, there has been a gradual decrease in this trend since 2001 in the United States, falling to only 31.5% by 2017.¹⁸⁸

Changes in allocation policies of deceased donor kidneys, which favor pediatric patients and shorten waiting times, were in part responsible for this decrease in living donation. The most current deceased donor allocation policy in the United States allocates the top 20% of kidneys in the kidney donor profile index to candidates in the top 20% of expected post-transplant survival. Priority is given to pediatric candidates, after candidates who were prior living donors, or candidates with zero-HLA mismatch kidneys, and calculated panel-reactive antibodies (PRAs) 98% and up.^{188,190}

In Europe, data collected from 32 countries demonstrated that 43% of pediatric kidney transplants were from living donors. Most countries implement an allocation scheme, which prioritizes pediatric patients on the waiting list, in particular with regard to younger donors.¹⁹⁴ Eurotransplant doubles the points for HLA-antigen mismatches and provides bonus points for waiting time for pediatric patients (<16 years at the time of listing in Europe, <18 years in Germany). Scandiatransplant gives priority to pediatric recipients if a donor is <40 years old. The Swiss Organ Allocation System gives preference to ABO-compatible candidates <20 years if a deceased donor is <60 years.¹⁹⁵

In Europe, pediatric kidney transplant rates vary widely between 0 and 13.5 per million children, higher in areas with a higher gross domestic product or a high pediatric priority policy.¹⁹⁶ Access to transplantation differed according to ethnic groups, with black and Asian children less likely to receive a kidney transplant than white patients, mostly due to decreased living donation.¹⁹⁷ Only a few reports emanate from middle- and low-income countries. The available data suggest that kidney transplant rates are 0 to 4 per million children.¹⁹⁸

Preemptive kidney transplantation, meaning no dialysis performed before transplantation, is the modality used

in 22% of pediatric ESKF patients in the United States and 27% in Europe.^{189,194} In contrast, rates of preemptive transplant are less than 6% in adult recipients.¹⁹⁹ There is a wide disparity in European countries in their overall rates of preemptive transplantation ranging from <5% to >60%.¹⁹⁴ Worldwide, preemptive transplantation is mostly from living donors. While socioeconomic factors that limit access to preemptive transplantation can confound the results, studies suggest advantages to preemptive transplants. In Eurotransplant data, preemptive transplants have a higher acute rejection-free proportion at 3 years post transplant (52% vs. 37%) and improved 6-year allograft survival when compared with dialysis patients (82% vs. 69%).²⁰⁰ In a recent meta-analysis of 22 studies ($n = 22,622$), both overall graft loss relative risk (RR; 0.57, 95% CI: 0.49–0.66) and acute rejection risk (RR: 0.81, 95% CI: 0.75–0.88) were lower in preemptive kidney transplants compared with transplantation after dialysis.²⁰¹

Kidney transplantation in infants younger than the age of 1 year is uncommon, reported at 0.8% of pediatric transplants in the NAPRTCS database.¹⁸⁹ As noted earlier, deceased donor kidneys from younger donors are often allocated preferentially to children. However, utilization of deceased donor kidneys from infants and small children holds unique challenges as they are more prone to vasospasm and graft thrombosis, urinary complications, and possibly glomerular hyperfiltration. Such grafts do well when transplanted en bloc into adult recipients.²⁰² Recent data have shown good results in children receiving kidneys from small pediatric donors, either a single kidney or both donor kidneys en bloc.^{203–205} One study from Europe reported a significant increase in allograft size after transplantation and superior graft function in the recipients of pediatric kidneys.²⁰⁴

Although allocation systems usually try to match pediatric recipients with the best brain-dead donors, a recent study showed excellent 3-year graft survival in children receiving a kidney after even circulatory death of the donor.²⁰⁶

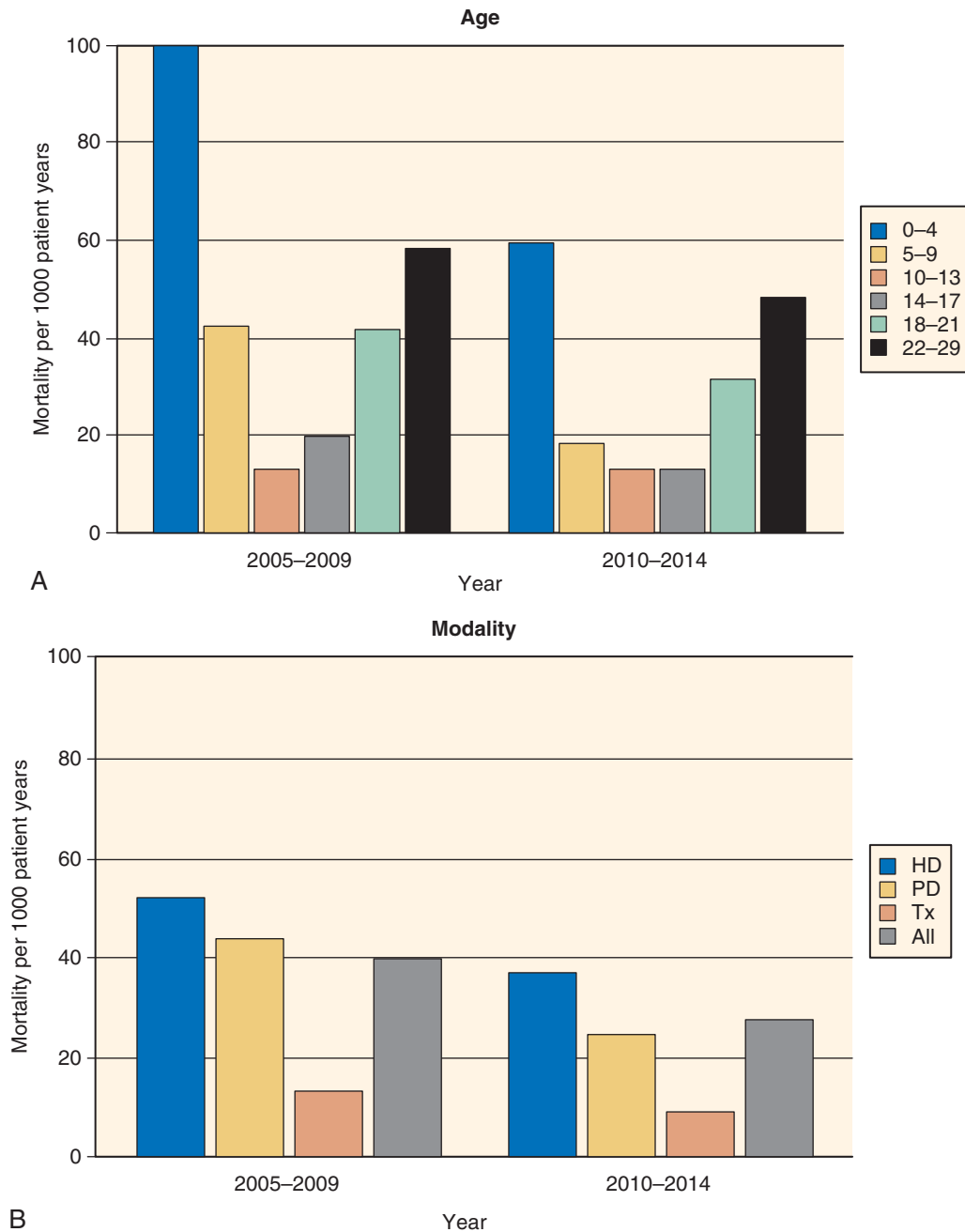


Fig. 72.6 One-year adjusted all-cause mortality rates in incident pediatric patients with end-stage kidney disease by (A) age with comparison to young adults (aged 0–29 years) and (B) modality (aged 0–21 years only), 2005 to 2009, and 2010 to 2014. *HD*, Hemodialysis; *PD*, peritoneal dialysis; *Tx*, kidney transplantation. (From 2017 USRDS Annual Data Report. Chapter 7: ESKF among children, adolescents and young adults. *Am J Kidney Dis.* 2018;71(S1):S383–S416.)

PREPARATION OF THE RECIPIENT FOR TRANSPLANTATION

CHOICE OF DONOR

Living donor kidney transplantation offers better graft survival than deceased donor transplantation, though other factors, such as donor age, HLA matching, and sensitization, play a significant role.²⁰⁷ Kidney grafts from middle-aged living donors afford improved long-term allograft survival compared with younger deceased donor kidneys.²⁰⁸ However, data from

the Collaborative Transplant Study showed that well-matched deceased donor kidneys compare favorably with living donor grafts with many HLA mismatches.²⁰⁹ The question of whether it is advantageous to perform living or deceased donor transplantation first has been studied, as most pediatric kidney transplant patients will eventually require a second transplant. For most candidates, living donor transplantation as the first treatment option provides the best long-term benefit, except possibly for children who are highly sensitized, for whom deceased donor transplantation first, while they have priority as children on the waiting list, may be an advantage.²¹⁰

Table 72.11 Age, Population, Gender, and End-Stage Kidney Disease Etiology at Time of Transplant in the United States

	Age at Transplantation				
	0-1 Years (Percent)	2-5 Years (Percent)	6-12 Years (Percent)	13-17 Years (Percent)	≥18 Years (Percent)
Gender					
Male	70.0	65.2	58.0	56.0	54.3
Female	30.0	34.8	42.0	44.0	45.7
Population					
Non-Latino European	73.1	62.3	60.0	55.7	50.9
African- American	7.9	14.6	14.9	19.7	24.5
Hispanic (or Latino)	12.7	16.1	17.8	17.9	16.8
Other	6.3	7.0	7.2	6.7	7.7
Primary Diagnosis					
Renal dysplasias	27.5	23.3	16.4	11.4	9.8
Obstructive uropathy	19.6	20.2	15.6	13.2	9.9
Other	52.4	48.7	55.6	62.3	63.9
FSGS	0.6	7.7	12.4	13.1	16.4

Since 1987, data have been collected on a total of 12,920 renal transplants in 11,870 pediatric patients. This report represents 1317 new transplants and 1238 new patients entered into the transplant registry since the 2010 Annual Report published in 2014. Of the 12,920 transplants, 10,032 were primary transplants and 2888 were repeat transplants.

Adapted from the 2018 Transplant Report of the North American Pediatric Renal Trials and Collaborative Studies (NAPRTCS).¹⁸⁸

TIMING

When the patient approaches KDOQI stage 4 CKD, preparation of the patient for transplant and evaluation of potential living donors should be pursued. However, graft survival is not indefinite and no benefit has been demonstrated for transplantation before reaching ESKF.²¹¹ Preemptive transplantation is associated with better patient and graft survival when compared with children who receive dialysis for longer than 1 year.²¹¹ Other considerations, particularly morbidity on dialysis, neurocognitive development, and decreased linear growth, also favor transplantation as the first modality of KRT. Certain situations may preclude preemptive transplantation, such as active nephrotic disease, which is also a hypercoagulable state, presenting a high risk of thrombosis during transplantation. To be a candidate for transplantation without dialysis, the patient must have some minimal residual kidney function and urine volume.

Two age groups require special consideration.²¹² Small children may have inferior early results due mainly to technical difficulties and graft thrombosis, though long-term results are best in these recipients—the infant paradox. Some specialized centers, with greater experience in this age group, have demonstrated excellent results.²⁰⁷ The alternative, remaining on dialysis for a prolonged period, also carries a high morbidity and mortality, particularly in

this young age group.¹⁸⁹ Most units refer young patients for transplantation when they reach a weight close to 10 kg or length >65 cm,²¹² but the risks and benefits must be carefully considered in each case. The other potentially complex age group is adolescents. While surgical issues are simpler in this age group compared with younger children, there is a higher rate of acute rejection episodes, shorter graft survival,²¹³ and lower creatinine clearance in this group—the adolescent paradox.¹⁸⁹ This is mostly due to nonadherence, which is more frequent in teenagers and may manifest as missed doses of immunosuppressive medications, more extensive “drug holidays,” missed clinic visits, and other high-risk behaviors.²¹⁴⁻²¹⁶

CONTRAINDICATIONS

Some clinical situations may present a temporary or permanent contraindication to transplantation. Malignancy should postpone plans for transplantation until complete remission is achieved, chemotherapeutic medications are discontinued, and no relapse is evident. High-dose immunosuppressive therapy after transplantation may increase the risk of cancer recurrence due to enhanced development of micrometastases. The period between completion of therapy for cancer and listing for transplant will also depend on the type of malignancy and its

Table 72.12 Evaluation of Pediatric Candidates for Renal Transplantation

Medical history	End-stage renal disease etiology, family history of renal or other disease, biopsy results, previous transplant history, dialysis access and prescription, medication list, compliance with medical regimen, urologic interventions, urine output, other comorbidities, allergies, previous procedures, transfusions, growth charts
Physical examination	Weight, height, body mass index and percentiles, blood pressure and pulse, general well-being, complete physical examination
Specialist assessments	Pediatric nephrologist, transplant surgeon, urologist, anesthesiologist, transplant coordinator, nurse, social worker, dietitian, pediatric dentist. As indicated: psychologist, cardiologist, hematologist, pulmonologist
Laboratory tests	Blood type, complete blood count, electrolytes, blood urea nitrogen, creatinine, calcium, phosphorus, liver enzymes, protein, albumin, lipids, iron, parathyroid hormone, thyroid, fasting glucose and HbA1C, prothrombin time/partial thromboplastin time, urinalysis, 24-h urine collection for creatinine clearance and proteinuria
Serology	Epstein-Barr virus (immunoglobulin [Ig]G Epstein-Barr nuclear antigen and IgM), cytomegalovirus (IgG and IgM), hepatitis B virus (surface Ag and Ab), hepatitis C virus, human immunodeficiency virus, varicella, purified protein derivative test
Histocompatibility testing	Human leukocyte antigen typing—class I (A,B), class II (DR, DQ), panel-reactive antibodies, donor-specific antibodies, crossmatch with donor (T- and B-cell crossmatch, by complement-dependent cytotoxicity with antihuman globulin or flow cytometry)
Imaging	Chest radiograph, abdominal ultrasound including the kidneys, bladder, and postvoiding volume Cardiac echo and electrocardiogram. Doppler study of abdominal and pelvic arteries and veins Voiding cystourethrogram, urodynamic testing, or other if indicated
Vaccinations	Diphtheria, tetanus, pertussis (DTaP), hemophilus influenza type B (HiB), inactivated polio vaccine (IPV), hepatitis A virus (HAV), hepatitis B virus (HBV), varicella, measles, mumps, and rubella (MMR), pneumococcus conjugate and polysaccharide, influenza
Social	Selected patients: meningococcus, human papillomavirus Evaluation of the family, support systems, financial issues, school

characteristics.²¹⁷ Immunosuppressive medications also impair the body's ability to fight infection; therefore active infection is a temporary contraindication to transplantation and latent infections should be sought out and addressed before proceeding to transplant. Severe comorbidities or profound neurodevelopmental delay may render a child unsuitable for transplant. However, intellectual disability does not impact patient or graft survival and should not be a barrier to transplantation in itself.^{218,219} Each child should be evaluated individually to assess the impact of transplantation on life expectancy, quality of life, and rehabilitation, taking into account the wishes of the family. Recalcitrant nonadherence to medication or dialysis schedule, diet, and clinic appointments may also be a temporary contraindication to transplantation.

RECIPIENT EVALUATION

A comprehensive evaluation of the pediatric kidney transplant candidate is crucial to achieve the best results. The goals of this workup are to provide an optimal plan for each individual patient, reduce complications, and increase long-term patient and graft survival. The main elements of the pretransplant evaluation are summarized in [Table 72.12](#). Some of the important issues are discussed in the following sections.

INFECTION

The history, physical signs, or laboratory findings suggesting active infection need to be addressed before subjecting the patient to major surgery and high-dose immunosuppression. The respiratory tract, teeth, skin,

dialysis access exit site, and other sites of possible chronic infection should be carefully examined, as even a minor infection may be exacerbated by immunosuppressive therapy. Testing of serology for cytomegalovirus (CMV) and Epstein-Barr virus (EBV) to assess the risk of post-transplant viral disease are important in planning monitoring and antiviral prophylaxis post-transplant. Many pediatric patients are seronegative for these viruses, and organs from adult donors are often positive. Vaccinations should be reviewed and completed 6 to 8 weeks before transplantation because patients will be at increased risk of infection after transplantation. In addition, response to vaccines may be attenuated by immunosuppressive medications, and live vaccines are not recommended post transplant.²²⁰ However, even with pretransplantation vaccination, immunity may wane in some recipients when they receive immunosuppression.²²¹ Higher pretransplant titers of hepatitis B antibodies may protect against titer loss after transplantation.²²² The regular pediatric schedule should be given including vaccines for pneumococcus, influenza, and varicella, as well as other vaccines depending on age and exposure. Inactivated polio vaccine (IPV) is indicated, rather than oral polio vaccine (OPV). Children should receive immunization for pneumococcal infection, with the 13-valent vaccine followed by the 23-valent vaccine at a minimum of 8 weeks' interval.²²³ Close contacts and household members of these children should complete age-appropriate vaccination schedules to increase the child's indirect protection.²²⁴ Despite these recommendations, the vaccination coverage in pediatric kidney transplant candidates is incomplete and efforts need to be made to improve vaccination rates before kidney transplantation.²²²

MALIGNANCY

Although less common in children, a basic workup to rule out malignancy includes history, physical examination, chest radiograph, and abdominal ultrasound, as well as routine laboratory tests. Children with a history of cancer should be evaluated by a pediatric oncologist to assess remission and risk of recurrence and to help determine the timing of transplantation.²¹⁷

UROLOGIC ISSUES

The most frequent cause of ESKF in younger children is CAKUT.¹⁸⁹ Structural and functional evaluation of the urinary tract is important before transplant, as the kidney allograft may be compromised by an abnormal urinary tract, with obstructed drainage, massive reflux, or a small defunctionalized bladder. Investigation should begin with an ultrasound of the kidneys and urinary tract including imaging of the full and postvoid bladder. In patients with an abnormal urinary tract, further tests like a voiding cystourethrogram and urodynamic evaluation provide further information. When the child has prolonged anuria, it is sometimes difficult to assess bladder capacity and function. Surgical intervention to alleviate obstruction, correct massive reflux if needed, or occasionally bladder augmentation can be performed before or during transplantation. However, the capacity of a small defunctionalized bladder may increase after renal transplantation and enable normal voiding without compromising graft function, thus avoiding the need for bladder augmentation.²²⁵ Nephrectomy of native kidneys is indicated in cases of persistent nephrotic syndrome of presumed immunological etiology (to enable prompt and accurate diagnosis of recurring nephrotic syndrome), chronically infected kidneys, large kidneys in polycystic kidney disease (to enable more physical space for the graft and alleviate discomfort), and sometimes for uncontrolled HTN.²²⁶ In cases of congenital nephrotic syndrome, particularly the Finnish type, mortality due to infection and thrombosis is high; therefore bilateral nephrectomy is usually performed when the child has reached a size compatible with transplantation, even if renal function is good or even normal.²²⁷

CARDIOVASCULAR ISSUES

Congenital heart disease may coexist with CKD in some patients, which requires consultation with a pediatric cardiologist and consideration of surgical correction before transplantation. HTN is common in ESKF, in many cases causing LVH, with diastolic and occasionally systolic dysfunction. Treatment with antihypertensive medications and avoiding chronic volume overload in dialysis patients are important, though optimal control of HTN is often difficult to achieve. However, FGF-23, which is elevated in CKD, induces LVH independent of BP. Therefore LVH may not be completely correctible as long as CKD persists.^{228,229} Cardiomyopathy may improve significantly in children undergoing renal transplantation and LVH may improve.^{230,231}

Evaluation of abdominal vasculature before transplantation is beneficial in planning anastomoses and anticipating significant surgical challenges. Children who have had previous abdominal or pelvic surgery, those with CVCs in the lower body including neonatal umbilical arterial or venous

catheterization, and patients with hypercoagulable states should have Doppler studies of the large arteries and veins before transplantation. Abnormal studies can be followed by formal venography.²³²

RISK OF RECURRENCE

The disease process that caused ESKF may affect the graft. Many immune-mediated kidney diseases tend to recur due to continuation of the underlying immunologic process or persistence of a humoral factor (FSGS, membranoproliferative glomerulonephritis [MPGN], systemic lupus erythematosus, immunoglobulin A [IgA] nephropathy). Genetic mutations causing formation of an abnormal protein product or enzymatic deficiency are highly likely to lead to rapid disease recurrence in the transplanted kidney (atypical HUS [aHUS], primary hyperoxaluria).

Recurrence is particularly frequent in idiopathic nongenetic FSGS, with a rate in children of 30% to 80%, depending on the report.^{233,234} Nephrotic syndrome and graft dysfunction may appear immediately after transplantation, and prompt intervention is needed. In these cases pretransplant nephrectomy of native kidneys, if the patient still has proteinuria, will assist in timely diagnosis of recurrence and enable early treatment.²²⁶ However, FSGS is a histologic diagnosis, which may be due to one of several underlying pathophysiologic processes. Conversely, SRNS may have histologic findings compatible with minimal change nephropathy or other nonspecific findings and yet have clinical features indistinguishable from FSGS. The PodoNet registry, a large multinational study of childhood-onset SRNS, demonstrated post-transplant disease recurrence in only 4.5% of children with a genetic diagnosis compared with 25.8% of patients without.²³⁵ Children who developed secondary steroid resistance were more likely to have post-transplantation recurrence than those who never responded to steroid treatment.²³⁶ Higher risk of recurrence is seen in young children and in those who already had recurrence in a previous graft.^{233,237} To help quantify the risk of recurrence, it is important to clarify, before transplantation, whether FSGS is genetic, due to mutations in genes encoding components of the glomerular filtration barrier.

MPGN also carries a high risk of recurrence, which affects graft survival. The 5-year graft losses were significantly higher for children with glomerular pathology (FSGS [25.7%] MPGN [32.4%]) when compared with children with CAKUT (14.4%) as the primary diagnosis.²³⁸

Other glomerular diseases, such as lupus nephritis and IgA nephropathy, may recur but are not usually associated with graft loss.

aHUS can cause ESKF and has a high rate of recurrence in the graft.²³⁹ It is often caused by various disorders of complement regulation including mutations in complement factor H or I and membrane cofactor protein. The specific cause of the disease should be investigated before considering transplantation to assess the risk of recurrence and guide treatment. Living related donors should be screened for the causative mutation, if known, before approval for donation. Disease recurrence can be prevented or treated with eculizumab (discussed later).²⁴⁰

PH1 is a disorder of glyoxylate metabolism in which an enzymatic defect in the liver causes overproduction of

oxalate and deposition of calcium oxalate in the kidneys and in other organ systems once advanced renal failure develops. Presentation in infancy is associated with nephrocalcinosis and early CKD. Renal transplantation alone usually results in massive oxalate deposition in the new kidney and early graft failure. Combined liver and kidney transplantation (or liver transplantation alone if renal function is not severely affected) is curative, correcting the underlying enzymatic defect.²⁴¹ The natural history of PH1 may change with the approval of the RNA interference therapeutic lumasiran, which permits kidney-alone transplantation.²⁴²

SENSITIZATION

Blood type, HLA typing, and panel-reactive antibodies should be tested in each transplant candidate. If a living donor is being considered, donor-specific antibodies (DSAs) should be identified and quantified if panel reactive antibody is positive. A negative direct crossmatch by complement-dependent cytotoxicity with antihuman globulin is crucial to avoid hyperacute rejection. Crossmatch by the flow cytometry method is more sensitive, but the clinical implications of a positive test are less clearcut. Low-level DSA may be overcome with desensitization protocols, particularly in a planned living donor transplantation.

TRANSPLANT IMMUNOLOGY

HLA MATCHING AND SENSITIZATION

Transplantation immunobiology is covered in detail in Chapter 68. The importance of HLA matching between donor and recipient was recognized in the 1960s and 1970s and became the cornerstone of deceased donor allocation policy. A clear association between number of HLA mismatches and graft survival was shown in several large studies.²⁴³ With the introduction of more potent immunosuppression, the significance of HLA matching may have diminished, while other factors such as peak PRAs, cold ischemic time, and donor age retained their relative importance.²⁴⁴ However, other recent publications have shown that HLA mismatches continue to adversely affect renal allograft outcomes in the present era, as shown by a study from ANZDATA that included children.²⁴⁵ Studies that examined the effect of HLA matching specifically in children have also shown that an increase in number of HLA mismatches is associated with decreased graft survival.²⁰⁹ New allocation algorithms that have been implemented prioritize better HLA-matched kidneys in addition to favoring pediatric patients.¹⁹⁰ Both donor age and HLA matching are important in determining deceased donor graft survival and may be offset by each other.²⁴⁶

Most pediatric patients with CKD will need to undergo more than one transplant over their lifetime, and HLA matching of the first kidney graft may have an impact on subsequent transplants. More HLA or eplet mismatches at first transplant are associated with HLA sensitization, longer waiting time for a second transplant, and decreased graft survival.²⁴⁷⁻²⁴⁹ However, the benefits of better HLA matching at first transplant on lifetime with graft function, although significant, are relatively small.²⁵⁰

In addition to HLA matching, PRAs are measured in renal transplant candidates and, if positive, should be characterized by solid-phase assays.²⁵¹ The presence of DSA is associated with increased risk of acute antibody-mediated rejection (ABMR), as well as chronic AMR and decreased graft survival. When DSAs are further characterized, patients with complement-binding DSA after transplantation are at highest risk of graft loss and ABMR.²⁵² The deleterious impact of DSA has been demonstrated in children, even when the direct crossmatch between donor and recipient is negative by flow cytometry, though not in all series.^{253,254} Although good HLA matching without DSA is desirable, not all patients will find an appropriate donor. Annual reports show that about 5% of children listed for deceased donor transplantation have PRA >80%.¹⁹⁰ Desensitization protocols, including treatment with combinations of intravenous immunoglobulin (IVIG), plasma pheresis, and rituximab, have shown acceptable intermediate-term outcomes in adults undergoing living or deceased donor transplantation.^{255,256} There are remarkably few case series and clinical trials of desensitization in children. One study used IVIG and rituximab desensitization with alemtuzumab induction in highly sensitized children who underwent deceased donor kidney transplantation. Although a higher rate of acute rejection was seen in the sensitized patients, there was no significant decrease in 6-year graft survival and no increase in infectious complication.²⁵⁷

De novo DSA appears in 35% of children in the first years after transplantation, with a high prevalence of DSA to class II and specifically HLA-DQ antigens.²⁵⁸ These antibodies are associated with increased risk of ABMR and graft dysfunction, regardless of whether they develop in the first year after transplantation or later.²⁵⁹ Serial monitoring of DSA in children after transplant may help to identify patients at risk for adverse outcomes and guide tailoring of immunosuppression, though benefits are not clear if no ABMR is present and allograft function remains stable.

ABO COMPATIBILITY

In the past, ABO incompatibility was considered a contraindication to transplantation because of the high risk of hyperacute ABMR. However, in the past few years there have been reports of successful ABO-incompatible renal transplantation in children. ABO-incompatible transplantation in children is rare and usually involves patients with blood type O or B and low titers of anti-A antibodies, receiving an organ from blood type A donors. ABO-incompatible transplantation is more common in Japan, where most transplants are from living donors. Treatment of these patients with a desensitization protocol has shown long-term outcomes equivalent to the general pediatric transplant population.²⁶⁰ In Europe, data from the Collaborative Transplant Study, which included children, reported that 1420 ABO-incompatible living donor kidney transplants were performed in 2005–2012, with similar long-term patient and graft survival. A slight increase in death due to infection was seen in the first year after ABO-incompatible transplantation.²⁶¹ Treatment protocols initially used pretransplant plasmapheresis in order to remove anti-A/anti-B antibodies, together with splenectomy or rituximab. Newer regimens include immunoadsorption,

using either blood group-specific columns or nonspecific columns, to remove the antibodies, together with rituximab to prevent ongoing antibody production after transplantation. Addition of other antibody induction in these patients did not result in altered graft survival.²⁶¹ The risk of ABMR is associated with higher titers of anti-A/B isoagglutinins, and the intensity of desensitization may be modifiable according to pretreatment titers, thus potentially opening this option in deceased donor transplantation as well.²⁶² Immunoabsorption therapy for ABO-incompatible renal transplantation has been associated with surgical bleeding in some cases.²⁶³ Late ABMR in the setting of ABO incompatibility is less frequent than AMR in HLA sensitization.^{251,264,265} Despite more intense immunosuppressive regimens, a study in adults did not demonstrate a higher risk of cancer in patients undergoing ABO-incompatible transplantation.²⁶⁶

IMMUNOSUPPRESSION

The goal of immunosuppressive therapy in pediatric kidney transplantation is to prevent graft rejection while minimizing adverse effects. Clinical transplantation is discussed in more detail in Chapter 69. A wide range of immunosuppressive medications are used in pediatric renal transplantation and are a key factor in preventing rejection and enabling long-term graft function. Special considerations in children include altered drug metabolism, as well as adverse effects particular to children, including specific infections and effects on growth and development.

Post-transplant immunosuppression includes two phases: A) Induction immunosuppression, in the perioperative period to prevent early rejection; and B) Maintenance immunosuppression, to promote long-term graft survival. Maintenance immunosuppression starts during the perioperative period and continues indefinitely.

INDUCTION IMMUNOSUPPRESSION

The risk of acute rejection is highest in the immediate post-transplant period, which is why in addition to higher doses of maintenance immunosuppressive drugs, many transplantation protocols include a biological induction agent. These antibody preparations are aimed at depleting T cells or preventing their activation and are administered in the perioperative period.

Antilymphocyte preparations are T-cell depleting-agents that can be polyclonal or monoclonal. The polyclonal antibody antithymocyte globulin (ATG) contains antibodies to a wide variety of lymphocyte antigens in serum of animals (most commonly rabbit) immunized with human lymphocytes.

The anti-CD52 antibody preparation alemtuzumab binds to the CD52 receptor, found on both T and B cells, as well as monocytes and natural killer cells, causing cell lysis. IL-2 receptor antibodies such as basiliximab block T-cell activation without depletion.

In the United States 94.3% of pediatric kidney transplant recipients reported some induction use in 2020. ATG is the most frequently used antibody induction in the United States, particularly in steroid-free protocols and in

highly sensitized patients.^{267,268} In NAPRTCS data, T-cell depleting agent usage is steadily increasing to reach 63.6% in 2018 and IL-2-RA therapy remained stable at 35.1%.²⁶⁹ However, induction therapy is used in only approximately 50% of patients in European countries reporting to the Cooperative European Pediatric Renal Transplant Initiative (CERTAIN), with IL-2 receptor antagonists therapy being the most common.²⁷⁰

Alemtuzumab induction has been described in several small series of children, enabling steroid avoidance or tacrolimus monotherapy, with good results.²⁷¹⁻²⁷³ Alemtuzumab has also been used for highly sensitized children in combination with desensitization. Outcomes were similar to nonsensitized patients who received IL-2 receptor blockers. A significant reduction in lymphocyte counts was observed in the alemtuzumab group up to 1 year from transplantation.²⁵⁷ Concerns regarding the risk for post-transplant lymphoproliferative disease (PTLD) have been raised regarding the use of lymphocyte-depleting agents, particularly alemtuzumab. In a large OPTN database study, T-cell depleting agents, alemtuzumab and ATG, were compared with nondepleting induction agents or no induction, and the incidence of PTLD without induction (0.43%) was shown to be not different from that with basiliximab (0.38%) or alemtuzumab (0.37%) but was slightly higher in patients who received ATG (0.67%).²⁷⁴ The population in the study had received higher doses of ATG than employed today, higher doses of maintenance immunosuppression, and less antiviral prophylaxis, suggesting that the ATG induction may not currently be a significant risk factor for PTLD.²⁷⁵ In two recent retrospective multicenter studies, low rATG induction dose ≤ 7.5 mg/kg (NAPRTCS data) or ≤ 4.5 mg/kg (Pediatric Nephrology Research consortium data) provided safe and effective outcomes in low immunologic risk pediatric cohorts.^{276,277}

Pediatric studies using the IL-2 receptor blocker basiliximab as induction therapy combined with triple-drug maintenance immunosuppression showed a good safety profile but minimal or no additional benefit compared with no antibody induction.^{278,279} A large ANDATA analysis showed that induction with IL-2 receptor antibodies in pediatric patients was associated with at least a 40% reduction in the rate of acute rejection, though no significant decrease in graft loss was found. Propensity scoring was used to account for confounding factors.²⁸⁰

Retrospective data from the OPTN database found that lymphocyte-depleting induction (ATG or alemtuzumab) is more effective at lowering acute rejection rates in African-American pediatric kidney transplant recipients, but no difference was found in other ethnic groups.²⁸¹

MAINTENANCE IMMUNOSUPPRESSION

CALCINEURIN INHIBITORS

Cyclosporine

Cyclosporine was introduced as an immunosuppressant for renal transplantation in 1982, revolutionizing the treatment of transplant recipients. Whereas in 1996, 82% of children were treated with cyclosporine 1 month after renal transplantation, consistently less than 3% received cyclosporine since 2009, as it was largely replaced by

tacrolimus.¹⁹⁰ But low-dose cyclosporine has been used in pediatric kidney transplant patients together with the proliferative signal inhibitor everolimus, with good 3-year outcomes.²⁸²

Cyclosporine doses in children need to be higher per body weight, as it is more rapidly metabolized than in adults. One suggestion is to dose cyclosporine every 8 hours in young children and infants; however, this may actually result in lower dose-normalized area under the curve (AUC) throughout the day, in addition to being a drug regimen that is more difficult to follow.²⁸³ Trough levels are usually kept at 150 to 300 mcg/L initially and 75 to 125 mcg/L after the first 6 months. Monitoring AUC or 2-hour postdrug level, as an indicator of AUC, may be more effective in avoiding toxic effects. Absorption of different formulations may vary, and dosing should be guided by serum drug levels.

Cyclosporine is nephrotoxic in a dose-dependent manner. It is associated with HTN, hyperlipidemia, and hyperuricemia, all of which may contribute to cardiovascular disease, as well as their deleterious effect on graft function.²⁸⁴ Cosmetic side effects are a major issue in young patients, including hirsutism, gum hyperplasia, and less frequently coarse facial features, which may contribute to nonadherence, particularly in teenagers. Several drugs interact with cyclosporine, in many cases decreasing its metabolism and potentially causing toxic concentrations. Common examples in children include macrolide antibiotics or azole antifungal medications, which should be given with care, while monitoring calcineurin inhibitor levels. Ingestion of grapefruit also decreases calcineurin inhibitor metabolism and increases serum levels, mostly by inhibition of enteric CYP3A4, and should be avoided.

Tacrolimus

The use of tacrolimus as part of the immunosuppression regimen has continuously increased since its introduction in the 1990s. In 2018, 92.7% of pediatric kidney transplant recipients received tacrolimus initially, either with mycophenolates and steroids (58%) or mycophenolates alone (37%).^{190,269} Metabolism varies greatly among individuals; younger children may require higher doses by weight, and all patients should have regular drug-level monitoring.²⁸⁵ Studies have shown that frequency of various cytochrome P450 CYP3A5 allele variants may account for >50% of variance of tacrolimus blood concentrations.²⁸⁶ Moreover, the frequency of these alleles is different in African-American patients, which can explain higher doses needed to achieve therapeutic drug concentrations in this population.²⁸⁷ Initial target tacrolimus levels are high (8–12 ng/mL), gradually decreasing to 3 to 6 ng/mL after the first 6 months, depending on the specific protocol.

An 18-center European study from 9 countries compared cyclosporine and tacrolimus administered concomitantly with AZA and steroids across 196 children. The incidence of acute rejection was lower at 36.9% in the tacrolimus group versus 59% in the cyclosporine group. The mean GFR at 1 year was higher at 62 mL/min/1.73 m² in the tacrolimus group versus 56 mL/min/1.73 m² in the cyclosporine group.²⁸⁸ Similarly, a NAPRTCS data analysis of 986 pediatric kidney transplant recipients treated with either cyclosporine or tacrolimus, in conjunction with mycophenolates and

steroids, found no difference in 1-year patient and graft survival but a significantly higher mean GFR at 1 year in patients treated with tacrolimus.²⁸⁹ A comparison of protocol biopsies in children demonstrated more subclinical acute rejection in those treated with cyclosporine.²⁹⁰ There is a lower prevalence of cardiovascular risk factors in children treated with tacrolimus, compared with cyclosporine or sirolimus.²⁸⁴

Tacrolimus's adverse effects are similar to those of cyclosporine, with a higher rate of post-transplant diabetes mellitus and neurotoxicity for tacrolimus, manifested as tremors or seizures.²⁹¹ A frequent reason for changing treatment from cyclosporine to tacrolimus in children and adolescents is the lack of cosmetic side effects of tacrolimus. A once-daily modified-release formulation of tacrolimus has been shown to improve patient compliance.²⁹²

ANTIMETABOLITES

Azathioprine

While in the first years, most children were treated with azathioprine as part of their immunosuppression regimen, in the 2018 NAPRTCS report, only 2.1% to 3% of children received azathioprine since 2008.^{188,189} The largest pediatric study of 140 patients comparing mycophenolate with azathioprine found a higher 5-year graft survival with mycophenolate (90.7% vs. 68.5%; $P < .001$).²⁹³ Adverse effects of azathioprine include myelosuppression and hepatotoxicity, which might require dose reduction. The risk of myelosuppression can be reduced in pretesting for Thiopurine S-methyltransferase (TMPT) activity and avoiding use in those with reduced activity/genetic mutations. The myelosuppressive action of azathioprine is potentiated by xanthine oxidase inhibitors such as allopurinol, and if necessary, the dose of azathioprine must be reduced by 50% to 75%. Although azathioprine is still classified by the FDA as category D for use in pregnancy, suggesting positive evidence of risk to the fetus, data accumulated over the past few decades, especially from the National Transplantation Pregnancy Registry, have shown that the incidence of birth defects is similar to the general population.²⁹⁴ Therefore azathioprine might present as an alternative to mycophenolate formulations for female kidney transplant recipients planning pregnancy. In addition, azathioprine is significantly less expensive than mycophenolate.

Mycophenolate

Mycophenolate mofetil, a reversible inhibitor of inosine monophosphate dehydrogenase that downregulates specifically T- and B-cell proliferation, was developed as an alternative antimetabolite to azathioprine, with less bone marrow toxicity. It is rapidly metabolized to mycophenolic acid in vivo. Some data suggest that in children with chronic allograft nephropathy (CAN), adding mycophenolate to the immunosuppressive regimen improves graft function.²⁹⁵ In 2018 mycophenolate formulations were used as part of the initial maintenance immunosuppression regimen in 87.7% of pediatric kidney recipients in the United States.^{188,190} Although initially dosing was 600 mg/m² BSA twice daily, some protocols give a lower dose (300 mg/m²) to low-risk patients, with fewer adverse effects.²⁹⁶

The main adverse effects of mycophenolate mofetil are gastrointestinal including abdominal discomfort, nausea, and diarrhea. Mycophenolate mofetil is available as capsules, tablets, and oral suspension, as well as intravenous preparation. Enteric-coated mycophenolic acid cannot be made into a suspension, which limits its use in younger children. There is debate over whether mycophenolic acid monitoring is beneficial in prevention of acute rejection. Data suggest that mycophenolate exposure is associated with time from transplantation, graft function, and age, with younger patients requiring higher doses per BSA in order to reach therapeutic AUC levels.²⁹⁷ However, trough C0 levels show only a weak correlation with AUC and do not associate well with adverse effects, so trough-level measurements are not commonly performed.²⁹⁸ Lower mycophenolate exposure has been shown to be associated with de novo DSA formation.²⁹⁹ Mycophenolate has been associated with neutropenia in pediatric kidney transplant recipients, often necessitating dose reduction or temporary interruption.³⁰⁰ Pregnancies in patients exposed to mycophenolic acid are associated with a high rate of birth defects; therefore adolescent girls and women of childbearing potential must be cautioned to use reliable contraception if sexually active while treated with mycophenolate.³⁰¹

CORTICOSTEROIDS

Corticosteroids have been a cornerstone drug in prevention of rejection of transplanted organs for decades. However, chronic corticosteroid therapy is associated with many adverse effects including HTN, hyperlipidemia, cardiovascular disease, glucose intolerance, osteoporosis and aseptic bone necrosis, cataracts, and glaucoma, as well as weight gain, cushingoid appearance, acne, and psychological effects. The cosmetic effects are often a cause of nonadherence in many patients, particularly adolescents. Corticosteroids also inhibit linear growth by several mechanisms of action including interference with the action of GH and inhibition of bone formation.³⁰² Efforts to reduce the adverse effects of corticosteroids in children undergoing renal transplantation have included using lower doses of prednisone, alternate-day regimens, steroid-withdrawal protocols, and steroid avoidance.

According to NAPRTCS, corticosteroid use was almost universal in children undergoing renal transplantation until the year 2000. Because of the previously mentioned dose-related side effects, different steroid discontinuing strategies have been used across various centers: rapid discontinuation of steroids (within 1 week post-transplant), early withdrawal (within 7–14 days post transplant), or late withdrawal (within the first year).

In OPTN data, the use of steroid minimization protocols increased from 8.7% in 2002 to 37% in 2009.³⁰³ However, centers that continue to use triple immunosuppression in low-risk patients generally use lower doses than previously used, tapering corticosteroid dose within a few weeks to doses of 0.1 mg/kg or 5 mg/m² or even lower.¹⁸⁹

Studies comparing maintenance immunosuppression regimens with and without steroids showed no difference in terms of patient and graft survival.^{296,304,305}

Large prospective, multicenter, randomized controlled studies, in both the United States and Europe, found improved linear growth with no increase in biopsy-proven

acute rejection (BPAR) in children on steroid avoidance or early steroid-withdrawal protocols.^{296,305} Improvement in height standard deviation scores was significant only in prepubertal children. Other benefits of steroid avoidance, such as lower rates of HTN and post-transplant diabetes, were inconsistent. Of note, most of these studies used daclizumab induction, which is no longer available, and although small studies have shown similar results with thymoglobulin induction, prospective studies using other IL-2 receptor blockers have not been performed.²⁶⁷ A meta-analysis showed an increase in height-standardized Z-scores but no increase in acute rejection with steroid-avoidance/withdrawal regimens compared with steroid-based regimens. The magnitude of post-transplant growth in the prepubertal recipients was greater than that in the pubertal recipients.³⁰⁶

Late withdrawal of steroid treatment in children treated with cyclosporine–mycophenolate immunosuppression also demonstrated improved linear growth with no increase in the rate of graft rejection.³⁰⁷ Corticosteroid avoidance protocols have not been well studied in high-risk pediatric transplant recipients including non-Caucasians, retransplants, and sensitized patients.

MAMMALIAN TARGET OF RAPAMYCIN INHIBITORS

Mammalian target of rapamycin (mTOR) inhibitors block the proliferative response of T cells to IL-2 stimulation. This group of immunosuppressive agents also inhibits proliferation of vascular muscle and some cancer cells and therefore can reduce transplant vasculopathy. This, together with lack of nephrotoxicity, made this group of drugs a promising treatment option for pediatric transplantation. Adverse effects of mTOR inhibitors include impaired wound healing, which can present a problem when used immediately after transplantation, hyperlipidemia, proteinuria, decreased levels of testosterone in some adolescent males, aphthous ulcers, interstitial pneumonia (including *Pneumocystis jirovecii* infections), anemia, and thrombocytopenia.³⁰⁸

Sirolimus

The first drug in this group was approved for use in adult solid organ transplantation in 2000. Initially, it was used to avoid calcineurin inhibitors completely, by combination with IL-2 receptor blocker induction, mycophenolate, and steroids. A higher incidence of acute rejection was seen in this protocol and increased adverse events.³⁰⁹ Subsequently, sirolimus was used in steroid-sparing protocols, together with low-dose cyclosporine in a study that was terminated due to a higher incidence of PTLT.³¹⁰ Conversion from calcineurin inhibitor to sirolimus to treat calcineurin inhibitor–induced nephrotoxicity has also been attempted with improved GFR in some children, but with an increase in adverse events.^{311,312} The use of sirolimus in pediatric kidney transplantation peaked in 2002 at 25% but subsequently decreased to <1% in the United States,¹⁸⁹ with a similar trend seen also in other registries.^{188,193}

Everolimus

Everolimus has been studied in several small, uncontrolled studies in children, generally instead of mycophenolate treatment and in combination with low-dose calcineurin inhibitors. A study based on data from the CERTAIN

registry compared children treated with everolimus with low-dose cyclosporine and steroids with those on standard immunosuppression. There was no increase in biopsy-proven rejection, and graft function was similar between the groups, although side-effect profiles differed. The everolimus group had a higher incidence of hyperlipidemia, HTN, and anemia than the control group.³¹³ Further analysis showed a significant decrease in CMV replication and disease in the group treated with everolimus, most likely due to the avoidance of mycophenolate.³¹⁴ Another study, with a similar treatment protocol with no control arm, also showed promising long-term results. To avoid wound-healing problems, most protocols delay sirolimus or everolimus introduction until a few weeks after transplantation.²⁸²

OTHER IMMUNOSUPPRESSIVE MEDICATIONS

In most cases, new immunosuppressive agents are studied in adults before trials in children are performed. Belatacept, a CD28 pathway costimulation inhibitor, has shown promise in adult renal transplant patients, demonstrating improved GFR compared with cyclosporine, though with a higher incidence of acute rejection. Only one single-dose pharmacokinetic study has been performed in children to date,³¹⁵ while a National Institutes of Health–funded multicenter conversion study is under way. A specific concern is a higher frequency of PTLD in EBV-seronegative patients, particularly of the central nervous system. As more children are EBV naïve going into transplantation, the potential risk of PTLD could be even higher in this population.³¹⁶ A potential advantage is intermittent intravenous administration, which could help overcome drug nonadherence, particularly in adolescent patients.

IMMUNOSUPPRESSION COMBINATIONS

From NAPRTCS data, the most common combination of immunosuppression for pediatric kidney transplantation in the past decade is tacrolimus, mycophenolate, and prednisone (48.9% at day 30 post transplantation), followed by tacrolimus and mycophenolate without corticosteroids (33.8% at day 30).^{188,189} Other combinations include tacrolimus with low-dose prednisone as dual therapy or with azathioprine. No difference was seen in the frequencies of the immunosuppression combinations between deceased versus living donor recipients.

PERIOPERATIVE MANAGEMENT

Pediatric renal transplantation should always be managed by an experienced team, which includes pediatric experts in the fields of transplant surgery, nephrology, anesthesiology, intensive care, and other specialties, to ensure an optimal outcome.

Technical causes are responsible for a higher proportion of early graft loss in young children, primarily from graft thrombosis. Although the graft kidney is usually placed extraperitoneally in the iliac fossa in adults and older children, a midline intraperitoneal approach is often used in youngest patients. Vascular anastomoses of the graft in small children are usually to the aorta and IVC rather than the iliac vessels used in older children and adults, to provide

sufficient renal perfusion, particularly when the graft is an adult-sized kidney. The donor renal artery may be the same size as the recipient's aorta, and the renal vein may be significantly larger than the recipient's IVC. A straight and direct course for each renal vessel to the aorta and vena cava will avoid kinking and decrease the risk of thrombosis. Stenosis or thrombosis of the IVC is a rare finding but may be seen in children who have had CVCs in infancy, abdominal surgery, or a hypercoagulable state, posing a problem for graft anastomosis. This should be evaluated pretransplant by Doppler study or venography, and in case of significant IVC pathology an alternative vein should be selected. There may be an advantage for transplanting a pediatric (deceased) donor kidney in such cases.^{317,318}

The adult-sized kidney graft requires a major increase in CO to maintain good perfusion and avoid acute tubular necrosis (ATN) or vascular thrombosis. Even with optimal maintenance of recipient intravascular volume, blood flow to the adult-sized donor kidney is significantly reduced at transplantation. Further reduction in blood flow could result in graft ischemia; therefore BP values, which may be relatively high for the young child, and high-normal central venous pressure need to be maintained during transplant surgery and postoperatively. After anastomosis is completed, release of vascular clamps will cause immediate flow of a large portion of the child's blood volume into the graft, a critical time point when intravascular volume must be controlled. Maintaining relative hypervolemia immediately post transplant may delay weaning from mechanical ventilation for 1 to 2 days but is nonetheless of utmost importance to preserve graft perfusion.³¹⁹ Aggressive hydration to optimize intravascular volume and graft blood flow is important in the following weeks and months and may require fluid supplements by the nasogastric route.³¹⁷ Avoiding ATN in the postoperative period also enables early full dosing of calcineurin inhibitors, thus preventing early acute rejection.

IMMEDIATE POSTOPERATIVE PERIOD

In the first few days after transplantation, the smaller child is usually treated in the intensive care unit. Close attention is paid to vital signs and fluid management with the goal of optimizing graft perfusion. Brisk urine output, particularly in a patient previously oliguric or anuric, is a useful marker of graft function. Oligoanuria or a significant decrease in urine output, by contrast, needs to be evaluated and treated immediately. Bladder drainage must be ensured by flushing or exchanging the bladder catheter. Ultrasound can assess the kidney, the ureteral stent, the blood vessels, ureter, and bladder and provide information on organ perfusion, and the presence of a fluid collection suggesting leak or bleeding. Absent blood flow may be due to thrombosis and necessitate prompt surgical intervention, as do significant bleeding or a urinary leak. Intravenous fluids are given to replace insensible losses and urine output, as well as additional fluids to account for third space losses and to keep central venous pressure on the high side of normal. This approach should maintain good graft perfusion, which in turn will increase urine output. However, if urine output does not increase, intravascular volume overload may

result, manifesting as respiratory compromise and HTN.³¹⁹ Volume overload may respond to diuretics, but dialysis may be necessary in some patients.

Immunosuppression is started at transplantation including intravenous corticosteroids perioperatively in most cases (except for complete steroid-avoidance protocols). Initiation of calcineurin inhibitors may be delayed for several hours or longer if there is postoperative oliguria. Prophylactic treatment for infections is started postoperatively. Perioperative antibiotics are usually given to prevent wound infection.³²⁰ *P. jirovecii* prophylaxis, usually with trimethoprim-sulfamethoxazole (TMP-SMX), may also decrease the incidence of urinary tract infection (UTI). Antiviral medication is given to prevent CMV disease, usually oral valganciclovir, with the duration of therapy depending on pretransplant donor and recipient serologic status and on whether lymphocyte-depleting induction immunosuppression was given.³²¹ Some centers prescribe antifungal prophylaxis, especially in patients with prolonged intensive care unit stays or antibiotic treatment. Low-molecular-weight heparin or aspirin may be given after transplantation to prevent graft thrombosis, particularly in younger patients and in patients with abnormal pretransplant coagulation workup.³²²

EARLY COMPLICATIONS OF KIDNEY TRANSPLANTATION

DELAYED GRAFT FUNCTION

Delayed graft function (DGF) is usually defined as the need for dialysis during the first week after renal transplantation and is sometimes referred to as ATN.¹⁸⁹ The most common cause of DGF is postischemic acute injury of the graft, which is associated with deceased donor kidney, prolonged ischemic time, and hemodynamic compromise during and immediately after transplantation. Changes in allocation policies in the United States resulted in deceased donor organs being shipped over longer distances, leading to increased cold ischemic times, with an increase in DGF rates.³²³ The incidence of DGF is higher in children receiving kidneys from donors after cardiac death (DCD), although pediatric DCD kidney recipients have good renal allograft survival at 3-year follow-up.²⁰⁶ DGF has been shown to adversely affect long-term graft survival and function, with children who required post-transplantation dialysis having a twofold risk of overall graft loss.³²⁴ DGF is also associated with a higher risk of subsequent acute rejection in both living and deceased donor pediatric kidney transplantation.¹⁸⁹

Other causes of DGF include graft thrombosis, which can be diagnosed by ultrasound or radionuclide scan, and may require surgical revision. Thrombosis can be prevented by meticulous surgical technique and attention to fluid balance, as well as identifying and treating patients with thrombophilic risk factors before transplantation. Use of antibody induction may decrease the incidence of postischemic DGF, although the reasons for this are not clear.³²⁵

Urologic complications such as urinary leak or obstruction may also cause DGF or acute decrease in urine output, and diagnosis is aided by ultrasound and radionuclide scan.

Ureteral stents are often placed at transplantation for 4 to 6 weeks to prevent obstruction, but their use may be associated with increased risk of early UTI or later BK virus complications.³²⁶⁻³²⁸

Hyperacute rejection or accelerated acute rejection superimposed upon postoperative ATN may also present as DGF. Hyperacute rejection is usually due to preformed antidonor ABO or HLA antibodies, manifesting as irreversible rapid destruction of the graft within minutes to hours of transplantation. Because blood type and HLA matching have been routinely performed, these reactions are exceedingly rare. Pretreatment of recipients with grafts mismatched for ABO antigens or with DSA with desensitization protocols generally prevents this disastrous outcome.³²⁹

Recurrence of FSGS can manifest as primary nonfunction and requires a graft biopsy for diagnosis to guide treatment.²²⁵

ACUTE REJECTION

Acute renal allograft rejection is a common complication of pediatric renal transplantation, though its frequency has decreased significantly from >70% in 1987 to 1990 to 13% in the 2012–2017 era.^{188,189} In NAPRTCS, a rejection episode is defined by the physician's decision to initiate specific antirejection therapy rather than BPAP. However, in the latest cohort the frequency of biopsy in reported acute rejections has increased from 45% in 1987 to 94%.

The lower incidence of acute rejection in recent years is thought to be due to newer, more potent immunosuppressive medications now used. The highest risk of acute rejection is in the first few weeks to months; however, it may occur at any time, even years after transplantation.³³⁰ It is more frequent after deceased donor transplantation compared with living donor transplantation, suggesting susceptibility of the graft with longer ischemic time (e.g., due to exposure of antigens promoting an immune reaction). Although in earlier patient cohorts risk factors for acute rejection included African-American race and no induction therapy, in the latest NAPRTCS cohort (2012–2017) these factors were no longer significant. This may be a result of differential practices in choice of immunosuppression protocol in African-American children.²⁸¹ As noted earlier, DGF due to ATN continues to be a significant risk factor for subsequent acute rejection.¹⁸⁹

Acute rejection may often be clinically asymptomatic. Graft tenderness and fever as manifestations are unusual in the present era of immunosuppression, and acute decrease in GFR may be the only sign. GFR may decrease significantly before an increase in serum creatinine is noted, particularly when baseline creatinine is normal. A prompt workup should include urinalysis and urine culture to rule out infection and glomerular disease, BK virus (BKV) polymerase chain reaction (PCR), and ultrasound to look for obstruction, fluid collection, or vascular compromise. Biopsy should be performed whenever possible to confirm a diagnosis of acute rejection, as other causes of graft dysfunction, such as calcineurin inhibitor toxicity or viral infection including BK nephropathy, may be difficult to distinguish from acute rejection on clinical grounds alone. Graft biopsy is safely performed under ultrasound guidance and light general

anesthesia or conscious sedation is helpful in reducing anxiety and pain. The Banff criteria are used to grade acute rejection.³³¹ Acute cellular rejection is characterized by lymphocyte infiltration of tubules (tubulitis) and interstitial inflammation, with involvement of arterial walls in more severe cases. In acute ABMR there is acute tissue injury including findings of microvascular inflammation and often also acute tubular injury, glomerulitis, arteritis, or fibrin thrombi in glomerular capillaries. In addition, there is evidence of antibody-mediated complement activation—C4d deposition in the peritubular capillaries of the allograft, although C4d-negative ABMR is recognized. The biopsy findings together with circulating DSAs complete the diagnosis of ABMR. The two types of acute rejection may coexist.³³¹ The question of whether to perform routine surveillance graft biopsies to help tailor immunosuppressive therapy and detect subclinical rejection is unresolved.³³² Serial monitoring of DSA may help identify high-risk patients, adjust immunosuppression, and prevent overt antibody-mediated injury.³³³ Several novel biomarkers such as peripheral blood donor-derived cell-free DNA or gene expression panels may allow for more targeted allograft biopsies.^{334,335}

The cornerstone of therapy for acute cellular rejection is high-dose corticosteroids, most often pulse intravenous methylprednisolone. In steroid-resistant rejection, lymphocyte-depleting agents, particularly thymoglobulin, may be administered. Premedication with methylprednisolone and antihistamine usually prevents a cytokine release syndrome, which may be seen particularly with the first dose of thymoglobulin. Approximately 50% to 60% of children treated for acute rejection will have a complete response, with GFR returning to baseline, and another 30% to 40% will have a partial response. Complete reversal of acute rejection is less likely in repeated episodes.¹⁸⁹ When acute rejection is diagnosed, the possibility of underimmunosuppression must be addressed. This may be due to underdosing of medications or, more commonly, particularly in teenage patients, nonadherence to immunosuppressive drugs.²¹⁶ Attention must be given to maintenance immunosuppression, considering adding prednisone to a steroid-free protocol, switching cyclosporine to tacrolimus, or increasing the antimetabolite dose or the target levels of tacrolimus to improve response to therapy and reduce the incidence of recurrent acute rejection.³³⁶

ABMR is often refractory to conventional antirejection therapy and has a high rate of graft loss. Few randomized controlled studies have been performed regarding treatment of ABMR, in adults or in children, and there is insufficient evidence to guide therapy.^{337,338} The first line of treatment usually includes high-dose corticosteroids, plasma exchange, and IVIG. The goal of plasma exchange and IVIG is removal of DSA from the circulation.³³⁹ Second-line therapies target antibody production by depleting B cells (rituximab) or plasma cells (bortezomib, carfilzomib). Recent studies have shown stabilization of graft function in children with refractory ABMR after treatment with bortezomib.^{340,341} Inhibition of complement with eculizumab to decrease the antibody-mediated injury has been described in a small number of pediatric patients.³⁴² Less severe cases of ABMR (e.g., the presence of DSA with mild findings of ABMR on graft biopsy, but without significant deterioration of

graft function) might be treated with corticosteroids and IVIG with or without rituximab. This approach avoids the need for CVC insertion and plasma exchange in lower-risk patients.³⁴⁰

Acute rejection is the most common cause of graft loss in the first year after transplantation and has a significant negative impact on long-term graft survival.^{207,330,343} Close monitoring is imperative to avoid early acute rejection and enable its prompt detection and treatment. Serum creatinine is monitored daily in the first few days after transplantation, followed by at least twice weekly for 2 to 3 months, together with drug concentration monitoring. Later, if there are no complications, the frequency of testing is decreased to once a month after the first 6 to 12 months. DSA can be monitored regularly during the first months after transplantation and periodically after that. This is especially important in children with pretransplantation sensitization or at any time if there are signs of graft dysfunction.³⁴⁴ Even stable children and adolescents require regular follow-up at the pediatric transplant clinic to help prevent nonadherence and the resulting risk of graft rejection, as well as long-term management of many other aspects of their health.

INFECTIONS

As more potent immunosuppressive agents were introduced into the treatment of renal transplantation, the frequency of acute rejection decreased, as did hospitalizations for rejection episodes. In the pediatric population in particular, the price for this has been a higher rate of infections and hospitalization for infection.³⁴⁵ The use of antibody induction and presence of indwelling vascular or urinary catheters and presence of post-transplantation neutropenia increase the risk of infection.³⁰⁰ Hospitalization for infection was higher in children after renal transplantation at 47% in the first 3 years than in adults.³⁴⁶ In the immediate post-transplant period, the most frequent infections are surgical, UTIs, and pneumonia. Subsequently, in the early months after transplantation, opportunistic viral infections are common, particularly CMV, EBV, and BKV nephropathy, which cause significant morbidity and may adversely affect graft function. In the prospective multicenter Immune Development in Pediatric Transplantation (IMPACT) study, 73% of all children developed viral DNAemia (EBV, 34%; CMV, 23%; BK virus, 23%; and JC virus, 21%) in the first year after kidney transplant.³⁴⁷

In pediatric kidney transplant recipients hospitalized for fever, the rate of bacterial illness is high (50%) and warrants a full workup, with consideration for empiric antibiotic treatment.³⁴⁸

URINARY TRACT INFECTION

UTI is a common complication in children who have undergone renal transplantation, with hospitalization of 16.8% of patients in the first 3 years.³⁴⁶ The risk of UTI after pediatric kidney transplant has increased over the past 2 decades.³⁴⁵ Risk factors include a pretransplant diagnosis of obstructive uropathy or VUR and bladder abnormalities.³⁴⁹ Ureteral stents placed during transplantation may also increase the risk of early UTI.³²⁶ Although graft dysfunction may occur during allograft pyelonephritis, long-term graft function may not be affected.³⁵⁰ TMP-SMX prophylaxis for

P. jiroveci infection may also reduce the incidence of UTI, but bacterial susceptibilities vary between medical centers and may render this strategy ineffective. Many clinics routinely culture urine from transplant patients at every visit; however, the indication for treatment of asymptomatic bacteriuria without pyuria remains unclear, as resistant organisms may emerge.³⁵¹ Recurrent UTI after kidney transplantation may be associated with VUR, which should be addressed (see the “Urologic Complications” section).

CYTOMEGALOVIRUS

CMV infection in pediatric kidney transplantation has decreased significantly since the widespread use of antiviral prophylactic treatment, yet it remains an important and occasionally severe complication. Up to 20% of pediatric kidney recipients in the current era will develop CMV infection (defined as evidence of viral replication) and 3% to 10% have CMV disease (viral replication with clinical disease).³⁵²

The most vulnerable group is children who are seronegative for CMV pretransplant, receiving a kidney from a seropositive donor (D+/R–), who develop a primary infection. Primary infection may also be community acquired in children whose donor was seronegative for CMV.³⁵² Children who receive thymoglobulin induction are also at a higher risk of CMV infection.³⁵³ According to OPTN data, more than half of children undergoing kidney transplantation from 2011 to 2015 had negative CMV serology, whereas only a quarter of adults were CMV negative at transplantation.¹⁹⁰ Secondary infection may also occur, either as reactivation of the host strain or as infection from the donor strain, and may be less severe clinically. The risk of acute graft rejection is higher following CMV infection in kidney transplant recipients and may be associated with increased graft loss.³⁵⁴ Decline in kidney graft function was more pronounced in children with CMV replication.³¹⁴

Valganciclovir has been shown to be at least as effective as oral ganciclovir in children for CMV prevention and has better bioavailability and easier once-daily dosing.³⁵⁵ Dosing should be based on BSA and corrected for renal function.³⁵⁶ There are two CMV prevention strategies: prophylaxis and preemptive antiviral treatment. In adults, the incidence of CMV infection and disease is lower in patients receiving prophylaxis; however, there is no significant difference in graft outcome or death between the strategies.³⁵⁷ Most prophylaxis protocols are 3 to 6 months long, depending on immunosuppression regimen and patient serostatus, but the optimal length of prophylaxis is unknown.³⁵² Viral monitoring is important during prophylaxis and particularly in patients being managed preemptively or after valganciclovir discontinuation.³⁵⁸ Antiviral prophylaxis should also be considered when there is intensification of immunosuppression (e.g., when treating acute graft rejection). Adverse effects of valganciclovir include hematological toxicity—mainly leukopenia and neutropenia, as well as gastrointestinal discomfort or diarrhea, occasionally necessitating dose reduction or discontinuation of treatment.³⁵⁹

CMV infection is most common after cessation of prophylactic antiviral therapy. Asymptomatic CMV infection should be treated with intravenous ganciclovir; however, in stable older children and adolescents, oral valganciclovir

can be used as initial therapy.³⁵² CMV disease is defined as viremia with clinical symptoms, most commonly fever and malaise, with tissue-invasive disease in some cases, involving the respiratory, hepatic, and gastrointestinal tracts. Allograft dysfunction, as well as leukopenia and thrombocytopenia, are frequently observed laboratory findings. In invasive disease, treatment is usually started with intravenous ganciclovir, followed by oral valganciclovir, when clinical improvement and reduction in viral load are seen. Although the VICTOR study demonstrated equivalent efficacy of oral valganciclovir to IV ganciclovir in adult transplant recipients, it is not clear if these findings can be extrapolated to the pediatric population. To assist in viral eradication, immunosuppression should be reduced, usually by discontinuation of an antimetabolite drug—azathioprine or mycophenolate. Hyperimmune CMV globulin preparation may be given to the most severe cases, though its efficacy is questionable.³⁵² Treatment is continued until after viral eradication is demonstrated, followed by a prophylactic dose for several weeks. Relapse may occur after the end of therapy; therefore viral monitoring should be continued for several weeks after treatment cessation and after reintroduction of the antimetabolite, which is usually at a lower dose than before CMV infection. Failure to respond to therapy, either clinically or virologically, may be due to viral ganciclovir resistance, usually due to viral UL97 mutations. Treatment options include high-dose ganciclovir or switching to foscarnet, which is nephrotoxic, though studies in children are lacking.³⁶⁰

EPSTEIN-BARR VIRUS INFECTION

Similar to the situation with CMV serology, about half of the pediatric patients undergoing renal transplantation are seronegative for EBV, whereas only 10% of donors are EBV naïve.¹⁹⁰ Seronegative patients receiving grafts from EBV-positive patients are at a higher risk of infection, so EBV disease is more common in younger children. EBV infection may be asymptomatic, it can present as an acute viral syndrome, or it may be associated with PTLT. The risk of developing PTLT is higher when primary EBV infection occurs after transplantation. Children who are EBV negative before transplantation have a 2.9% incidence of PTLT in the first 5 years after transplantation, compared with 0.7% in children who are seropositive.¹⁹⁰ Of the patients with primary EBV infection, adolescents may be more likely to develop PTLT and have a poorer prognosis.³⁶¹

Monitoring viral load in peripheral blood by quantitative PCR, particularly in high-risk patients in the first year after transplantation, is important to identify infection and allow early intervention. A prospective study in Germany found that primary infection occurred in 63% of seronegative pediatric patients and reactivation or reinfection in 44% of seropositive patients in the first year, though many of the infections were asymptomatic.³⁶² More intensive immunosuppression is associated with a higher rate of symptomatic EBV infection. Prophylaxis with ganciclovir or valganciclovir in high-risk patients was shown to decrease the incidence of EBV, according to some but not all studies.³⁶³⁻³⁶⁵ Clinical disease usually presents with protracted fever, lymphadenopathy, tonsillitis, and hepatosplenomegaly. Laboratory findings may include leukopenia, atypical lymphocytes, elevated liver enzymes, and IgM antibodies

for EBV, and allograft dysfunction may occur. Diagnosis is confirmed by EBV nucleic acid viral load, which is also used to monitor the course of the disease and response to therapy.

EBV-associated PTLT is most commonly a B-cell proliferation induced by infection with EBV. There is continuum between EBV-associated diseases from mononucleosis to full-blown PTLT. Imaging will help in detecting lymphoid masses, which ideally requires biopsy to rule out malignant PTLT. Other than a chest radiograph and abdominal sonography, computed tomography and positron emission tomography can assist in staging of the disease and can guide biopsy.³⁶⁶ The cornerstone of treatment of EBV infection and prevention of PTLT is immunosuppression reduction. Strategies include discontinuation of the antimetabolite and lowering the target levels of tacrolimus. Antiviral therapy for persistent EBV viremia with valganciclovir remains controversial, with efficacy shown in the CERTAIN registry, but a meta-analysis did not show efficacy.^{363,365,367} Some children after renal transplantation may have persistent high EBV viral load without ever developing PTLT, though the risk of PTLT is higher.^{363,368}

BK VIRUS INFECTION

BK polyomavirus infection emerged as an important clinical entity in patients undergoing renal transplantation in the late 1990s, coinciding with the introduction of more potent immunosuppression regimens. BKV infection may result in BKV-associated nephropathy (BKVAN), manifesting as tubulointerstitial nephritis with graft dysfunction and potential graft loss, as well as ureteral stenosis and hemorrhagic cystitis.³⁶⁹ Although previously thought to be rare in children, recent data from NAPRTCS showed a prevalence of 4.6%, similar to that in adults.³⁷⁰ Some recent studies have shown a higher incidence (10%–27%) of pediatric kidney transplant patients with BK viremia detected by screening tests, though overt nephropathy is less common.^{371,372} BK transmission occurs in childhood when the virus remains latent in the tubular epithelial cells. Immunosuppression after transplantation may cause reactivation and high-rate viral replication in the kidney graft and is therefore considered a consequence of immunosuppression. Young recipients may develop primary BK infection transmitted from the donor kidney. Risk factors for BKVAN include BK donor/recipient D+/R-seromismatch (not commonly tested), antibody induction, DR mismatch, ureteral stent placement, or treatment for rejection.^{326-328,373}

The virus appears first in the urine (viruria) and, if progression occurs, is detected in the blood (viremia) by quantitative PCR for viral DNA. A diagnosis of BKV nephropathy is made in the presence of high-level viremia with an elevation in serum creatinine or with histologic changes (inflammation in areas of BKV-positive immunostaining).³⁶⁹ Detection of viruria has questionable clinical implications and is not recommended in current guidelines, as it is not correlated with graft damage, but does identify patients at risk for subsequent viremia. Monitoring plasma quantitative PCR periodically, particularly in the early months after transplantation, after acute rejection, and at any time when there

is an unexplained elevation of serum creatinine, is recommended in both adults and children.^{374,375} Immunohistochemical staining for SV40 viral antigen is highly specific for BKV nephropathy.³⁶⁹

The cornerstone of management of BKV infection is reduction of immunosuppression when significant BK viremia is detected, to expand BKV-specific immunity and promote viral clearance. This is usually done by discontinuation of the antimetabolite or decrease in the target levels of the calcineurin inhibitor, with the goal of prevention of progression to overt nephropathy.³⁷⁰ Treatment of established BKVAN is often less successful. Several medications with antiviral activity have been used to treat patients with BKVAN unresponsive to immunosuppression minimization, with varying degrees of success, but no randomized controlled studies have been performed.³⁶⁹ These include high-dose IVIG, quinolone antibiotics, cidofovir, and leflunomide, none with any rigorous evidence of efficacy.^{372,375-377}

A related polyoma family virus, the JC virus, can also create complications in kidney transplant patients. JC virus is associated with a rare but often fatal demyelinating disease, progressive multifocal leukoencephalopathy, in immunosuppressed states³⁷⁸; however, routine monitoring is controversial.

VARICELLA

Immunization against varicella-zoster virus is recommended before transplantation, as the use of live attenuated viral vaccine is controversial in immunocompromised patients.²²⁴ Loss of antibody response after transplantation has been reported in 25% of children who were vaccinated before transplantation.²²² In addition, these antibodies may have lower avidity and the cellular immune response may also be decreased in solid organ transplant recipients compared with healthy controls.³⁷⁹ Varicella infection in an unimmunized renal transplantation recipient may result in severe disease, with visceral involvement in some cases. Postexposure prophylaxis of unimmunized patients with varicella-zoster immunoglobulin should be administered as early as possible, and if clinical disease develops, treatment with intravenous acyclovir is initiated. Temporary discontinuation of the antimetabolite, or dose reduction, is usually recommended.³⁸⁰

OTHER INFECTIONS

Respiratory viruses

Respiratory viruses, in particular influenza virus infection, may be associated with severe morbidity in the immunosuppressed host. Disease severity may be related to the degree of immunosuppression and clinical presentation may be atypical. Diagnosis of presumed respiratory viral infection can be made by nasopharyngeal swab, wash, or aspirate testing for pathogens. Neuraminidase inhibitors such as oral oseltamivir can be started empirically for suspected influenza infection before virologic confirmation. Pretransplant vaccination is recommended (2 doses 4 weeks apart if the child is younger than 9 years old), as well as yearly seasonal inactivated influenza vaccine after transplantation. Vaccine immunogenicity is variable, but recent data suggest

that immunization reduces mortality and clinical severity, as well as infection-related acute rejection.³⁸¹

Cutaneous Warts

Cutaneous warts, usually due to infection with human papillomavirus, are common in children after renal transplantation, as are molluscum contagiosum lesions. Although malignant potential is low, these lesions can be numerous and cause distress due to cosmetic issues. Warts may be refractory to treatment or recurrent and may respond to adjustment of the immunosuppressive regimen if possible.³⁸² Topical treatment includes cryotherapy, but surgical removal may be required.

Pneumocystis Jirovecii Pneumonia

P. jirovecii pneumonia is a classic example of an opportunistic infection, which is rare in the immunocompetent host. With the widespread use of prophylaxis, usually with TMP-SMX, the rate of invasive disease is low. Intravenous pentamidine can be given if TMP-SMX is contraindicated.³⁸³ Antipneumocystis prophylaxis is usually given for 6–12 months after kidney transplantation, after which net immunosuppression is lower.

Infectious Diarrhea

Diarrhea is a common symptom experienced by children after kidney transplantation. In some cases, it can be attributed to medication, such as mycophenolate or valganciclovir. It is usually mild and resolves with a decrease in dose or change of formulation.³⁵⁹ Rarely, mycophenolate may cause severe colitis, which improves with discontinuation of the drug.³⁸⁴ In up to 60% of children with diarrhea, an infectious cause can be diagnosed, whether viral, bacterial, or parasitic.

In immunosuppressed patients, diarrhea from the parasite cryptosporidium may be protracted and can cause hypovolemia, perturbations in calcineurin inhibitor serum concentrations, and AKI. Treatment with nitazoxanide is often necessary, as well as supportive measures.^{385,386}

Organ transplant recipients can also have persistent norovirus infection. Chronic infection with viral shedding may continue for several months.³⁸⁷ In many cases hospitalization for fluid therapy is necessary, and 30% still have diarrhea after 2 weeks.³⁸⁸

RECURRENCE OF PRIMARY RENAL DISEASE

According to NAPRTCS data, 7% of grafts in children are lost to disease recurrence.^{188,190} The most common recurring kidney disease is FSGS, but MPGN (type II in particular), aHUS, PH1, and others can recur in the graft. A study from the ANZDATA database demonstrated that younger patients have a higher risk of recurrence of glomerular disease after transplantation, in comparison with older adults.³⁸⁹ It is important to identify patients at risk for recurrence to enable early diagnosis and therapeutic intervention (see also in “Recipient Evaluation” section earlier).

FSGS is responsible for approximately 10% of ESKF in children in the United States, presenting as SRNS, and carries an overall risk of 30% for recurrent nephrotic syndrome.¹⁸⁹ Native kidney nephrectomy before or during transplantation surgery, in children with persistent proteinuria, may allow prompt diagnosis of recurrence without confounding.

Clinical presentation may include severe proteinuria immediately after transplantation or later in the course, but DGF or anuria may also be a sign of disease recurrence. Humoral factors cause damage to the glomerular barrier, which is the rationale for treatment strategies aimed at removal of such a factor. Serum-soluble urokinase-type plasminogen activator receptor has been identified as a possible permeability factor responsible for recurrent FSGS, though this has not been supported in pediatric studies.³⁹⁰ Preemptive and therapeutic plasmapheresis or low-density lipoprotein apheresis for disease recurrence, as well as treatment with anti-CD-20 antibody rituximab or the CTLA4 blocking fusion proteins abatacept or belatacept, have shown some efficacy in achieving remission of proteinuria.^{235,391-393} Recurrent FSGS in the graft, particularly if unresponsive to therapy, is associated with a higher rate of graft failure.

MPGN is the underlying diagnosis in 1.7% of children receiving their initial kidney transplant, may recur in 20% to 45% after transplantation, and is a cause of graft failure in many of the patients.³⁹⁴ When studying MPGN type I and type II separately, the risk of 5-year graft loss was 23.5% for MPGN type I and 67.5% among patients with type II, but the number of children with MPGN type II was small.²³⁸ The newer classification of MPGN is based on glomerular injury mechanism rather than the location of and appearance of deposits.

aHUS may recur in up to 60% of patients undergoing kidney transplantation without specific intervention. Most will progress to graft loss unless promptly treated with plasmapheresis or complement inhibition.³⁹⁵ Eculizumab, a monoclonal antibody of complement factor 5, inhibits terminal complement activation and was found to be effective in the treatment of aHUS.²⁴⁰ It has become the treatment of choice to prevent and treat recurrence of aHUS after transplantation with excellent results, as shown in a case series from France. Three patients in whom treatment was discontinued experienced a relapse, suggesting that long-term therapy is needed.²⁴⁰ Patients treated with complement inhibitors are at risk for meningococcal infection, and preventive measures must be taken.³⁹⁵

In PH1, treatment of choice is combined or sequential liver–kidney transplantation or lumasiran with kidney transplantation. However, children who have been treated with dialysis for a long period before transplantation may have significant oxalate deposition, mostly in the bones. Despite the discontinuation of oxalate production, achieved by liver transplantation, mobilization of the systemic oxalate burden can cause damage to the kidney graft. Meticulous attention to hydration, use of crystallization inhibitors such as oral citrate, or even HD immediately after transplantation can help protect the new kidney.³⁹⁶

CHRONIC ALLOGRAFT NEPHROPATHY

The most common cause of graft loss after the first year is CAN, accounting for about 35% of failed kidney grafts, according to several pediatric databases worldwide.^{189,193} The term CAN replaces previously used terms including chronic rejection, transplant nephropathy, and chronic renal allograft dysfunction or injury. It is primarily a histopathologic diagnosis, used to denote features of

chronic interstitial fibrosis and tubular atrophy within the renal allograft without evidence of any specific etiology. Pathologic involvement also typically includes the blood vessels with arteriolar hyalinosis and fibrointimal hyperplasia, as well as chronic transplant glomerulopathy.³⁹⁷ The pathogenesis is not completely understood, but numerous etiological factors have been implicated, both immune and nonimmune. Clinically, CAN is suspected when there is a slow deterioration in allograft function beyond the first 3 months after transplantation, with no evidence of acute rejection, drug toxicity, infection, or other pathological process. There may also be increasing proteinuria and worsening HTN. The prevention and management of chronic renal allograft rejection remains one of the major challenges facing clinicians. Adequate long-term immunosuppression is important in prevention of immunologically mediated CAN, although chronic calcineurin inhibitor toxicity may be difficult to distinguish from CAN in some cases. Nonadherence to medical therapy in pediatric renal transplant recipients is associated with a higher rate of interstitial fibrosis and tubular atrophy.³⁹⁸ Changes in immunosuppression may occasionally ameliorate CAN, for example, treatment with mycophenolate in children with CAN may improve graft function.²⁹⁵ Optimal control of HTN, with possible preference for angiotensin-converting enzyme (ACE) inhibitors, and treatment of hyperlipidemia are also important. One future direction is clinical utilization of urinary biomarkers to help differentiate CAN from other graft injury phenotypes.³⁹⁹

PATIENT SURVIVAL

Survival of children after renal transplantation has improved in 2007 to 2017 compared with 1987 to 1996 and is 96.4% for recipients of deceased donor allografts and 97.4% for recipients of living donor allografts, at 5 years.¹⁸⁸ Survival is lower in infants younger than 24 months at transplantation; however, this group of patients is small and has also shown the most significant improvement in the past 2 decades.¹⁸⁹ The most common cause of death is infection (27.9%), followed by cardiopulmonary causes (14.5%) and malignancy (11.3%). Adjusted all-cause mortality is approximately fourfold higher in children on dialysis than in children after renal transplantation.¹⁸⁶

GRAFT SURVIVAL

Graft survival has improved greatly over the past 3 decades, and the gap between living donor versus deceased donor sources has narrowed. In NAPRTCS data, 5-year deceased donor graft survival in the 2012–2017 cohort is 9.49% for living donor source and 90.1% for deceased donor source, compared with 78.1% in the cohort of 1996 to 2004 and only 62.4% in patients transplanted in 1986 to 1995.¹⁸⁸ Graft failure was defined by return to dialysis, retransplantation, or death. When assessing graft loss from 2008 to 2017, the most common cause of graft failure was CAN or chronic rejection (21.8%), with various types of acute rejection responsible for 15% of graft loss. Disease recurrence was the cause of graft loss in 9.8% of pediatric

patients, with FSGS accounting for most of these. In comparison with adults, death with functioning graft was less common (12%), whereas vascular thrombosis or technical difficulties were more frequent (10.5%).¹⁸⁹ This is due to better general health in pediatric kidney recipients on the one hand, and smaller blood vessels, which make vascular anastomoses technically challenging, on the other. In the 2008–2017 cohort, for living donor recipients, significant covariates impacting allograft survival include history of prior dialysis (worse survival) and more recent transplant year (better survival). For DD recipients, the only significant covariate was male gender (better survival). Notably, recipient age <24 months, black recipient race, and prolonged cold storage time, which were previously associated with worse survival, did not impact graft survival in recent years.

GFR, estimated by serum creatinine–based formula has improved over each successive decade: the median eGFR at 7 years post transplant now ranges from 65 to 77 mL/min 1.73 m² across the recipient age groups.¹⁸⁸

Irreversible graft failure may be associated with an inflammatory reaction, which may be severe enough to necessitate graft nephrectomy. This is more common in early graft loss or high-grade rejection by biopsy. The impact of graft nephrectomy on alloimmunization is difficult to determine due to multiple potential confounding factors.⁴⁰⁰

LONG-TERM FOLLOW-UP OF CHILDREN WITH KIDNEY TRANSPLANT

HYPERTENSION

BP monitoring is an integral part of regular follow-up of pediatric kidney transplant recipients. Appropriate size cuff should be used, and at least two or three measurements should be performed when the child is calm and seated for a few minutes. BP measurements are compared with normal value for gender, age, and height percentile, as published in the fourth report on the diagnosis, evaluation, and treatment of high BP in children and adolescents, which provides 90th and 95th percentiles.⁴⁰¹ The vast majority of children reaching ESKF have HTN, defined as values above the 95th percentile, and it remains a problem for many patients after kidney transplantation. In a European study reporting data from 10 countries, about two thirds of children receiving KRT had HTN, but uncontrolled HTN was more common in children on dialysis than in transplanted patients.⁴⁰²

In the early period after transplantation, HTN is common due partly to volume overload, early graft dysfunction, higher doses of corticosteroids and calcineurin inhibitors, or kinking of the transplant renal artery. HTN may appear months or even years after transplantation and is more common in older recipients. In a study from The Netherlands, the prevalence of uncontrolled HTN, defined according to Kidney Disease Improving Global Outcomes guidelines, was 75.8% in adolescents and young adults compared with 38.3% in younger children and was not found to be affected by transition to adult care facilities.⁴⁰³ Ambulatory 24-hour BP monitoring is a useful tool in diagnosing masked or nocturnal HTN, which are phenomena frequently seen in children with kidney transplantation.⁴⁰⁴

Any injury of the allograft, such as acute rejection or recurrence of glomerular disease, can cause or worsen HTN. CAN, like CKD of any cause, is also strongly associated with HTN. Increased prevalence of HTN is seen with use of calcineurin inhibitors, particularly cyclosporine, but also tacrolimus.⁴⁰⁵ The most common etiology of secondary HTN after transplantation is transplant renal artery stenosis. This entity may present as new-onset resistant HTN, occasionally with hypokalemia or worsening allograft function. A bruit may be heard over the kidney allograft and duplex ultrasonography often detects the stenosis, although computed tomography, MRI, or angiography may be necessary for diagnosis. Treatment of transplant renal artery stenosis depends on the severity and response to medication; however, angioplasty with or without stenting is the definitive treatment.⁴⁰⁵

Although post-transplantation HTN may be partly due to native kidneys, it often improves with a well-functioning graft and only rarely warrants native nephrectomy. Obesity, which predates transplantation or develops subsequently, is also associated with HTN and metabolic syndrome.⁴⁰⁶

Treatment of post-transplantation HTN involves lifestyle modification including exercise, reducing salt intake, weight control, and smoking avoidance. Antihypertensive medication is prescribed as needed, with target BP measurements under the 90th percentile for age and height, as in other high-risk pediatric groups.⁴⁰⁴ Choice of therapeutic agent depends on the underlying cause; ACE inhibitors and angiotensin receptor blockers may be appropriate in patients with HTN, particularly associated with proteinuria, but possibly hazardous in uncorrected allograft artery stenosis. Diuretics are helpful in treatment of early volume overload-associated HTN but are not ideal agents later because they induce a state of relative volume depletion. Calcium channel blockers may minimize calcineurin inhibitor-related renal vasoconstriction.⁴⁰⁴ Uncontrolled HTN may adversely affect graft function, as shown in adults in the Collaborative Transplant Study, which demonstrated that lowering systolic BP was associated with improved graft function.⁴⁰⁷

CARDIOVASCULAR DISEASE

LVH is the most frequent cardiac abnormality in children after renal transplantation, with a prevalence of 10% to 50% in various studies.⁴⁰⁸⁻⁴¹⁰ LVH is correlated with HTN and also with the presence of metabolic syndrome.⁴¹¹ In most children with LVH it appears before transplantation, and some studies have shown improvement in left ventricular mass index after transplantation.²³⁰

Cardiovascular mortality is much higher in children with CKD compared with the general pediatric population. Although mortality is lower in children after kidney transplantation than in children on dialysis, it remains unacceptably high. According to the NAPRTCS database, 13.9% of mortality is due to cardiopulmonary causes, second only to infection, and ahead of malignancy.^{188,189} Traditional risk factors such as HTN, dyslipidemia, diabetes mellitus, and obesity are more frequent due to impaired renal function, as well as the adverse effects of immunosuppressive medications. Nontraditional risk factors, such as abnormalities in calcium-phosphorus metabolism,

anemia, hyperuricemia, and hyperhomocysteinemia, are also common in children with CKD and are correlated with arterial stiffness and carotid intimal-medial thickness.⁴¹² Some of these risk factors may persist after transplantation, particularly if graft function is impaired. Arterial stiffness, increased pulse wave velocity, and endothelial dysfunction are common in pediatric kidney transplant recipients.⁴¹³ While vascular lesions progress in children with CKD, kidney transplantation can bring about improvement in some measures of endothelial function. However, cIMT may remain abnormal even years after kidney function is restored.^{414,415}

Interventions to reduce premature cardiovascular disease for young patients after renal transplantation include lifestyle modification, dietary modifications, and encouraging physical activity. Optimization of graft function and BP remain major factors in cardiovascular risk after kidney transplantation. One of the main goals of steroid avoidance or withdrawal protocols is mitigating risk factors for cardiovascular disease. Several studies in children have shown decrease in the frequency of post-transplant diabetes mellitus, hyperlipidemia, and HTN in children on steroid-free protocols compared with triple immunosuppression.^{307,416} Children treated with tacrolimus were less hypertensive and had better lipid profiles than those on cyclosporine-based protocols.^{289,417} Treatment with β -hydroxy β -methylglutaryl-coenzyme A (HMG-CoA) reductase inhibitors has been shown to be effective and safe in reducing total and LDL-cholesterol in children.⁴¹⁸ Although the ALERT study showed a reduction in cardiac deaths and nonfatal myocardial infarctions in adults, this has not been shown in other studies.⁴¹⁹ In children and adolescents, movement among study groups of CKD, dialysis, and transplantation is rapid and makes long-term study of statins challenging. Currently, there are no pediatric data on post-transplantation statin therapy with endpoints of cardiovascular events; therefore except in special circumstances, statin treatment is not recommended.⁴²⁰

METABOLIC SYNDROME AND DIABETES MELLITUS

Metabolic syndrome is less well defined in children and adolescents than in adults but is based on the finding of at least three abnormalities, such as obesity, elevated BP, low-HDL cholesterol, hypertriglyceridemia, and impaired glucose tolerance. It is increasingly common, and its frequency correlates with age. In the United States, prevalence is estimated at 1.2% in 12-year-olds and up to 7.1% in 16- to 17-year-olds.⁴²¹ Screening and prevention of metabolic syndrome in childhood is important because it frequently continues into adulthood and carries an increased risk for cardiovascular disease. Even in childhood, there is evidence of significant atherosclerotic changes. The frequency of metabolic syndrome in children in the United States after kidney transplantation is high, up to 37% in the first year, most of which is of new onset.⁴¹¹ Dietary intervention and physical activity are the mainstay of prevention and treatment. In patients treated with corticosteroids, late steroid-withdrawal has been shown to improve metabolic parameters and decrease the frequency of metabolic syndrome from 39% to 6% 2 years later.³⁰⁷

New-onset diabetes mellitus after transplant (NODAT) is less common in children than adult patients, with an incidence of 7.1% at 3 years. However, its frequency may be increasing. It is associated with older age (adolescents vs. younger children), higher BMI, and treatment with tacrolimus.⁴²² A study comparing early corticosteroid withdrawal with low-dose chronic prednisone treatment in adults did not show a significantly lower incidence of NODAT.⁴²³ Hypomagnesemia has emerged recently as a significant risk factor for NODAT in adults and children. Although tacrolimus is known to cause renal magnesium wasting and is also a risk factor for NODAT, it appears that hypomagnesemia is independently associated with NODAT.⁴²⁴ Further studies are necessary to elucidate the effect of magnesium supplementation. Fasting blood glucose should be monitored regularly in the first months after transplantation; hemoglobin A1C is also helpful in diagnosing insidious-onset diabetes. When diabetes is diagnosed, optimizing tacrolimus levels and minimizing corticosteroid dose should be considered. Referral to a pediatric diabetes specialist clinic for patient and parent education and a combination of dietary modification and pharmacologic therapy are needed. NODAT is not associated with decreased graft or patient survival in pediatric patients, and some cases may resolve spontaneously.³⁴³

CANCER

Immunosuppressive therapy for prevention of rejection after solid organ transplantation carries a risk for malignancy. In children, 2.6% of renal transplants develop a malignancy, of which 85% were classified as lymphoproliferative cancers.¹⁸⁹ The 3-year incidence of malignancy increased from 1% in 1987 to 1991 to 2.97% in 1997 to 2001, dropped to 1.51% in 2007 to 2011, and increased again to 2.31 in 2012–2017 for unclear reasons.¹⁸⁸ Compared with the general pediatric population, the incidence was significantly increased for hematological malignancies, particularly non-Hodgkin lymphoma, as well as other types of cancer including of the genitourinary tract.⁴²⁵ Long-term follow-up showed a sixfold increased rate of cancer compared with the general population. A larger single-center series in the United States found 13% of patients contracting some form of malignancy by 20 years after transplantation, with similar findings in a large study from the ANZDATA database—a cumulative incidence of nonskin cancer in 14% of pediatric patients at 25 years.^{426,427} A higher hazard of death or graft loss was seen in patients with post-transplantation cancer, compared with those with no malignancy.

PTLD is an uncontrolled proliferation of lymphocytes in the constellation of post-transplant immunosuppression, often associated with EBV infection. The cumulative incidence of PTLD in children in the United States is 1.7% at 5 years post-transplant but is close to 3% in children who are EBV seronegative at transplantation.¹⁹⁰ The risk of PTLD is 30-fold higher than in healthy controls and is higher in EBV-negative recipients of EBV-positive donor kidneys.^{426,427} A high EBV viral load is associated with the probability of developing PTLD.⁴²⁸ Diagnosis is made by histologic examination of involved tissue and is classified by the World Health Organization (WHO) 4th edition 2016 into four categories⁴²⁹: early lesions, polymorphic PTLD,

monomorphic PTLD, and Hodgkin lymphoma, while a 2022 International Consensus Conference has lumped post-transplant and immunodeficiency lymphoproliferations into a common category.⁴³⁰ Early lesions are usually managed by immunosuppression reduction in order to restore the host's virus-specific immunity and by monitoring of viral loads to assess response. Further treatment options include rituximab in CD20-positive disease, either alone or in conjunction with chemotherapy, either with a reduced chemotherapy regimen⁴³¹ or in risk-adjusted sequential fashion,⁴³² to limit morbidities. Non-B-cell lineage, Burkitt-type lymphoma, or central nervous system locations can require early chemotherapy. The source of the lymphoma is almost always the recipient rather than the donor kidney.⁴³³ Patients with PTLD are at higher risk for graft dysfunction and loss due to acute rejection or CAN.⁴³⁴

Other types of cancer in children after renal transplantation include the urinary tract, primarily renal cell or urothelial cell carcinoma, which may arise from the native kidneys or allograft.⁴³⁵ Nonmelanoma skin cancer is common in transplant recipients, though more frequently diagnosed in young adults than children. Regular use of high-protection factor sunscreen can reduce the incidence of skin cancer, and referral for an annual dermatological examination may help in detection of malignant and premalignant lesions.⁴³⁶

ANEMIA

Blood transfusion is necessary in up to half of pediatric kidney transplantations, predicted by pretransplant hemoglobin.⁴³⁷ Anemia is common in children after renal transplantation, ranging from 25% to 50% of patients.^{438–440} Incidence is highest in the early months after transplantation, after which many patients have a gradual increase in hemoglobin.⁴³⁸ Decreased graft function is strongly associated with post-transplant anemia because anemia is a marker of allograft dysfunction due to impaired erythropoietin production. However, immunosuppressive and antiviral drugs are also implicated in its pathogenesis.^{441,442} Other causes of anemia include nutritional deficiencies and blood loss, particularly in the perioperative period; viral infections; chronic inflammation; severe hyperparathyroidism; and treatment with ACE inhibitors. CMV infection is associated with anemia, as well as leukopenia and thrombocytopenia in some patients, and the anemia resolves with antiviral treatment. Parvovirus B19 infection has been described in pediatric kidney transplant recipients, presenting as severe anemia with reticulocytopenia and bone marrow findings compatible with pure red cell aplasia.⁴⁴³ Parvovirus infection can be confirmed by bone marrow and sometimes by peripheral blood PCR testing for the virus, and treatment with IVIG is successful in viral eradication and correction of anemia.⁴⁴⁴

Post-transplant anemia should be treated to optimize oxygen delivery to organs and improve patient well-being. It is important to avoid unnecessary blood transfusions, which may cause alloantigen sensitization. An iron-rich diet is insufficient, and oral iron supplementation is usually required (often ESAs). In some patients, parenteral iron sucrose or sodium ferric gluconate may be beneficial because it results in faster repletion of iron stores and is not

dependent on patient compliance.⁴⁴⁵ Complete blood count and iron stores parameters should be frequently monitored in pediatric kidney transplant recipients, and other causes of anemia, including B₁₂ and folate deficiencies, occult infection, inflammation, or blood loss, must be ruled out.

Leukopenia or neutropenia may also be seen after kidney transplantation and is usually noted in the first few months after transplantation.³⁰⁰ These conditions are associated with immunosuppressive medications, most frequently mycophenolate, though both viral infections and antiviral prophylactic therapy may also induce neutropenia.³⁵⁹ Drug withdrawal may increase the risk of rejection or viral infection, respectively, and granulocyte-colony stimulating factor therapy may accelerate recovery in severe cases.³⁰⁰

BONE HEALTH AND GROWTH

Successful kidney transplantation corrects many of the metabolic abnormalities that characterize CKD; however, some may persist, particularly when allograft function is impaired. New mineral disturbances may appear, such as hypophosphatemia, due to residual hyperparathyroidism and elevated levels of FGF-23, a phosphaturic hormone produced by osteocytes, particularly in the early post-transplant period.^{54,446} Vitamin D deficiency is common among children with CKD and does not significantly improve after transplantation, whereas PTH levels decrease in most patients and are inversely correlated with 25-OH vitamin D concentrations.^{447,448} Therapeutic interventions to promote bone health include steroid minimization or avoidance, promoting good nutrition, correction of metabolic abnormalities, and vitamin D deficiency. Weight-bearing physical activity can increase bone mass, especially in children and adolescents, who have not yet reached their peak bone mass and should be encouraged in pediatric kidney transplant recipients.⁴⁴⁹

At the time of transplantation, most children have linear growth retardation, with a mean height deficit of −1.62 standard deviation in the most recent cohort according to NAPRTCS data, improved from prior cohorts.^{188,189} After transplantation, reversal of the uremic milieu should allow normal GH/IGF-1 secretion and function; however, catch-up growth is not seen in all patients. Improved height deficit is observed in children younger than 6 years at transplantation, whereas other age groups tend to maintain a relatively normal growth rate without improvement in Z-scores. The increase in height Z-score in children undergoing kidney transplantation over the past 20 years is a result of many factors including optimizing rhGH use before and after transplantation, corticosteroid minimization and avoidance, and a lower incidence of acute rejection (necessitating additional steroid therapy).

Steroid avoidance or early steroid-withdrawal protocols have shown improved linear growth, particularly in younger children.²⁹⁶ Several studies have demonstrated the efficacy of rhGH treatment in pediatric kidney transplant recipients. Treatment with rhGH improves final adult height without adversely affecting graft function.⁴⁵⁰

UROLOGIC ISSUES

Ureteral stent placement at the time of transplantation has become standard of practice and may reduce early

obstruction or leak.⁴⁵¹ The stent is usually removed within a month to 6 weeks, but earlier removal may prevent stent-related adverse events including UTI, pain, and hematuria.⁴⁵² Late-onset ureteral obstruction is less frequent and has been reported to be associated with BKV infection.⁴⁵³ Children with ESKF due to posterior urethral valves are at a higher risk for ureteral stenosis and VUR, most likely due to abnormal bladder wall.^{454,455} VUR to the allograft is not uncommon, as shown in a study in adults, which found early VUR to the graft in 40.7% of patients, despite using extravesical mucosal tunneling technique to prevent reflux. There was no increased incidence of UTI or impact on graft function.⁴⁵⁶ In children with recurrent UTI after transplantation, voiding cystourethrography is recommended. In these patients, antibiotic prophylaxis together with behavioral treatment of voiding dysfunction, if present, may be helpful. Endoscopic correction of reflux in symptomatic children is a feasible option; however, postprocedural obstruction and graft dysfunction have been reported.^{457,458} Open surgical correction of VUR is an alternative approach that has been found safe.^{459,460}

Boys who undergo renal transplantation due to posterior urethral valves often have ongoing bladder dysfunction. Small defunctionalized bladders can increase in capacity and compliance after transplantation, but others might develop more severe bladder dysfunction over time and require later surgical intervention.⁴⁶¹ Bladder dysfunction may manifest as increased postvoid volume; daytime frequency, urgency, or incontinence; and nighttime incontinence. Inadequate bladder drainage may adversely affect allograft function. Some children require intermittent catheterization, particularly if they have undergone bladder augmentation.⁴⁶²

PUBERTY AND REPRODUCTION

Delayed onset of puberty, menstrual irregularity, and infertility are common in patients with CKD, particularly ESKF. After transplantation many of the hormonal disturbances resolve and fertility is usually restored. Most adolescents who underwent kidney transplantation in childhood have normal onset of puberty, although a minority have delayed sexual maturation.⁴⁶³ No difference in growth and pubertal development was noted between patients treated with mTOR inhibitors compared with conventional calcineurin inhibitor-based immunosuppression.⁴⁶⁴ A study of young men who had undergone kidney transplantation in childhood suggested that although reproductive hormone levels were normal, testicular volume and sperm counts were lower than in healthy controls, raising concerns about fertility.^{463,465}

Restoration of fertility in adolescent girls after kidney transplantation exposes them to the risk of unplanned pregnancy. Counseling regarding reproductive health is important to avoid unplanned pregnancy and sexually transmitted diseases. The form of contraception used must take into account safety and efficacy of the method and patient adherence.⁴⁶⁶ Pregnancy after renal transplantation is high risk to both mother and fetus, but with meticulous prenatal care in selected patients with stable graft function and well-controlled BP, excellent outcomes can be achieved.²⁹⁴ Timing is important, as

pregnancy in the first 2 years after transplantation may adversely affect graft function.⁴⁶⁷ Changes in medications including immunosuppressive regimen may be necessary before considering pregnancy, to avoid teratogenic effects of medications, such as mycophenolate, sirolimus, and ACE inhibitors.⁴⁶⁸ No increase in congenital malformations has been observed in infants fathered by kidney allograft recipients.⁴⁶⁹

NONADHERENCE

Nonadherence to the medical regimen remains a major barrier to long-term allograft survival. Data from the U.S. Renal Data System and the Scientific Registry of Transplantation in some groups of recipients showed that the age group 17 to 24 years is at high risk of nonadherence and graft failure.⁴⁷⁰ In the case of young children, parents or caregivers are responsible for medical care. Other risk factors for nonadherence include health system factors, such as insurance coverage for care and medications and treatment-related factors, like regimen complexity, side effects, and even taste and size of pills. Psychosocial factors such as poor social support, family distress, and lower health literacy also affect adherence.⁴⁷¹ In adolescent patients, additional factors include the desire to be normal, poor planning, and insufficient parental monitoring.²¹⁴

Adherence can be measured by missed appointments, prescription-filling patterns, self-report or patient questionnaires, variance in immunosuppressive drug levels, or various electronic monitoring systems.⁴⁷² Adolescents are at high risk for nonadherence and have shorter graft survival; in fact, between the ages of 17 and 24 years the adjusted hazard ratio (HR) of graft loss was 1.61 compared with ages 3 to 17 years old and 1.28 compared with young adults aged older than 24 years.²¹⁵ Persistent nonadherence is associated with acute rejection, graft loss, and death, as well as increased medical costs.^{330,473} Attention to adverse effects and trying to minimize them, as well as simplifying the medication regimen, may help improve patient adherence. The Teen Adherence in Kidney Transplant Effectiveness of Intervention Trial (TAKE-IT) randomized to automated text message dose reminders, while both groups received electronic monitoring identification of personal barriers, problem solving, and action planning, in addition to social support to improve adherence in adolescent patients.⁴⁷⁴ The text message reminder group had better adherence over time.

TRANSITION

Transition of adolescents or young adults from the pediatric to adult nephrology or transplant unit is a challenging process, which may be associated with increased risk of nonadherence and graft loss.⁴⁷⁵ Although some clinics may use chronologic age alone to guide timing, other measures such as medical adherence, patient competency, and responsibility for care should be taken into account. Use of specialized transition clinics may assist in a smoother process, less medication changes, and higher patient satisfaction.⁴⁷⁶ Practices vary between centers as to age at transition (usually 16–21 years), the utilization of specialized transition clinics, structured assessment of patient readiness, and the use of

educational tools.⁴⁷⁷ Strategies for avoiding the pitfalls of transition might include initiating the transition process early and involving the patient in an age-appropriate manner. Coordination with all members of the team and good communication, as well as transfer during a stable period, are crucial for a successful process.⁴⁷⁸

REHABILITATION AND QUALITY OF LIFE

Measures of general health and well-being, psychosocial adjustment, school attendance, level of education, neurodevelopmental outcome, psychological morbidity, and other parameters are important in evaluating overall quality of life. Several studies have used various questionnaires and scales to quantify psychosocial outcomes in these patients. Although health-related quality of life is higher in pediatric kidney transplant recipients compared with those remaining on dialysis, it is still lower than in healthy controls.⁴⁷⁹ Many issues remain pertinent (e.g., distress arising from physical symptoms and body image, especially height, adverse effects of medications, nonadherence, and psychological problems such as anxiety and depression). Neurocognitive development may be impaired in children with CKD, some of whom have lower scores on IQ testing than the general population, particularly in executive function and memory domains. Although children after kidney transplantation compared favorably with their counterparts treated with dialysis, their scores were below those of matched healthy children. Mild deficits in academic skills were also found.⁴⁸⁰ This difference persisted when children with neurologic comorbidities were excluded.⁴⁸¹ School performance is also affected by prolonged absences for medical reasons, which affect academic achievements and peer relationships.

A study from France found a lower level of education than in the general population and a lower rate of independent living, marriage, and parenthood, although most of the patients were employed. A significant correlation was found among education level, employment, marriage, independent living, and final height.⁴⁸² A similar study from the United States found a high rate of employment, as well as education levels and overall satisfaction equivalent to the general population, despite significant physical morbidity.⁴⁸³ Of note, in both these studies a large percentage of the patients were severely short and therefore different from recent cohorts who are treated with lower doses of corticosteroids or steroid-avoidance protocols, as well as rhGH. One study compared young adults with ESKF that presented in childhood with a group with later-onset disease. Patients with later clinical presentation were more likely to have a higher educational attainment, to be employed, and to be living independently than young adults with early-onset disease.⁴⁸⁴ A study on long-term rehabilitation in kidney-transplanted children in France found lower rates of living with a partner and of parenthood than the general population, and this was correlated with disease severity. Approximately 10% had intellectual disability, and the remainder were within the normal range of intelligence. When corrected for the parental educational level, most educational levels were comparable with the general population, as were professional occupational categories. Adult height was much improved

in the pediatric kidney transplant recipients in comparison with the earlier cohort and was no longer associated with most social outcomes.⁴⁸⁵

Although most adults with CKD who underwent kidney transplantation in childhood have good long-term outcomes, some challenges persist in integrating into work, independent living, and building a family. A multidisciplinary team approach helps provide individualized care to children and adolescents with CKD on dialysis and after kidney transplantation. The goal in these young patients is to achieve the best outcomes, not just of kidney function, but in all aspects of health maintenance and rehabilitation.

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